



ग्रीड कंट्रोलर ऑफ इंडिया लिमिटेड
(भारत सरकार का उद्यम)
GRID CONTROLLER OF INDIA LIMITED
(A Government of India Enterprise)



[formerly Power System Operation Corporation Limited (POSOCO)]

राष्ट्रीय भार प्रेषण केन्द्र / **National Load Despatch Centre**

कार्यालय : बी-9, प्रथम एवं द्वितीय तल, कुतुब इंस्टीट्यूशनल एरिया, कटवारिया सराय, नई दिल्ली - 110016
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संदर्भ संख्या : - GRID-INDIA/NLDC/MR/

दिनांक : 26.12.2023

सेवा में,

वितरण सूची के अनुसार

विषय:- निष्पादन रिपोर्ट – नवंबर 2023

महोदय,

आई०ई०जी०सी०-2010 के प्रावधान के अनुसार, नवंबर 2023 माह की अखिल भारतीय प्रणाली की निष्पादन रिपोर्ट संलग्न है।

धन्यवाद

भवदीय

(सुरजित बनर्जी)

मुख्य महाप्रबन्धक (प्रणाली प्रचालन)

(रा०भा०प्रे०के०)

संलग्नक : मासिक प्रणाली रिपोर्ट

वितरण सूची Distribution List

- 1 सचिव, के.वि.नि.आ. तीसरा एवं चौथा तल, चंद्रलोक भवन, 36, जनपथ, नई दिल्ली-110001
Secretary, CERC, 3rd & 4th Floor, Chanderlok Building, 36, Janpath, New Delhi-110001
- 2 मुख्य अभियंता (जी एम), के.वि.प्रा., सेवा भवन, आर. के. पुरम, नई दिल्ली-110066
Chief Engineer (GM), CEA, Sewa Bhavan, R.K.Puram, New Delhi-110066
- 3 सदस्य सचिव, उ. क्षेत्र. वि. स., 18/ए, शहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Member Secretary, NRPC, 18/A, SJSS Marg, Katwaria Sarai, New Delhi-110016
- 4 सदस्य सचिव, प. क्षेत्र. वि. स., एफ-3, एम आई डी सी क्षेत्र, अंधेरी (पूर्व), मुंबई - 400093
Member Secretary, WRPC, F-3, MIDC Area, Andheri (East), Mumbai-400093
- 5 सदस्य सचिव, द. क्षेत्र. वि. स., 29, रेस कोर्स क्रॉस रोड, बंगलूरु - 560009
Member Secretary, SRPC, 29, Race Course Cross Road, Bangalore-560009
- 6 सदस्य सचिव, पू. क्षेत्र. वि. स., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Member Secretary, ERPC, 14, Golf Club Road, Kolkata-700033
- 7 सदस्य सचिव, उ. पू. क्षेत्र. वि. स., मेघालय राज्य आवासीय वित्त सहकारी समिति लिमिटेड भवन, नॉग्रिम हिल्स, शिलोंग - 793003
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- 9 कार्यपालक निदेशक, पू. क्षेत्र. भा. प्रे. के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14, Golf Club Road, Kolkata-700033
- 10 कार्यपालक निदेशक, प. क्षेत्र. भा. प्रे. के., एफ-3, एम आई डी सी क्षेत्र, अंधेरी (पूर्व), मुंबई - 400093
Executive Director, WRLDC, F-3, MIDC Area, Andheri (East), Mumbai-400093
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Executive Director, NRPC, 18/A, SJSS Marg, Katwaria Sarai, New Delhi-110016
- 12 कार्यपालक निदेशक, द. क्षेत्र. भा. प्रे. के., 29, रेस कोर्स क्रॉस रोड, बंगलूरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009
- 13 मुख्य महाप्रबंधक, पू. क्षेत्र. भा. प्रे. के., डोंगतिह, लोअर नॉग्रह, लापलंग, शिलोंग- 793006
Chief General Manager, NERLDC, Dongtiah, Lower Nongrah, Laplang, Shillong-793006



ग्रिड-इंडिया
GRID-INDIA

GRID CONTROLLER OF INDIA LIMITED

Formerly Power System Operation Corporation Limited

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

NATIONAL LOAD DESPATCH CENTRE –राष्ट्रीय भार प्रेषण केन्द्र

मासिक ऑपरेशन रिपोर्ट

Monthly
Operation Report



नवंबर 2023

ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड
राष्ट्रीय भार प्रेषण केंद्र, नई दिल्ली

GRID CONTROLLER OF INDIA LIMITED

Formerly Power System Operation Corporation Limited

NATIONAL LOAD DESPATCH CENTRE, NEW DELHI



ग्रिड-इंडिया
GRID-INDIA

माह नवंबर 2023 के लिए प्रचालन निष्पादन रिपोर्ट
OPERATIONAL PERFORMANCE REPORT FOR THE
MONTH OF NOVEMBER-2023

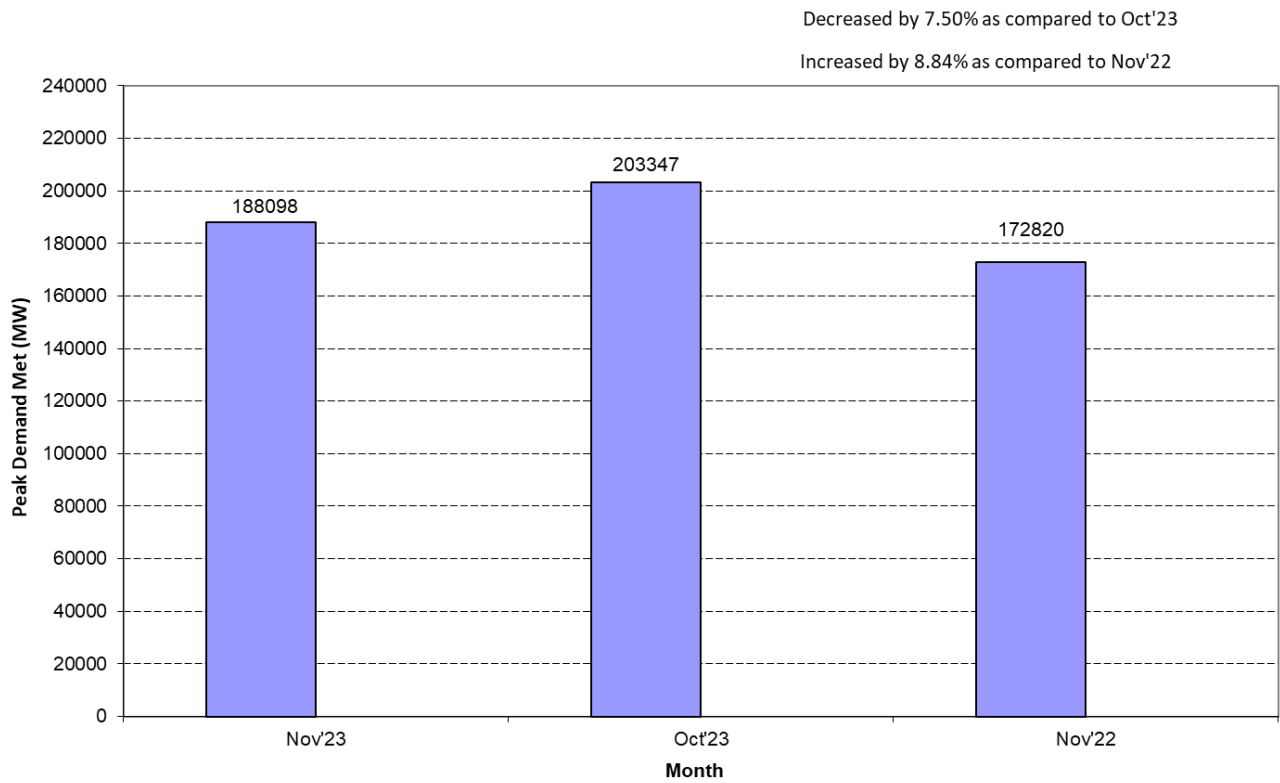
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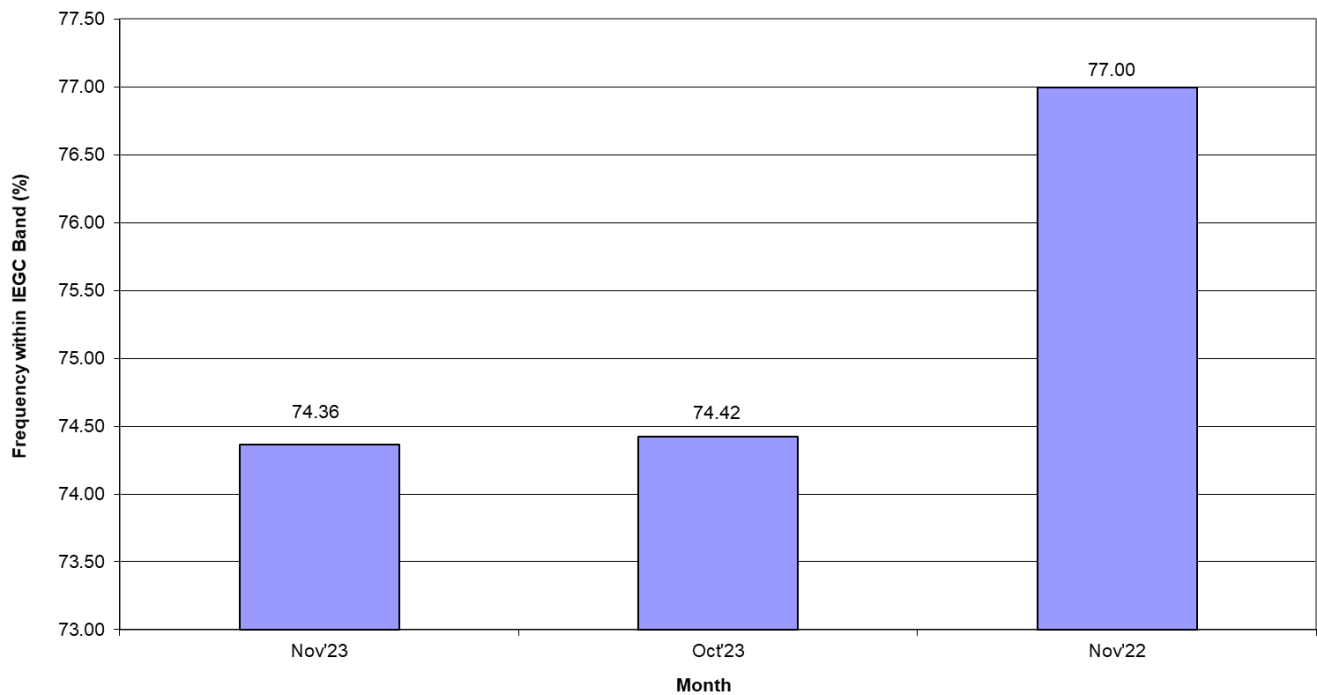
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1. SUMMARY OF REPORT FOR THE MONTH OF NOVEMBER-2023

EVENING PEAK HOUR (at 2000hrs) DEMAND MET AT NATIONAL LEVEL (MW)



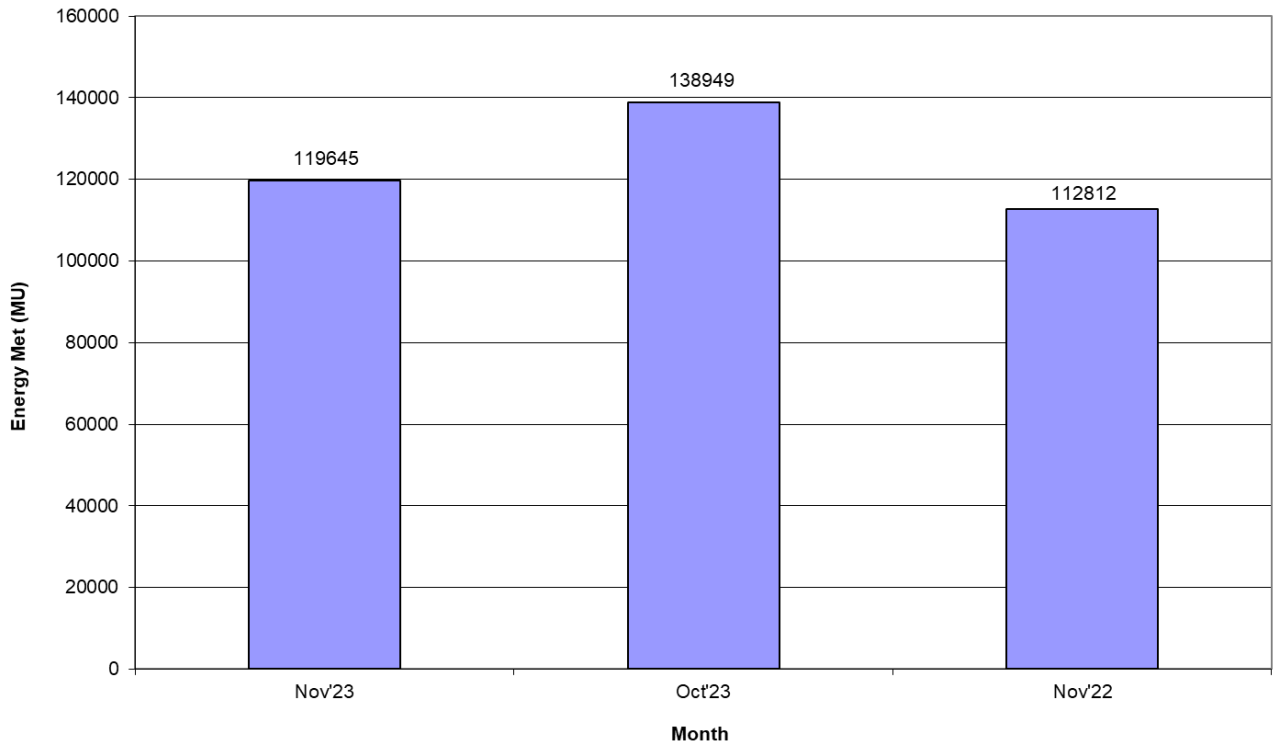
Frequency within IEGC Band



ENERGY MET AT NATIONAL LEVEL (MU)

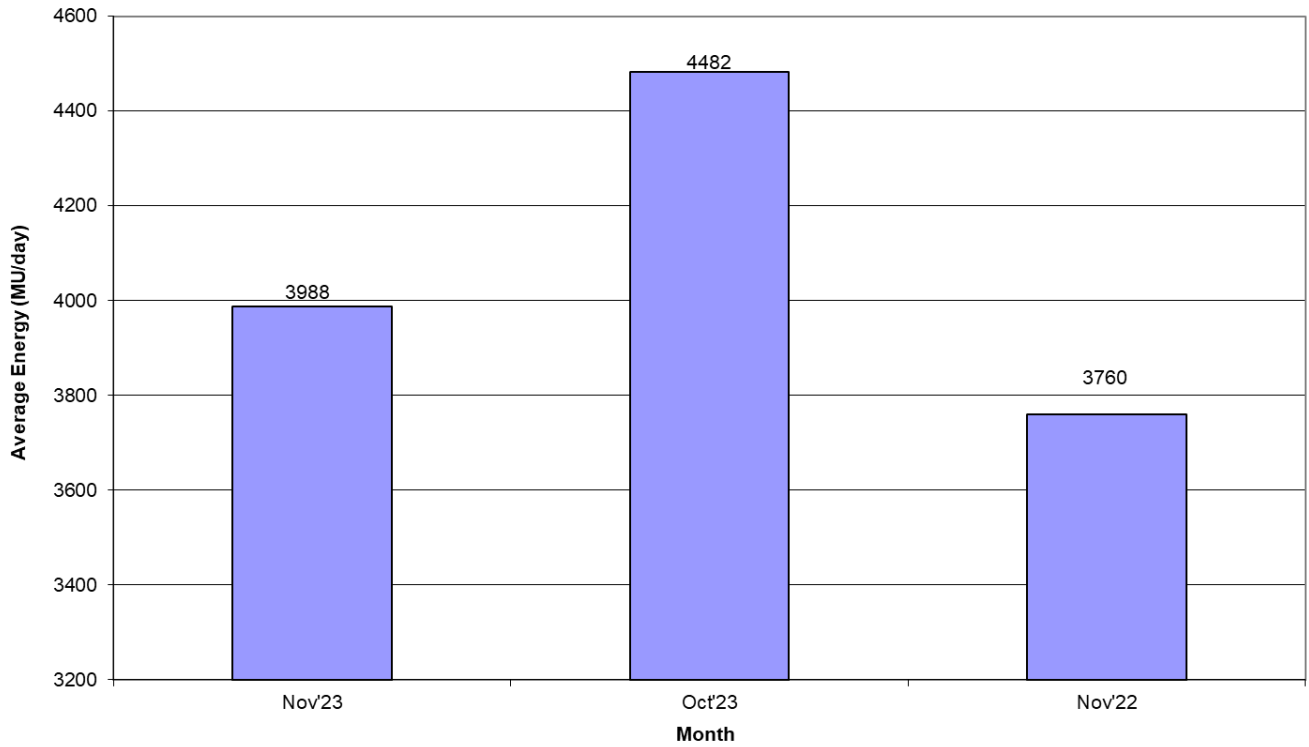
Decreased by 13.89% as compared to Oct'23

Increased by 6.06% as compared to Nov'22



AVERAGE ENERGY MET AT NATIONAL LEVEL (MU/Day)

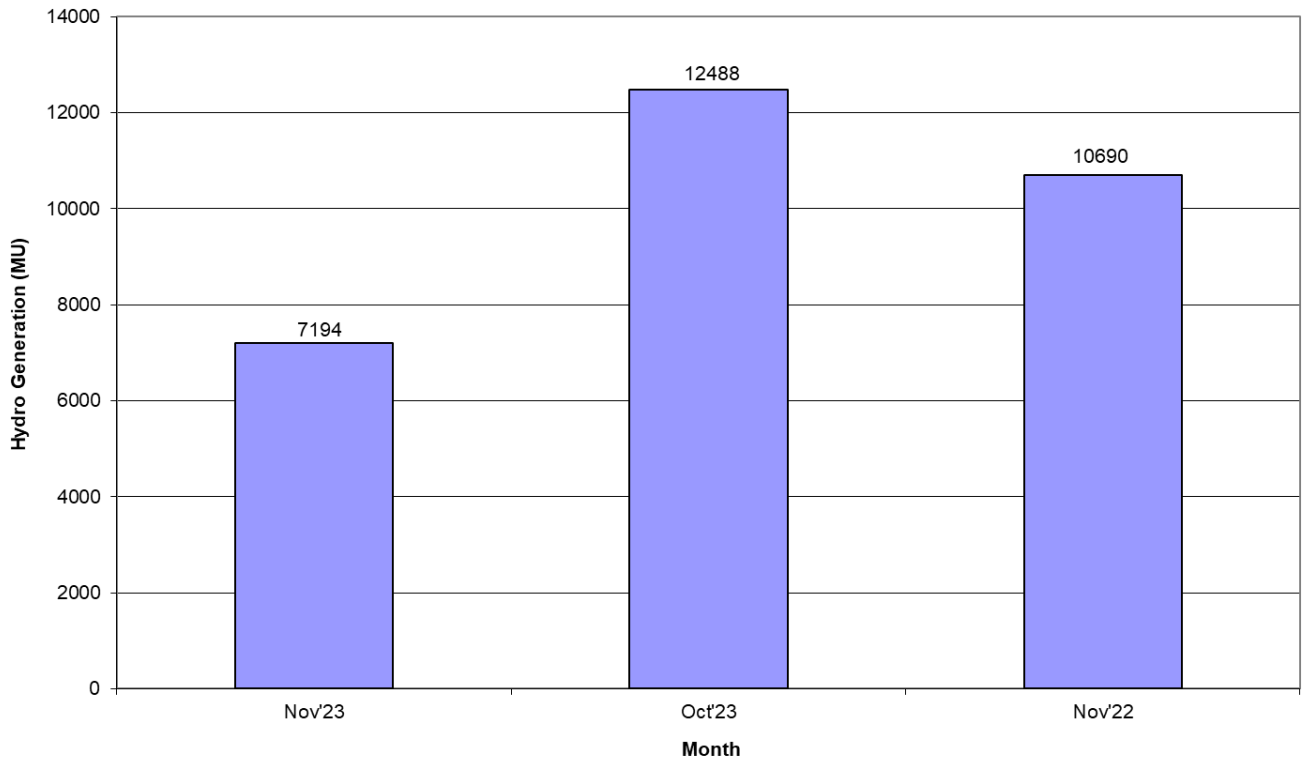
Increased by 6.06% as compared to Nov'22



HYDRO GENERATION AT NATIONAL LEVEL (MU)

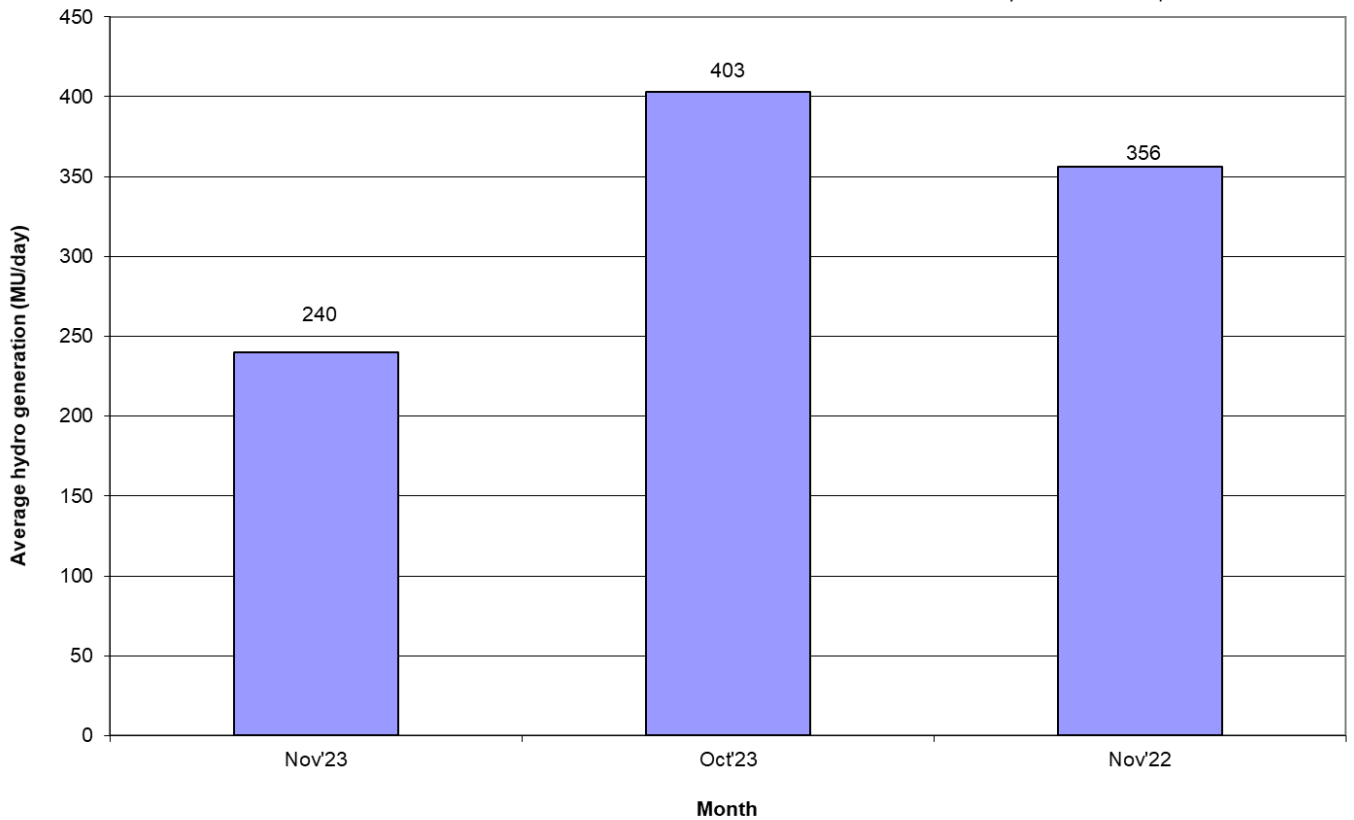
Decreased by 42.39% as compared to Oct'23

Decreased by 32.71% as compared to Nov'22



AVERAGE HYDRO GENERATION AT NATIONAL LEVEL (MU/Day)

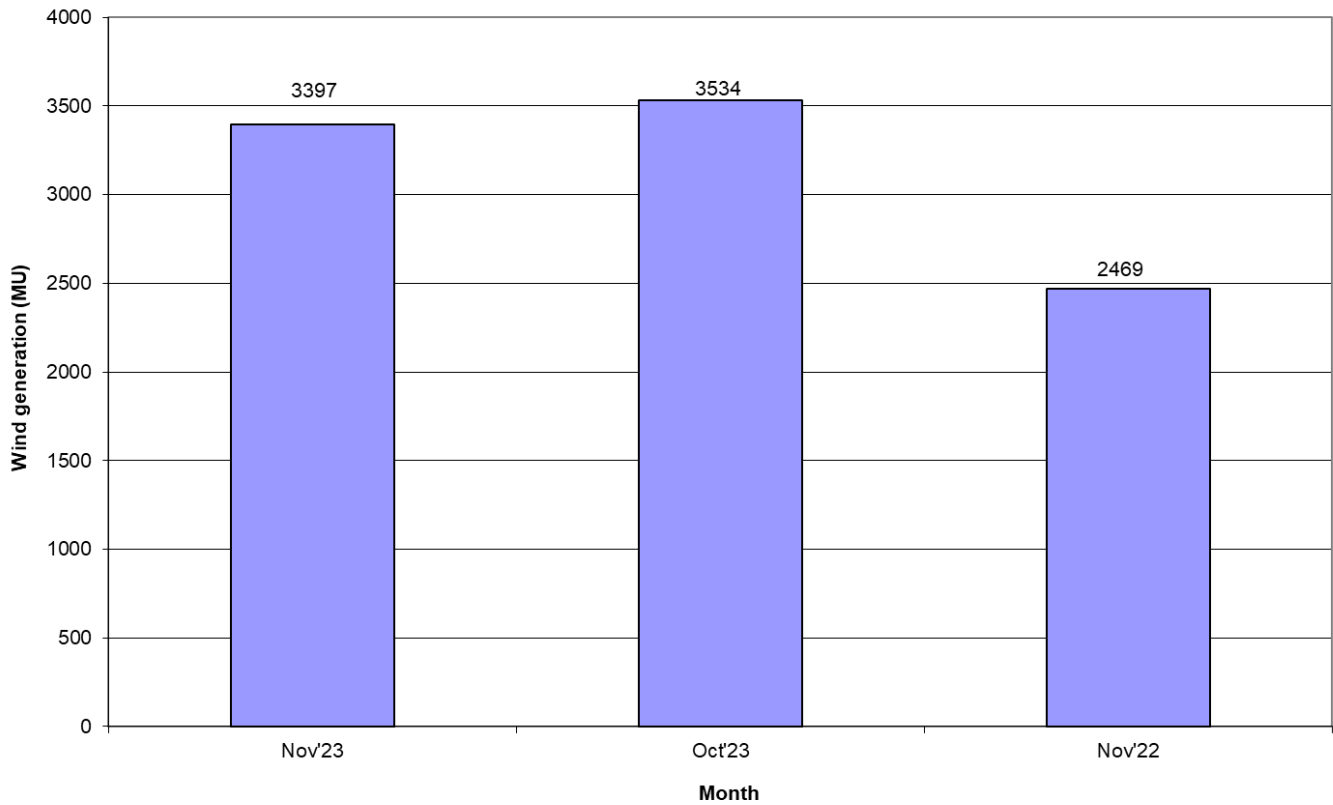
Decreased by 32.71% as compared to Nov'22



WIND GENERATION AT NATIONAL LEVEL (MU)

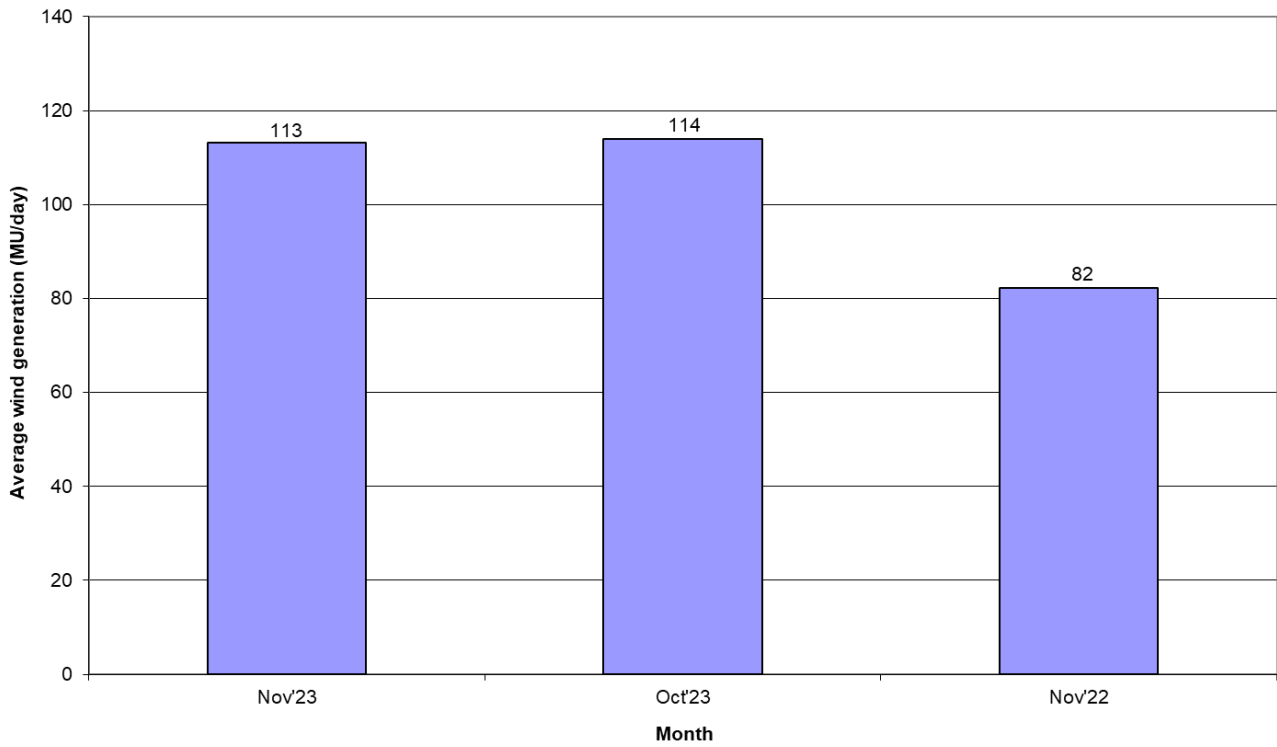
Decreased by 3.86% as compared to Oct'23

Increased by 37.59% as compared to Nov'22



AVERAGE WIND GENERATION AT NATIONAL LEVEL (MU/Day)

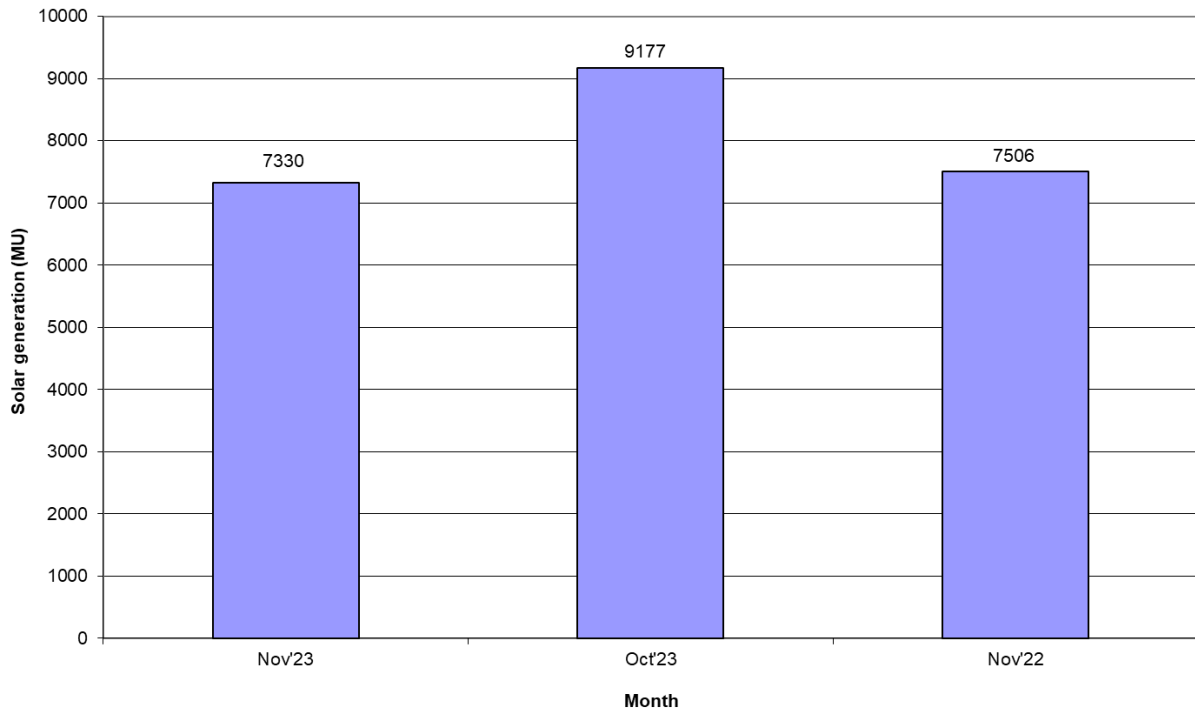
Increased by 37.59% as compared to Nov'22



SOLAR GENERATION AT NATIONAL LEVEL (MU)

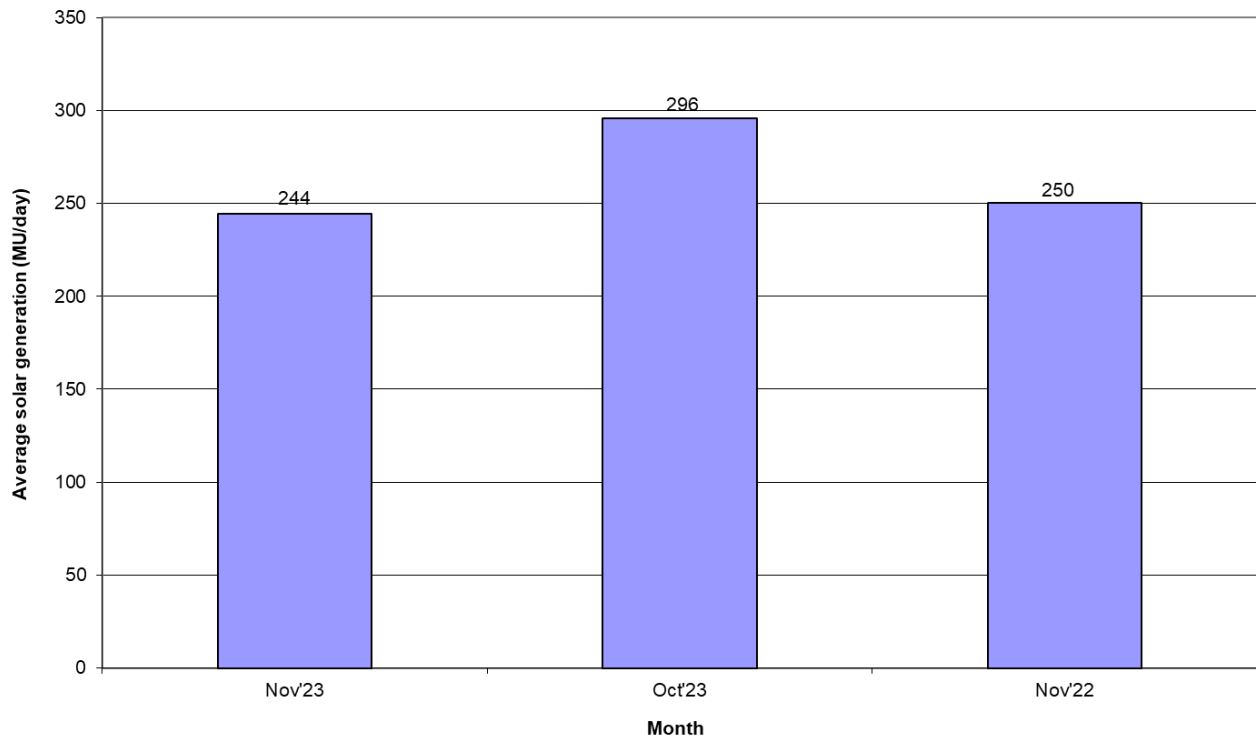
Decreased by 20.13% as compared to Oct'23

Decreased by 2.34% as compared to Nov'22



AVERAGE SOLAR GENERATION AT NATIONAL LEVEL (MU/Day)

Decreased by 2.34% as compared to Nov'22



2. ALL INDIA REGIONWISE INSTALLED CAPACITY

As on 30.11.2023
(All figures are in MW)

S No	Region	THERMAL					NUCLEAR	HYDRO	RES @ MNRE	GRAND TOTAL
		COAL	LIGNITE	GAS	DIESEL	TOTAL				
1	NR	55338	1580	5995	0	62913	1620	20782	34557	119872
2	WR	74513	1400	10806	0	86719	2540	7563	42870	139692
3	SR	47880	3640	6492	434	58446	3320	11827	52760	126354
4	ER	28145	0	80	0	28225	0	4764	1896	34886
5	NER	949	0	1665	36	2650	0	1944	575	5169
6	ISLANDS	0	0	0	120	120	0	0	40	159
	ALL INDIA	206826	6620	25038	589	239073	7480	46880	132699	426132

Remarks: Data from CEA website

3. राष्ट्रीय स्तर पर संध्याकालीन शिखर विद्युत मांग पूर्ति
EVENING PEAK HOUR DEMAND MET AT NATIONAL LEVEL

माह: नवंबर 2023 MONTH:- NOV 2023

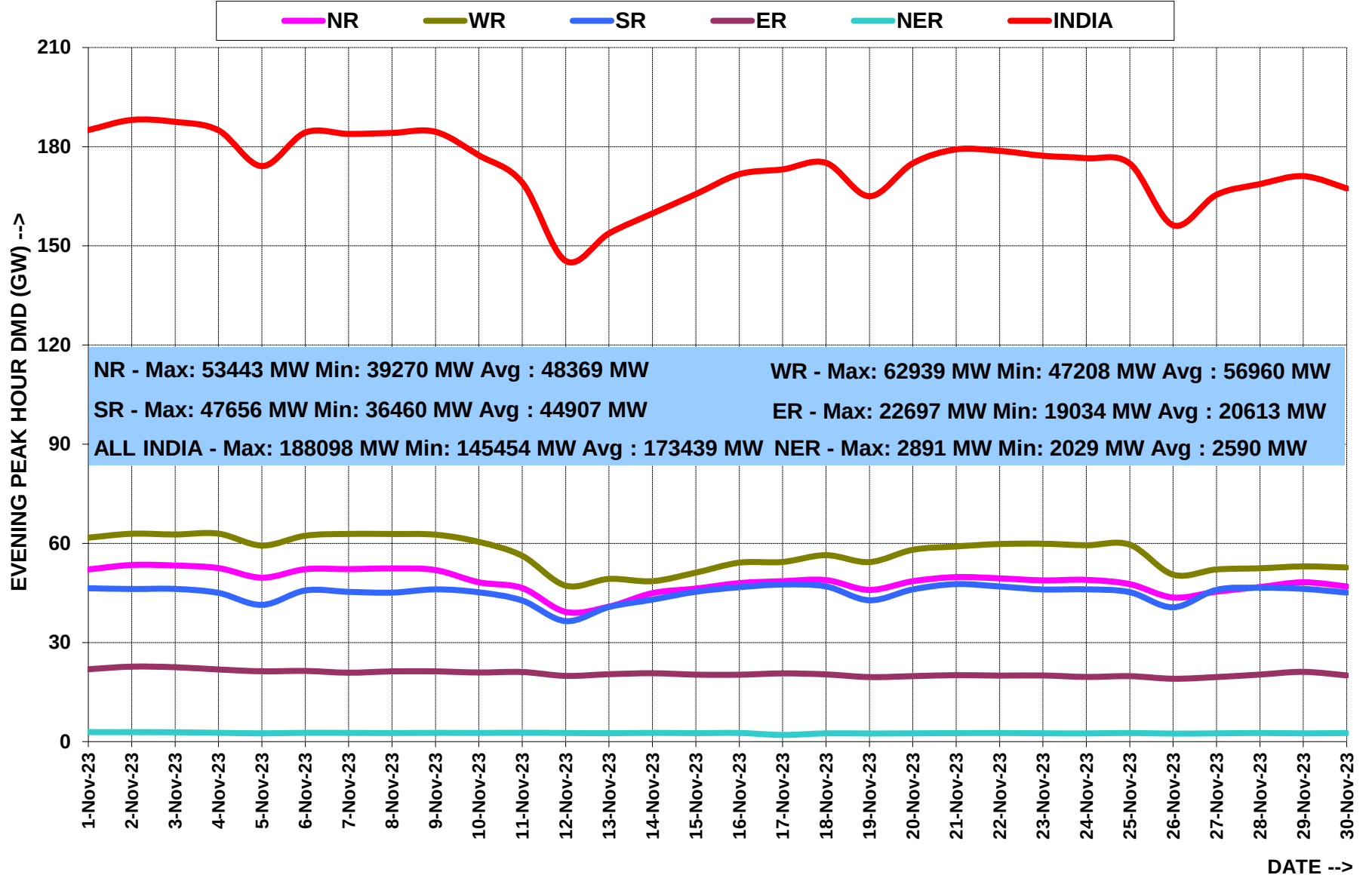
सभी आंकड़े मेगावाट मे All figures in MW

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
1-Nov-23	52099	61717	46405	21915	2891	185027
2-Nov-23	53443	62905	46186	22697	2867	188098
3-Nov-23	53288	62676	46207	22518	2829	187518
4-Nov-23	52502	62939	45006	21831	2675	184953
5-Nov-23	49606	59338	41374	21285	2516	174119
6-Nov-23	52175	62309	45739	21390	2673	184286
7-Nov-23	52199	62850	45318	20861	2642	183870
8-Nov-23	52405	62822	45084	21254	2598	184163
9-Nov-23	51921	62610	46062	21258	2658	184509
10-Nov-23	48219	60441	45182	20920	2633	177395
11-Nov-23	46472	56223	42692	21069	2683	169139
12-Nov-23	39270	47208	36460	19897	2619	145454
13-Nov-23	40857	49240	40731	20400	2553	153781
14-Nov-23	44934	48554	42969	20697	2647	159801
15-Nov-23	46308	51137	45348	20274	2578	165645
16-Nov-23	47955	54144	46713	20249	2616	171677
17-Nov-23	48547	54369	47537	20656	2029	173138
18-Nov-23	48880	56421	46958	20347	2509	175115
19-Nov-23	45919	54329	42774	19534	2458	165014
20-Nov-23	48541	58048	46056	19819	2518	174982
21-Nov-23	49790	59054	47656	20116	2576	179192
22-Nov-23	49408	59795	46949	19977	2611	178740
23-Nov-23	48788	59880	46041	20025	2529	177263
24-Nov-23	48947	59416	46080	19580	2492	176515
25-Nov-23	47652	59594	45223	19812	2620	174901
26-Nov-23	43597	50520	40644	19034	2432	156227
27-Nov-23	45387	52099	45929	19544	2511	165470
28-Nov-23	46778	52455	46550	20298	2614	168695
29-Nov-23	48212	53030	46227	21118	2510	171097
30-Nov-23	46957	52688	45107	20027	2606	167385
उच्चतम MAXIMUM	53443	62939	47656	22697	2891	188098
निम्नतम MINIMUM	39270	47208	36460	19034	2029	145454
औसत AVERAGE	48369	56960	44907	20613	2590	173439
अब तक का उच्चतम All Time Max.	76274	65518	52366	28579	3510	215592
दिनांक Date	04.09.23	13.10.23	18.04.23	18.04.23	12.08.22	28.08.23

नोट : सभी आंकड़े 2000 बजे की मांगपूर्ति पर आधारित है ।

Note: The above figures correspond to value at 2000 Hrs

EVENING PEAK HOUR DEMAND (2000hrs) MET DURING THE MONTH OF NOVEMBER 2023



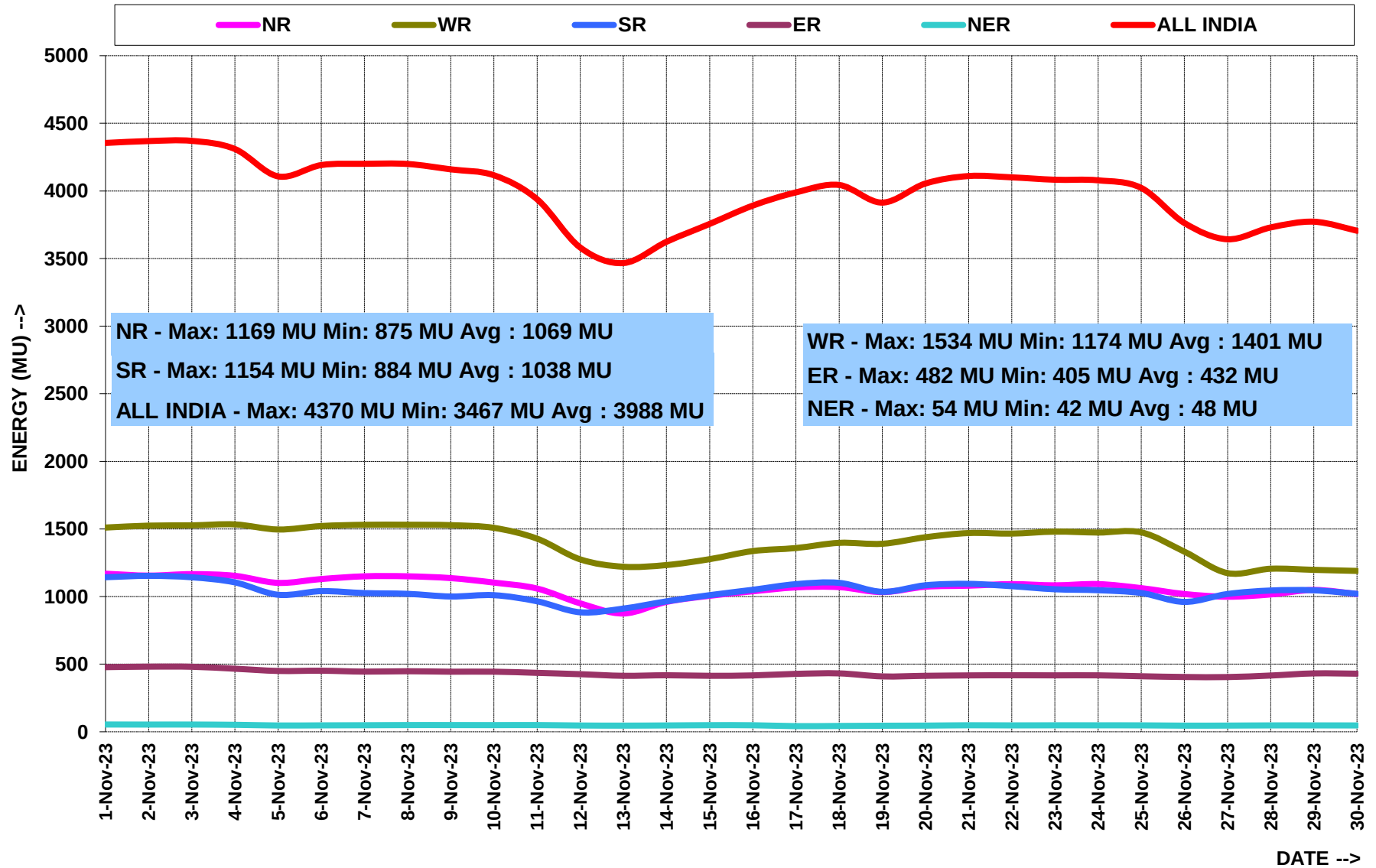
4. राष्ट्रीय स्तर पर विद्युत ऊर्जा आपूर्ति ENERGY MET AT NATIONAL LEVEL

माह: नवंबर 2023 MONTH:- NOV 2023

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Nov-23	1169	1511	1143	478	54	4355
02-Nov-23	1155	1525	1154	482	54	4369
03-Nov-23	1166	1527	1143	481	54	4370
04-Nov-23	1153	1534	1105	466	52	4310
05-Nov-23	1101	1496	1013	450	47	4107
06-Nov-23	1130	1522	1040	452	48	4191
07-Nov-23	1150	1531	1026	445	48	4200
08-Nov-23	1150	1532	1020	448	50	4199
09-Nov-23	1137	1528	1001	444	50	4159
10-Nov-23	1103	1508	1010	445	49	4116
11-Nov-23	1059	1428	965	436	50	3939
12-Nov-23	950	1275	884	426	47	3582
13-Nov-23	875	1221	911	414	46	3467
14-Nov-23	961	1233	965	418	47	3624
15-Nov-23	1006	1277	1011	414	49	3756
16-Nov-23	1039	1337	1050	417	48	3891
17-Nov-23	1068	1359	1092	429	42	3989
18-Nov-23	1070	1397	1101	432	43	4044
19-Nov-23	1033	1391	1035	410	45	3914
20-Nov-23	1073	1439	1083	414	46	4055
21-Nov-23	1081	1470	1094	417	48	4110
22-Nov-23	1092	1466	1077	418	47	4100
23-Nov-23	1083	1480	1054	417	48	4083
24-Nov-23	1092	1474	1047	418	48	4078
25-Nov-23	1062	1475	1026	411	48	4022
26-Nov-23	1019	1332	961	405	45	3763
27-Nov-23	999	1174	1019	405	46	3643
28-Nov-23	1016	1206	1045	416	47	3730
29-Nov-23	1049	1198	1046	432	47	3772
30-Nov-23	1019	1190	1020	430	47	3705
कुल TOTAL	32061	42033	31143	12970	1438	119645
उच्चतम MAXIMUM	1169	1534	1154	482	54	4370
निम्नतम MINIMUM	875	1174	884	405	42	3467
औसत AVERAGE	1069	1401	1038	432	48	3988
संचयी 2023-24 Cumulative 2023-24	332944	339862	276958	132122	13912	1095799
अब तक का उच्चतम All Time Max.	1796	1618	1329	639	72	5224
दिनांक Date	04.09.23	02.09.23	19.04.23	08.06.23	15.09.23	02.09.23

ENERGY MET DURING THE MONTH OF NOVEMBER' 2023



5. वर्ष 2023-24 के लिए आवृत्ति रूपरेखा FREQUENCY PROFILE FOR YEAR 2023-24

राष्ट्रीय ग्रिड NATIONAL GRID								
फ्रिक्वेंसी रूपरेखा (Hz)			<49.9	49.9-50.05	>50.05	उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी
% समय	Apr-23	All India Grid	10.54	67.90	21.56	50.33	49.49	49.99
	May-23	All India Grid	9.83	68.48	21.69	50.43	49.48	49.99
	Jun-23	All India Grid	6.47	67.82	25.71	50.41	49.51	50.01
	Jul-23	All India Grid	4.60	74.96	20.44	50.23	49.76	50.01
	Aug-23	All India Grid	7.08	77.28	15.63	50.20	49.74	50.00
	Sep-23	All India Grid	5.26	77.90	16.84	50.21	49.77	50.00
	Oct-23	All India Grid	8.87	74.42	16.71	50.21	49.71	49.99
	Nov-23	All India Grid	6.83	74.36	18.81	50.26	49.73	50.00
	2023-24 (upto Nov)	All India Grid	7.44	72.91	19.65	50.43	49.48	50.00

5.1 नवंबर 2023 के लिए आवृत्ति रूपरेखा FREQUENCY PROFILE FOR NOVEMBER 2023

फ्रिक्वेंसी रूपरेखा (Hz)		<49.9	49.9-50.05	>50.05	उच्चतम फ्रिक्वेंसी	निम्नतम फ्रिक्वेंसी	औसत फ्रिक्वेंसी	एफ़.वी.आई.
% समय	01-Nov-23	5.89	79.58	14.53	50.21	49.79	49.99	0.03
	02-Nov-23	10.67	79.49	9.84	50.20	49.63	49.98	0.04
	03-Nov-23	10.66	78.62	10.72	50.21	49.67	49.98	0.05
	04-Nov-23	10.51	76.18	13.31	50.19	49.69	49.99	0.05
	05-Nov-23	5.09	82.22	12.69	50.27	49.78	50.00	0.03
	06-Nov-23	2.47	77.51	20.02	50.22	49.84	50.01	0.03
	07-Nov-23	6.74	79.94	13.32	50.18	49.76	49.99	0.03
	08-Nov-23	10.71	75.64	13.66	50.15	49.71	49.98	0.05
	09-Nov-23	13.19	75.41	11.40	50.22	49.65	49.98	0.06
	10-Nov-23	1.97	72.43	25.60	50.30	49.83	50.02	0.04
	11-Nov-23	4.61	73.19	22.20	50.34	49.77	50.01	0.06
	12-Nov-23	4.16	67.88	27.96	50.37	49.84	50.02	0.06
	13-Nov-23	7.14	70.72	22.14	50.24	49.67	50.01	0.04
	14-Nov-23	7.88	71.74	20.38	50.26	49.69	50.00	0.05
	15-Nov-23	3.59	72.62	23.80	50.20	49.78	50.01	0.03
	16-Nov-23	5.49	71.27	23.24	50.24	49.74	50.01	0.04
	17-Nov-23	8.91	73.48	17.60	50.27	49.75	50.00	0.04
	18-Nov-23	15.63	74.09	10.29	50.20	49.71	49.97	0.06
	19-Nov-23	3.76	76.22	20.02	50.21	49.77	50.01	0.03
	20-Nov-23	1.94	78.29	19.77	50.24	49.83	50.01	0.03
	21-Nov-23	11.85	69.27	18.88	50.20	49.69	49.99	0.06
	22-Nov-23	11.04	72.82	16.13	50.29	49.67	49.99	0.06
	23-Nov-23	8.29	79.93	11.78	50.21	49.66	49.98	0.04
	24-Nov-23	2.94	75.50	21.56	50.26	49.73	50.01	0.04
	25-Nov-23	5.30	67.60	27.09	50.31	49.55	50.01	0.07
	26-Nov-23	1.84	75.25	22.91	50.32	49.84	50.01	0.03
	27-Nov-23	8.45	70.20	21.35	50.39	49.72	50.01	0.07
	28-Nov-23	5.67	67.26	27.07	50.37	49.69	50.02	0.06
	29-Nov-23	3.25	74.64	22.11	50.28	49.75	50.01	0.04
	30-Nov-23	5.25	71.86	22.88	50.35	49.80	50.01	0.04

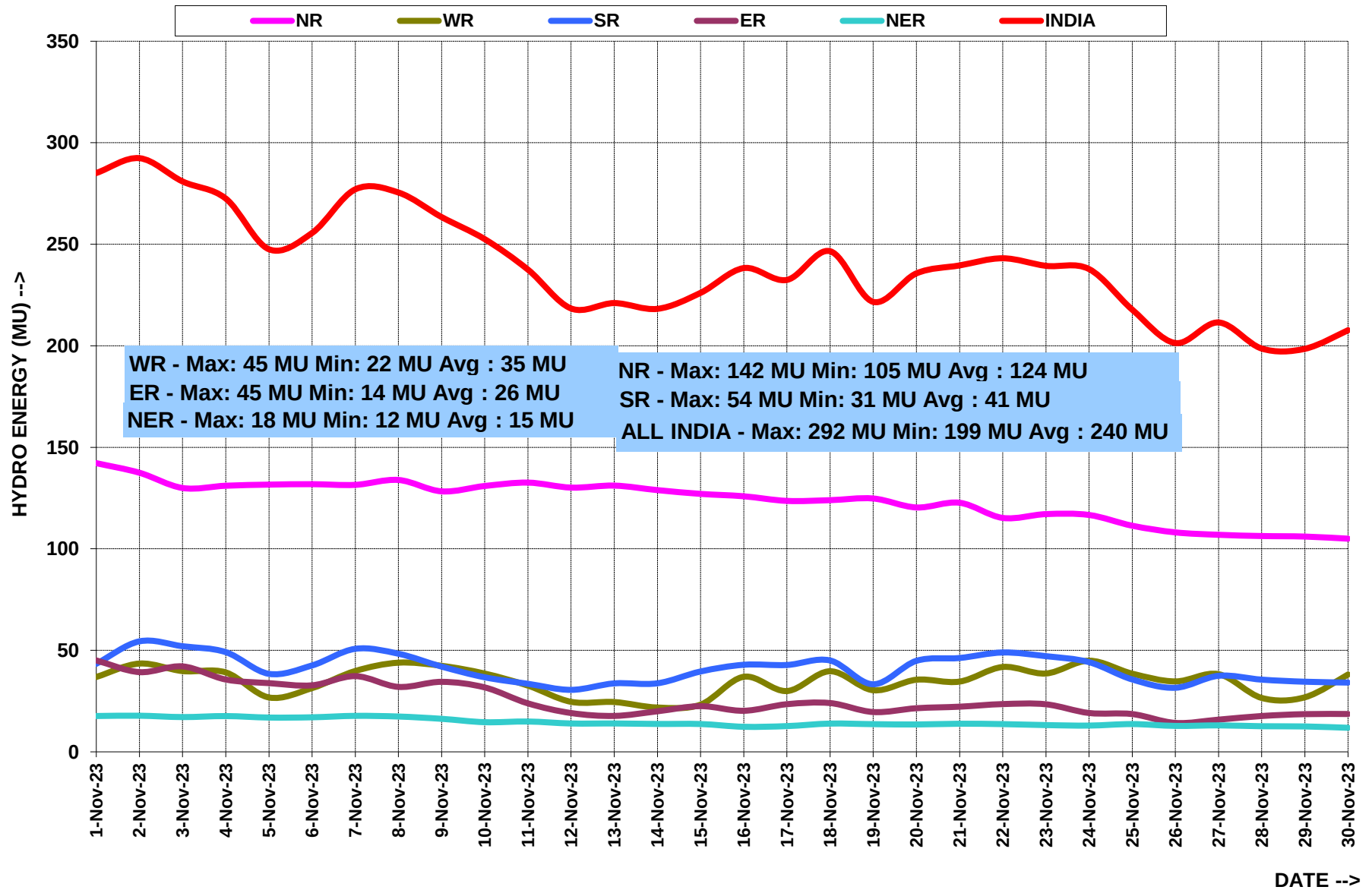
6.राष्ट्रीय स्तर पर जल विद्युत उत्पादन
HYDRO GENERATION AT NATIONAL LEVEL

माह: नवंबर 2023 MONTH:- NOV 2023

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Nov-23	142	37	43	45	18	285
02-Nov-23	137	43	54	39	18	292
03-Nov-23	130	40	52	42	17	281
04-Nov-23	131	39	49	36	18	273
05-Nov-23	132	27	38	34	17	248
06-Nov-23	132	31	43	33	17	256
07-Nov-23	132	40	51	37	18	277
08-Nov-23	134	44	48	32	17	276
09-Nov-23	128	42	42	34	16	263
10-Nov-23	131	39	37	32	15	253
11-Nov-23	133	33	34	24	15	238
12-Nov-23	130	25	31	19	14	218
13-Nov-23	131	25	34	18	14	221
14-Nov-23	129	22	34	20	14	218
15-Nov-23	127	23	40	23	14	226
16-Nov-23	126	37	43	20	12	238
17-Nov-23	124	30	43	23	13	232
18-Nov-23	124	40	45	24	14	247
19-Nov-23	125	30	33	20	14	222
20-Nov-23	120	36	45	21	13	236
21-Nov-23	123	35	46	22	14	240
22-Nov-23	115	42	49	24	14	243
23-Nov-23	117	39	47	23	13	239
24-Nov-23	117	45	44	19	13	238
25-Nov-23	111	39	36	19	14	218
26-Nov-23	108	35	32	14	13	201
27-Nov-23	107	38	37	16	13	211
28-Nov-23	106	27	36	18	13	199
29-Nov-23	106	27	35	19	13	199
30-Nov-23	105	38	34	19	12	208
कुल TOTAL	3712	1044	1233	768	436	7194
उच्चतम MAXIMUM	142	45	54	45	18	292
निम्नतम MINIMUM	105	22	31	14	12	199
औसत AVERAGE	124	35	41	26	15	240
संचयी 2023-24 Cumulative 2023-24	63144	12679	15723	20277	5496	117319
अब तक का उच्चतम All Time Max.	443	167	208	157	41	877
दिनांक Date	01.08.23	18.12.14	31.08.18	14.09.22	29.08.23	30.08.22

HYDRO ENERGY DURING THE MONTH OF NOVEMBER' 2023



7. राष्ट्रीय स्तर पर पवन ऊर्जा उत्पादन
WIND GENERATION AT NATIONAL LEVEL

माह: नवंबर 2023 MONTH:- NOV 2023

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Nov-23	10	33	36	-----	-----	79
02-Nov-23	5	31	34	-----	-----	70
03-Nov-23	4	33	37	-----	-----	74
04-Nov-23	4	38	33	-----	-----	75
05-Nov-23	4	41	34	-----	-----	79
06-Nov-23	4	38	24	-----	-----	67
07-Nov-23	5	25	23	-----	-----	53
08-Nov-23	2	19	25	-----	-----	47
09-Nov-23	8	18	21	-----	-----	47
10-Nov-23	19	32	40	-----	-----	91
11-Nov-23	20	102	63	-----	-----	186
12-Nov-23	4	88	55	-----	-----	147
13-Nov-23	7	60	60	-----	-----	126
14-Nov-23	16	64	44	-----	-----	124
15-Nov-23	21	87	20	-----	-----	128
16-Nov-23	15	91	19	-----	-----	126
17-Nov-23	20	96	33	-----	-----	149
18-Nov-23	12	55	37	-----	-----	104
19-Nov-23	15	74	49	-----	-----	138
20-Nov-23	28	94	41	-----	-----	163
21-Nov-23	21	87	33	-----	-----	142
22-Nov-23	3	35	28	-----	-----	67
23-Nov-23	4	26	25	-----	-----	55
24-Nov-23	17	84	40	-----	-----	142
25-Nov-23	26	102	60	-----	-----	188
26-Nov-23	19	105	95	-----	-----	218
27-Nov-23	15	51	59	-----	-----	125
28-Nov-23	11	68	49	-----	-----	129
29-Nov-23	8	67	55	-----	-----	131
30-Nov-23	7	68	56	-----	-----	130
कुल TOTAL	355	1814	1229	-----	-----	3397
उच्चतम MAXIMUM	28	105	95	-----	-----	218
निम्नतम MINIMUM	2	18	19	-----	-----	47
औसत AVERAGE	12	60	41	-----	-----	113
संचयी 2023-24 Cumulative 2023-24	6360	27465	28298	-----	-----	62123
अब तक का उच्चतम All Time Max.	86	273	298	-----	-----	611
दिनांक Date	07.08.23	07.08.23	21.07.23	-----	-----	02.08.23

***Source:** As reported by SLDCs. Limited visibility of embedded wind generator data.

8. राष्ट्रीय स्तर पर सौर ऊर्जा उत्पादन
SOLAR GENERATION AT NATIONAL LEVEL

माह: नवंबर 2023 MONTH:- NOV 2023

सभी आंकड़े मिलियन यूनिट में All figures in MU

दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL
01-Nov-23	101	61	96	5.0	1.3	264
02-Nov-23	106	59	103	5.2	1.3	275
03-Nov-23	104	56	95	2.9	1.2	259
04-Nov-23	102	57	90	3.0	1.0	253
05-Nov-23	106	58	89	2.6	1.2	257
06-Nov-23	105	51	91	2.6	1.2	251
07-Nov-23	99	57	81	2.9	1.0	241
08-Nov-23	88	54	82	3.2	1.1	228
09-Nov-23	74	54	89	6.2	1.1	224
10-Nov-23	110	52	102	5.2	1.2	270
11-Nov-23	109	61	108	2.6	1.2	282
12-Nov-23	122	63	113	2.4	1.2	301
13-Nov-23	106	56	106	2.7	1.2	273
14-Nov-23	103	57	76	2.4	1.3	239
15-Nov-23	103	60	98	2.5	1.1	265
16-Nov-23	108	58	109	2.5	0.9	278
17-Nov-23	84	57	110	5.3	0.2	256
18-Nov-23	90	55	106	5.2	0.8	257
19-Nov-23	84	51	95	2.5	1.2	234
20-Nov-23	94	54	110	2.3	1.2	261
21-Nov-23	93	55	89	2.4	0.8	240
22-Nov-23	104	57	79	2.5	1.0	244
23-Nov-23	104	56	71	2.3	1.1	235
24-Nov-23	105	54	71	2.1	1.2	233
25-Nov-23	85	50	83	5.1	1.2	223
26-Nov-23	29	27	94	5.2	1.1	157
27-Nov-23	44	31	80	2.3	1.2	159
28-Nov-23	72	42	92	2.5	1.1	209
29-Nov-23	79	47	94	2.8	0.9	224
30-Nov-23	91	47	96	2.9	1.2	238
कुल TOTAL	2804	1596	2798	99	33	7330
उच्चतम MAXIMUM	122	63	113	6	1.3	301
निम्नतम MINIMUM	29	27	71	2	0.2	157
औसत AVERAGE	93	53	93	3	1.1	244
संचयी 2023-24 Cumulative 2023-24	29914	13115	27214	782	236.0	71159
अब तक का उच्चतम All Time Max.	147	71	143	6.7	2.4	351
दिनांक Date	10.05.23	12.05.23	21.03.23	23.03.23	22.06.22	10.05.23

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

9. राष्ट्रीय स्तर पर दैनिक अधिकतम विद्युत मांग पूर्ति
DAILY MAXIMUM DEMAND MET AT NATIONAL LEVEL

माह: नवंबर 2023 MONTH:- NOV 2023

सभी आंकड़े मेगावाट में All figures in MW

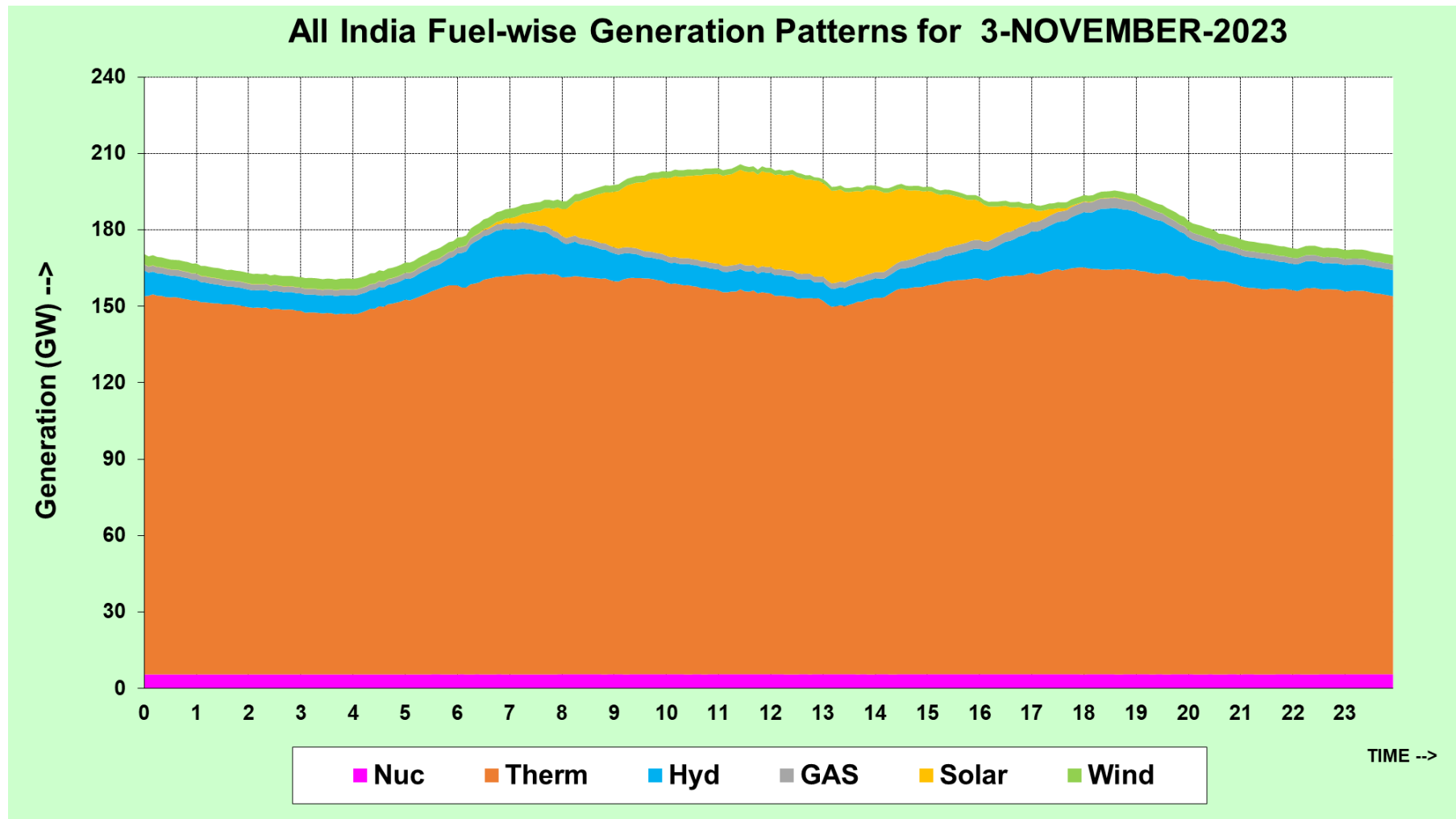
दिनांक Date	उत्तरी क्षे. NR	पश्चिमी क्षे. WR	दक्षिणी क्षे. SR	पूर्वी क्षे. ER	पूर्वोत्तर क्षे. NER	कुल TOTAL	विभिन्नता फैक्टर Diversity Factor*
01-Nov-23	55485	70714	56593	22723	3022	203839	1.023
02-Nov-23	55485	70714	56593	22723	3022	203839	1.023
03-Nov-23	55361	71160	56756	22831	3021	204605	1.022
04-Nov-23	56533	72377	54239	23110	2936	201946	1.036
05-Nov-23	52703	71147	49356	22201	2695	192944	1.027
06-Nov-23	56254	73028	50212	22991	2838	198571	1.034
07-Nov-23	56782	72848	48891	21898	2886	196615	1.034
08-Nov-23	56849	72868	48786	22663	2901	197001	1.036
09-Nov-23	55556	72913	46505	22516	2874	194095	1.032
10-Nov-23	53674	73369	46841	22112	2907	194379	1.023
11-Nov-23	52120	69047	44453	22132	2894	184825	1.031
12-Nov-23	47371	59577	42691	20254	2699	168985	1.021
13-Nov-23	44209	59381	44328	21453	2736	164063	1.049
14-Nov-23	48791	59048	46277	21537	2820	172997	1.032
15-Nov-23	50779	62249	47501	20858	2796	179300	1.027
16-Nov-23	51844	66488	49311	21094	2792	186130	1.029
17-Nov-23	53808	67488	52574	21977	2128	193831	1.021
18-Nov-23	53402	69383	52474	21489	2712	195832	1.019
19-Nov-23	52243	68553	49669	20846	2581	190047	1.020
20-Nov-23	53149	71490	52793	21390	2673	197068	1.022
21-Nov-23	53234	72159	52096	21366	2701	197845	1.019
22-Nov-23	54629	71616	50790	21444	2684	196989	1.021
23-Nov-23	53724	72402	49872	21226	2749	196282	1.019
24-Nov-23	55779	72241	49752	21238	2730	198154	1.018
25-Nov-23	53622	72880	48610	21224	2780	194855	1.022
26-Nov-23	53206	65872	45726	20726	2569	184198	1.021
27-Nov-23	52071	57427	48548	21121	2722	177636	1.024
28-Nov-23	51794	57413	49729	21533	2759	180012	1.018
29-Nov-23	53580	57463	49636	22036	2781	181442	1.022
30-Nov-23	50853	57164	48466	21894	2787	177582	1.020
उच्चतम MAXIMUM	56849	73369	56756	23110	3022	204605	1.049
निम्नतम MINIMUM	44209	57164	42691	20254	2128	164063	1.018
औसत AVERAGE	53163	67683	49669	21753	2773	190197	1.026
अब तक का उच्चतम All Time Max.	81220	74035	64314	29641	3712	239978	
दिनांक Date	04.09.23	01.09.23	15.03.23	28.07.23	04.09.23	01.09.23	

* Diversity factor = (Sum of regional max demands) / All India max demand

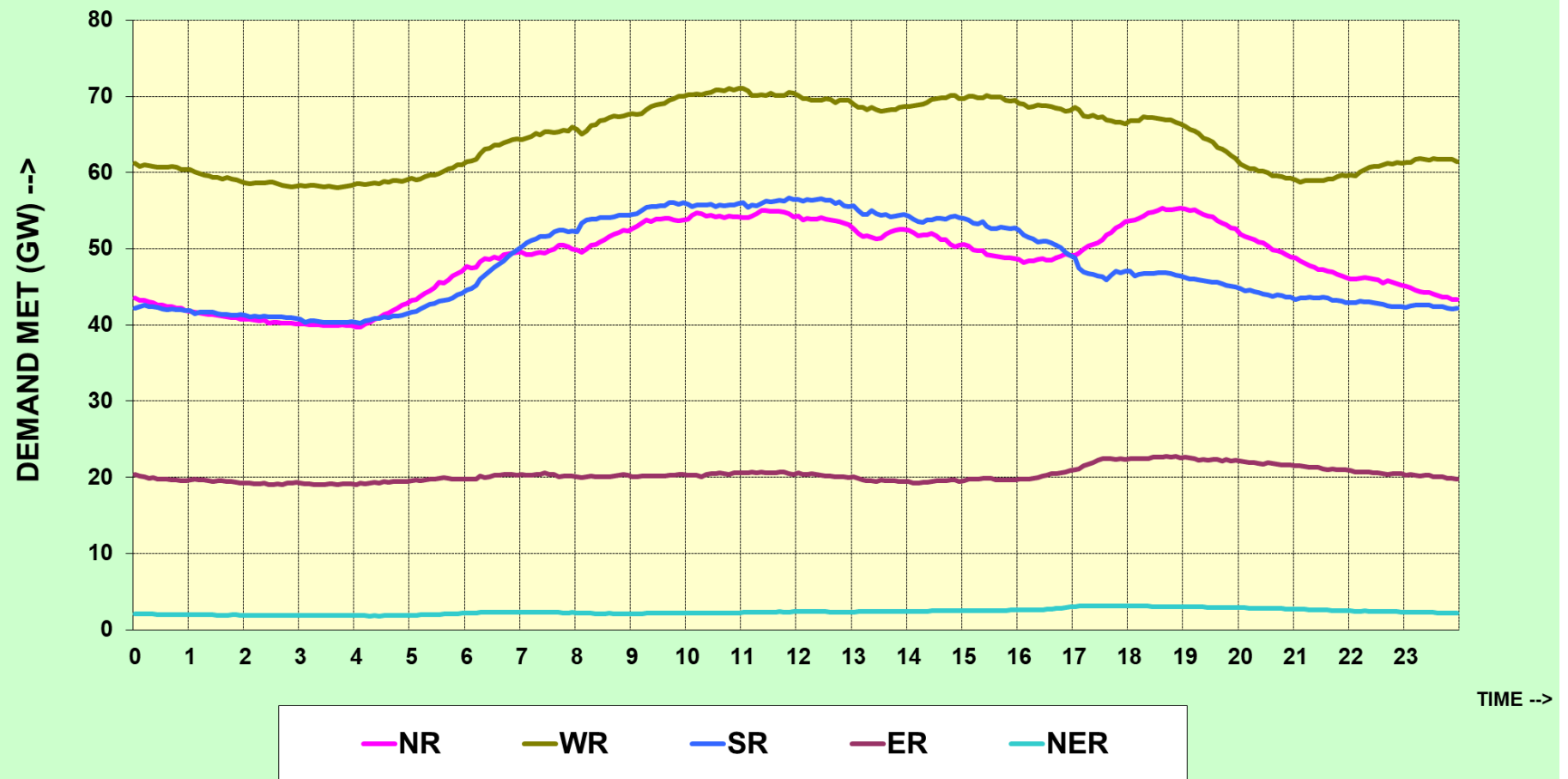
नोट : यह आंकड़े एनएलडीसी स्काडा प्रणाली में दर्ज दैनिक अधिकतम मांगपूर्ति दर्शाते हैं ।

Note: The above figures denote daily maximum demand met recorded in NLDC SCADA.

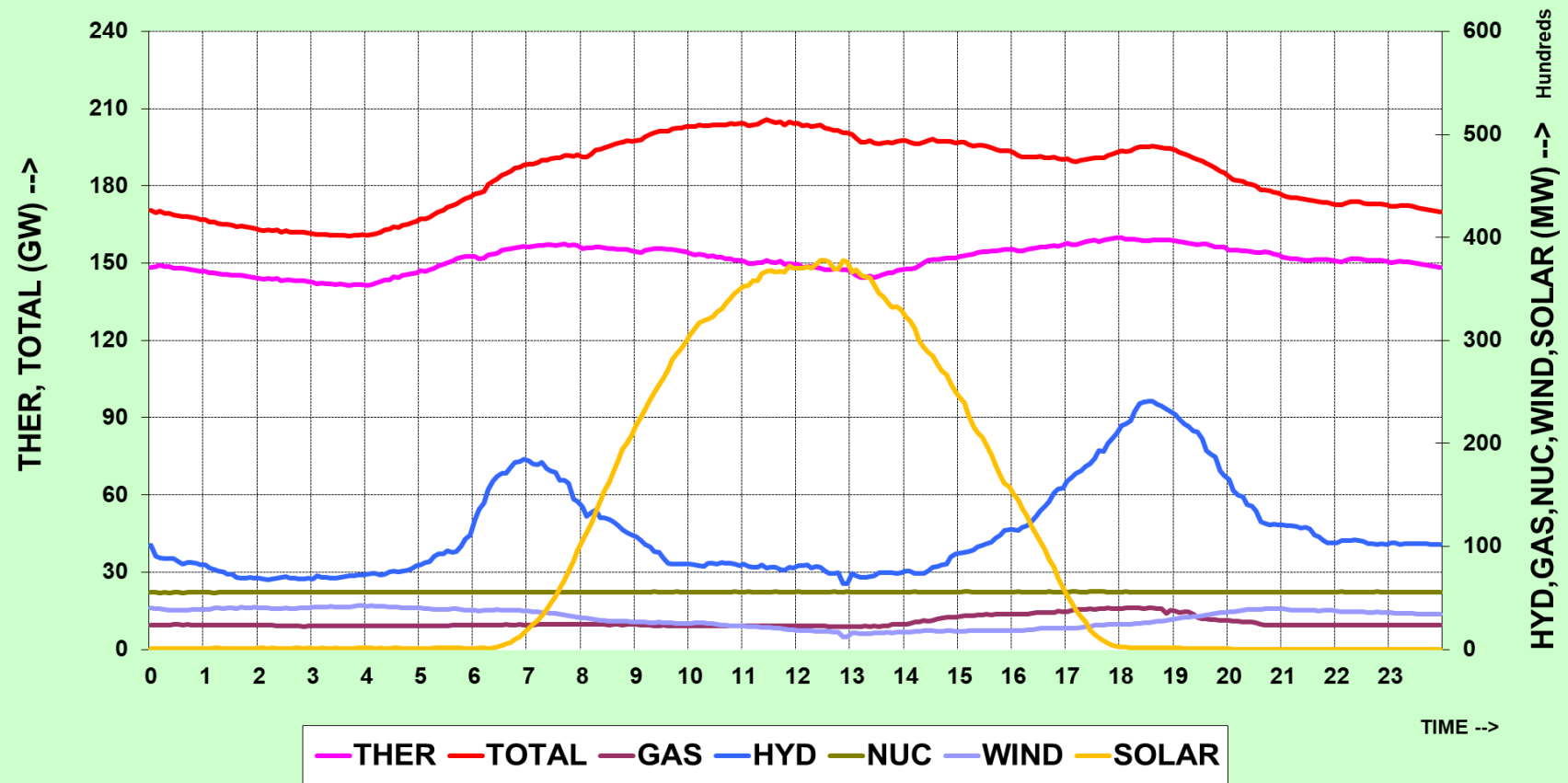
10. DEMAND AND GENERATION PLOTS FOR 03 NOVEMBER (MAXIMUM DEMAND MET)



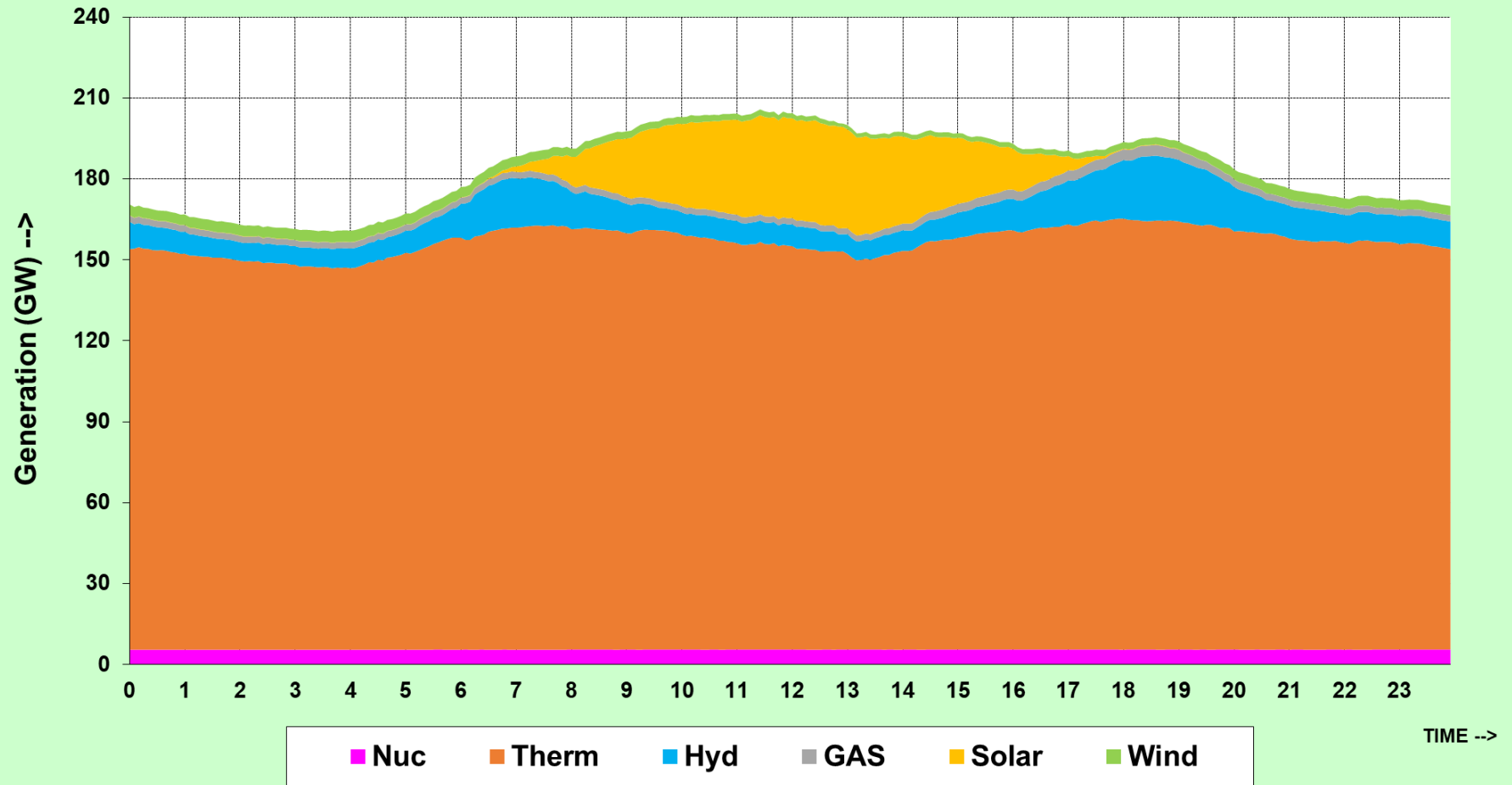
REGIONAL DEMAND MET PLOT FOR 3-NOVEMBER-2023



ALL INDIA GENERATION PLOT FOR 3-NOVEMBER-2023



All India Fuel-wise Generation Patterns for 3-NOVEMBER-2023



03 नवंबर 2023 (अधिकतम माँग) की आल इंडिया माँग आपूर्ति
ALL INDIA DEMAND MET FOR 03 NOVEMBER 2023 (MAXIMUM DEMAND*)

Time (hrs)	All India Demand (MW)		Time (hrs)	All India Demand (MW)
00:05	168595		12:05	202555
00:20	168064		12:20	202592
00:35	166893		12:35	202093
00:50	166167		12:50	200825
01:05	164873		13:05	197849
01:20	164062		13:20	196898
01:35	163009		13:35	196438
01:50	162467		13:50	197415
02:05	161421		14:05	196477
02:20	161520		14:20	196573
02:35	160783		14:35	196994
02:50	160505		14:50	196877
03:05	159691		15:05	196285
03:20	159565		15:20	195291
03:35	159357		15:35	193751
03:50	159372		15:50	192991
04:05	159572		16:05	191304
04:20	161405		16:20	190950
04:35	162462		16:35	191021
04:50	164041		16:50	190465
05:05	165699		17:05	189312
05:20	168504		17:20	189782
05:35	170898		17:35	190253
05:50	173601		17:50	192090
06:05	175783		18:05	192558
06:20	180585		18:20	194215
06:35	182883		18:35	194962
06:50	185837		18:50	194037
07:05	187099		19:05	192314
07:20	188686		19:20	190187
07:35	189961		19:35	187783
07:50	190301		19:50	185154
08:05	190090		20:05	181770
08:20	192921		20:20	179962
08:35	195159		20:35	177690
08:50	196692		20:50	176490
09:05	197311		21:05	174729
09:20	200099		21:20	174008
09:35	200988		21:35	173366
09:50	201982		21:50	172790
10:05	202559		22:05	171689
10:20	203032		22:20	172989
10:35	203500		22:35	172033
10:50	203605		22:50	171956
11:05	203142		23:05	171313
11:20	204033		23:20	171139
11:35	204129		23:35	170196
11:50	204568		23:50	169301
Maximum Demand of 204605 MW met@11:43 hrs (from 1 min. interval SCADA DATA)				

* 15 minute interval SCADA DATA of instantaneous demand

11. ACTUAL POWER SUPPLY POSITION - NOVEMBER 2023

REGION	STATE	Requirement/Availability in MU/DAY			Requirement/Availability in MU				Peak Demand/Peak Met in MW			
		Requirement	Energy met	Surplus(+)/ Deficit(-)	Requirement	Energy met	Surplus(+)/ Deficit(-)	% Shortage	Requirement	Peak Demand Met	Surplus(+)/ Deficit(-)	% Shortage
NR	Chandigarh	3	3	0.0	102	102	0	0.0	205	205	0	0.0
	Delhi	70	70	0.0	2114	2114	0	0.0	4320	4320	0	0.0
	Haryana	132	132	0.0	3951	3951	0	0.0	7685	7685	0	0.0
	Himachal Pradesh	32	32	-0.1	970	968	-2	-0.2	1977	1977	0	0.0
	J&K(UT) and Ladakh(UT)	53	52	-0.6	1586	1568	-19	-1.2	2710	2710	0	0.0
	Punjab	131	131	0.0	3925	3925	0	0.0	7572	7572	0	0.0
	Rajasthan	292	292	0.0	8760	8759	-1	0.0	16409	16409	0	0.0
	Uttar Pradesh	315	315	0.0	9445	9444	-1	0.0	19033	19033	0	0.0
	Uttarakhand	37	37	0.0	1114	1114	0	0.0	2141	2141	0	0.0
	Railway	4	4	0.0	111	111	0	0.0	206	206	0	0.0
	NFL	0	0	0.0	2	2	0	0.0	4	4	0	0.0
	TOTAL	1069	1069	-0.8	32080	32057	-23	-0.1	56366	56126	-240	-0.4
WR	Gujarat	372	372	0.0	11161	11161	0	0.0	21694	21694	0	0.0
	Madhya Pradesh	311	311	0.0	9323	9323	0	0.0	17117	17097	-20	-0.1
	Chhattisgarh	90	90	0.0	2694	2694	0	0.0	4648	4644	-3	-0.1
	Maharashtra	558	558	0.0	16726	16726	0	0.0	28386	28356	-30	-0.1
	Goa	14	14	0.0	422	422	0	0.0	733	733	0	0.0
	DNHDDPDCL	27	27	0.0	801	801	0	0.0	1267	1267	0	0.0
	AMNSIL	19	19	0.0	558	558	0	0.0	920	920	0	0.0
	BALCO*	12	12	0.0	373	373	0	0.0	536	536	0	0.0
	TOTAL	1402	1402	0.0	42058	42058	0	0.0	73288	73288	0	0.0
SR	Andhra Pradesh	210	210	0.0	6296	6296	0	0.0	12085	12085	0	0.0
	Telangana	207	207	0.1	6217	6220	3	0.0	11857	11857	0	0.0
	Karnataka	240	240	0.1	7191	7194	3	0.0	15203	15203	0	0.0
	Kerala	80	81	0.0	2415	2416	1	0.0	4300	4300	0	0.0
	Tamil Nadu	292	292	0.1	8759	8762	3	0.0	16099	16099	0	0.0
	Puducherry	9	9	0.0	256	256	0	0.0	454	454	0	0.0
	TOTAL	1038	1038	0.3	31134	31143	10	0.0	56925	56925	0	0.0
ER	Bihar	88	88	-0.2	2635	2630	-5	-0.2	5519	5257	-262	-4.7
	Jharkhand	31	30	-0.8	929	906	-24	-2.5	1699	1699	0	0.0
	DVC	70	70	0.0	2105	2105	0	0.0	3427	3427	0	0.0
	Odisha	100	100	0.0	3010	3010	0	0.0	5264	5264	0	0.0
	West Bengal	143	143	0.0	4276	4276	0	0.0	8481	8481	0	0.0
	Sikkim	1	1	0.0	38	38	0	0.0	101	101	0	0.0
	Railways_ER ISTS**	0	0	0.0	4	4	0	0.0	23	23	0	0.0
	TOTAL	433	432	-1.0	12998	12969	-29	-0.2	23216	23216	0	0.0
NER	Arunachal Pradesh	2	2	0.0	74	74	0	0.0	160	160	0	0.0
	Assam	28	28	0.0	847	847	0	0.0	1931	1931	0	0.0
	Manipur	3	3	0.0	80	80	0	0.0	207	207	0	0.0
	Meghalaya	6	6	0.0	190	190	0	0.0	363	363	0	0.0
	Mizoram	2	2	0.0	51	51	0	0.0	144	144	0	0.0
	Nagaland	2	2	0.0	71	71	0	0.0	147	147	0	0.0
	Tripura	4	4	0.0	123	123	0	0.0	299	299	0	0.0
	TOTAL	48	48	0.0	1434	1434	0	0.0	3022	3022	0	0.0
ALL INDIA		3990	3989	-1.4	119704	119661	-43	0.0				

* BALCO has been added as new entity wef 20.12.2022

** Railways_ER ISTS has been added as new entity wef 29.04.2023

NOTE: BASED ON THE DATA FURNISHED BY ENTITIES ON DAILY BASIS

12. ENERGY COMPARISON OF NOVEMBER 2023 vs NOVEMBER 2022

REGION	STATE	Energy Requirement (MU)					Energy Met (MU)				
		Nov-22	Nov-23	Difference	% Change	Average MU/day for Nov-23	Nov-22	Nov-23	Difference	% Change	Average MU/day for Nov-23
NR	Chandigarh	99	102	2	2	3	99	102	2	2	3
	Delhi	2030	2114	84	4	70	2030	2114	84	4	70
	Haryana	3875	3951	77	2	132	3871	3951	80	2	132
	Himachal Pradesh	968	970	2	0	32	967	968	0	0	32
	J&K(UT) and Ladakh(UT)	1649	1586	-62	-4	53	1628	1568	-60	-4	52
	Punjab	3809	3925	116	3	131	3809	3925	116	3	131
	Rajasthan	8788	8760	-29	0	292	8753	8759	6	0	292
	Uttar Pradesh	9033	9445	412	5	315	9025	9444	419	5	315
	Uttarakhand	1097	1114	17	2	37	1086	1114	28	3	37
	Railway/NFL	94	113	19	20	4	94	113	19	20	4
	TOTAL	31442	32080	638	2	1069	31364	32057	694	2	1069
WR	Gujarat	11719	11161	-558	-5	372	11719	11161	-558	-5	372
	Madhya Pradesh	8829	9323	494	6	311	8829	9323	494	6	311
	Chhattisgarh	2595	2694	99	4	90	2595	2694	99	4	90
	Maharashtra	15511	16726	1215	8	558	15511	16726	1215	8	558
	Goa	362	422	60	16	14	362	422	60	17	14
	DNHDDPDCL	800	801	1	0	27	800	801	1	0	27
	AMNSIL	496	558	62	12	19	496	558	62	12	19
	BALCO*		373	-	-	12		373	-	-	12
	TOTAL	40312	42058	1745	4	1402	40312	42058	1745	4	1402
SR	Andhra Pradesh	5500	6296	796	14	210	5484	6296	812	15	210
	Telangana	5098	6217	1119	22	207	5079	6220	1141	22	207
	Karnataka	6015	7191	1176	20	240	5996	7194	1198	20	240
	Kerala	2242	2415	173	8	80	2235	2416	180	8	81
	Tamil Nadu	8521	8759	238	3	292	8495	8762	267	3	292
	Puducherry	258	256	-2	-1	9	257	256	-1	-1	9
	TOTAL	27635	31134	3499	13	1038	27547	31143	3596	13	1038
ER	Bihar	2455	2635	180	7	88	2448	2630	182	7	88
	Jharkhand	958	929	-29	-3	31	853	906	52	6	30
	DVC	2087	2105	17	1	70	2087	2105	17	1	70
	Odisha	3022	3010	-12	0	100	3022	3010	-12	0	100
	West Bengal	3800	4276	476	13	143	3800	4276	477	13	143
	Sikkim	48	38	-9	-19	1	48	38	-9	-19	1
	Railways_ER ISTS**		4	-	-	0		4	-	-	0
	TOTAL	12371	12998	627	5	433	12258	12969	711	6	432
NER	Arunachal Pradesh	72	74	2	3	2	72	74	2	3	2
	Assam	812	847	35	4	28	812	847	35	4	28
	Manipur	84	80	-5	-6	3	84	80	-5	-6	3
	Meghalaya	200	190	-10	-5	6	200	190	-10	-5	6
	Mizoram	56	51	-6	-10	2	56	51	-6	-10	2
	Nagaland	65	71	5	8	2	65	71	5	8	2
	Tripura	121	123	2	2	4	120	123	2	2	4
	TOTAL	1410	1434	24	2	48	1410	1434	24	2	48
ALL INDIA		113170	119704	6534	6	3990	112891	119661	6770	6	3989

* BALCO has been added as new entity wef 20.12.2022

** Railways_ER ISTS has been added as new entity wef 29.04.2023

13. DEMAND COMPARISON OF NOVEMBER 2023 vs NOVEMBER 2022

REGION	STATE	Peak Requirement (MW)				Peak Met (MW)			
		Nov-22	Nov-23	Difference	% Change	Nov-22	Nov-23	Difference	% Change
NR	Chandigarh	254	205	-49	-19.3	254	205	-49	-19.3
	Delhi	3941	4320	379	9.6	3941	4320	379	9.6
	Haryana	7578	7685	107	1.4	7578	7685	107	1.4
	Himachal Pradesh	1965	1977	12	0.6	1965	1977	12	0.6
	J&K(UT) and Ladakh(UT)	3074	2710	-364	-11.9	2764	2710	-54	-2.0
	Punjab	7253	7572	319	4.4	7253	7572	319	4.4
	Rajasthan	16023	16409	386	2.4	16023	16409	386	2.4
	Uttar Pradesh	17387	19033	1646	9.5	17387	19033	1646	9.5
	Uttarakhand	2141	2141	0	0.0	2141	2141	0	0.0
	Railway/NFL	177	210	34	19.0	177	210	34	19.0
WR	Gujarat	19704	21694	1990	10	19704	21694	1990	10.1
	Madhya Pradesh	16085	17117	1032	6	16085	17097	1012	6.3
	Chhattisgarh	4098	4648	550	13	4098	4644	547	13.3
	Maharashtra	25936	28386	2449	9	25927	28356	2429	9.4
	Goa	679	733	54	8	679	733	54	8.0
	DNHDDPDCL	1205	1267	62	5.1	1205	1267	62	5.1
	AMNSIL	829	920	90	10.9	829	920	90	10.9
	BALCO*		536	-	-		536	-	-
SR	Andhra Pradesh	9605	12085	2480	25.8	9605	12085	2480	25.8
	Telangana	9971	11857	1886	18.9	9953	11857	1904	19.1
	Karnataka	12448	15203	2755	22.1	12448	15203	2755	22.1
	Kerala	3942	4300	358	9.1	3942	4300	358	9.1
	Tamil Nadu	15070	16099	1029	6.8	15070	16099	1029	6.8
	Puducherry	403	454	51	12.7	403	454	51	12.7
ER	Bihar	5077	5519	442	8.7	4752	5257	505	10.6
	Jharkhand	2024	1699	-325	-16.1	1498	1699	201	13.4
	DVC	3262	3427	165	5.1	3262	3427	165	5.1
	Odisha	5411	5264	-147	-2.7	5411	5264	-147	-2.7
	West Bengal	7933	8481	548	6.9	7929	8481	552	7.0
	Sikkim	118	101	-17	-14.4	118	101	-17	-14.4
	Railways_ER ISTS**		23	23	-		23	23	-
NER	Arunachal Pradesh	125	160	35	28.0	125	160	35	28.0
	Assam	1714	1931	217	12.7	1714	1931	217	12.7
	Manipur	219	207	-12	-5.4	219	207	-12	-5.4
	Meghalaya	381	363	-18	-4.7	381	363	-18	-4.7
	Mizoram	135	144	9	6.4	135	144	9	6.4
	Nagaland	165	147	-18	-10.8	165	147	-18	-10.8
	Tripura	282	299	17	6.0	282	299	17	6.0

* BALCO has been added as new entity wef 20.12.2022

** Railways_ER ISTS has been added as new entity wef 29.04.2023

14. SCHEDULE AND DRAWAL OF CONSTITUENTS - NOVEMBER 2023

REGION	STATE	SCHEDULE (MU)	ACTUAL DRAWAL (MU)	Over drawal(+)/ Under Drawal(-) (MU)	% OD/UD	SCHEDULE (MU/DAY)	ACTUAL DRAWAL (MU/DAY)	Over drawal(+)/ Under Drawal(-) (MU/DAY)
NR	Chandigarh	104.21	102.98	-1.23	-1.18	3.47	3.43	-0.04
	Delhi	1827.38	1821.71	-5.67	-0.31	60.91	60.72	-0.19
	Haryana	2767.76	2736.25	-31.51	-1.14	92.26	91.21	-1.05
	Himachal Pradesh	721.04	711.89	-9.15	-1.27	24.03	23.73	-0.31
	J&K(UT) and Ladakh(UT)	1341.19	1379.30	38.11	2.84	44.71	45.98	1.27
	Punjab	1382.38	1308.40	-73.98	-5.35	46.08	43.61	-2.47
	Rajasthan	3446.51	3352.81	-93.70	-2.72	114.88	111.76	-3.12
	Uttar Pradesh	3304.47	3262.09	-42.38	-1.28	110.15	108.74	-1.41
	Uttarakhand	814.08	829.08	15.00	1.84	27.14	27.64	0.50
	Railway	109.57	111.19	1.63	1.48	3.65	3.71	0.05
	NFL	1.90	1.76	-0.14	-7.39	0.06	0.06	0.00
	Total	15820.49	15617.46	-203.02	-1.28	527.35	520.58	-6.77
WR	Gujarat	5226.31	5212.75	-13.56	-0.26	174.21	173.76	-0.45
	Madhya Pradesh	5718.90	5583.26	-135.64	-2.37	190.63	186.11	-4.52
	Chhattisgarh	1031.00	1007.80	-23.20	-2.25	34.37	33.59	-0.77
	Maharashtra	6476.09	6410.64	-65.46	-1.01	215.87	213.69	-2.18
	Goa	388.22	406.26	18.04	4.65	12.94	13.54	0.60
	DNHDDPDCL	801.44	800.56	-0.88	-0.11	26.71	26.69	-0.03
	AMNSIL	267.10	267.77	0.67	0.25	8.90	8.93	0.02
	BALCO*	374.41	373.21	-1.20	-0.32	12.48	12.44	-0.04
	Total	20283.47	20062.24	-221.23	-1.09	676.12	668.74	-7.37
SR	Andhra Pradesh	2700.68	2676.52	-24.16	-0.89	90.02	89.22	-0.81
	Telangana	3012.47	3011.88	-0.59	-0.02	100.42	100.40	-0.02
	Karnataka	3054.84	2995.07	-59.77	-1.96	101.83	99.84	-1.99
	Kerala	1954.78	1984.48	29.70	1.52	65.16	66.15	0.99
	Tamil Nadu	5053.04	4995.54	-57.50	-1.14	168.43	166.52	-1.92
	Puducherry	253.37	243.29	-10.08	-3.98	8.45	8.11	-0.34
	Goa (SR)	67.28	60.88	-6.40	-9.51	2.24	2.03	-0.21
	Total	16096.46	15967.66	-128.80	-0.80	536.55	532.26	-4.29
ER	Bihar	2402.09	2364.88	-37.21	-1.55	80.07	78.83	-1.24
	Jharkhand	699.88	692.28	-7.60	-1.09	23.33	23.08	-0.25
	DVC	-1144.19	-1123.09	21.10	-1.84	-38.14	-37.44	0.70
	Odisha	857.78	810.66	-47.13	-5.49	28.59	27.02	-1.57
	West Bengal	567.16	478.25	-88.92	-15.68	18.91	15.94	-2.96
	Sikkim	39.51	38.50	-1.01	-2.56	1.32	1.28	-0.03
	Railways_ER ISTS*	3.85	3.52	-0.33	-8.67	0.13	0.12	-0.01
	Total	3426.09	3264.99	-161.10	-4.70	114.20	108.83	-5.37
NER	Arunachal Pradesh	73.99	74.24	0.25	0.34	2.47	2.47	0.01
	Assam	644.20	643.20	-1.00	-0.16	21.47	21.44	-0.03
	Manipur	79.89	79.50	-0.39	-0.49	2.66	2.65	-0.01
	Meghalaya	140.50	140.30	-0.20	-0.14	4.68	4.68	-0.01
	Mizoram	49.27	45.24	-4.03	-8.18	1.64	1.51	-0.13
	Nagaland	66.04	65.17	-0.87	-1.32	2.20	2.17	-0.03
	Tripura	111.09	105.92	-5.17	-4.65	3.70	3.53	-0.17
	Total	1164.98	1153.57	-11.41	-0.98	38.83	38.45	-0.38
All India Total		56791.48	56065.92	-725.57	-1.28	1893.05	1868.86	-24.19

* BALCO has been added as new entity wef 20.12.2022

** Railways_ER ISTS has been added as new entity wef 29.04.2023

15. INTER REGIONAL EXCHANGES 2023-24 (All figures in MU)									
	Apr'23	May'23	June'23	July'23	Aug'23	Sep'23	Oct'23	Nov'23	Fin. Year 2023-24
Name of Line									
Import of NR from WR (WR-NR)									
WR - NR 765 kV Gwalior - Agra 2xS/C	635.95	996.71	863.68	972.77	772.12	902.82	479.75	675.87	6299.67
WR-NR 765 kV Gwalior - Phagi 2xS/C	553.05	453.93	342.56	671.61	397.70	687.31	582.69	906.94	4595.79
WR-NR 765 kV Jabalpur-Orai D/C	430.29	804.53	778.33	964.77	658.94	991.15	602.03	622.44	5852.48
WR-NR 765 kV Satna-Orai	505.37	598.85	574.83	615.77	543.29	593.66	548.69	623.24	4603.70
WR-NR 765 kV Gwalior-Orai	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
WR - NR HVDC Champa - Kurukshetra	804.26	1126.68	2069.11	2235.18	1955.13	2230.56	739.22	674.04	11834.18
WR - NR HVDC Vindhyachal	0.49	3.23	3.13	94.78	21.55	7.26	50.51	28.33	209.28
WR - NR 400 kV Zerda Kankroli	0.00	0.00	1.48	15.36	13.10	8.58	0.00	1.30	39.82
WR - NR 400 kV Zerda Bhinmal	0.00	0.00	0.00	12.54	13.56	23.81	0.34	3.28	53.53
WR - NR 400 kV RAPP C - Shujalpur D/C	2.28	31.04	16.05	164.49	80.51	147.53	18.12	25.82	485.84
WR - NR 220 kV Bhanpura - Ranpur	9.67	65.53	68.13	70.02	74.29	61.03	65.71	43.01	457.39
WR - NR 220 kV Bhanpura - Modak	0.00	0.00	0.00	0.00	0.00	38.00	62.92	72.56	173.48
WR - NR 220 kV Auraiya - Malanpur/Mehgaon	0.00	0.15	0.00	0.00	0.00	0.00	0.00	0.00	0.15
WR - NR 400 kV Vindhyachal-Rihand D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
WR - NR HVDC Mundra - Mohindergarh	132.55	340.75	234.27	124.89	667.23	827.57	920.93	511.53	3759.72
WR-NR 765 kV Banaskata-Chittorgarh D/C	0.00	3.26	16.02	169.31	260.02	66.60	0.00	4.54	519.75
WR-NR 765 kV Varanasi-Vindhyachal	1030.05	1663.34	1854.30	1854.38	1551.38	1697.01	1174.50	1049.13	11874.09
Total WR - NR	4103.96	6088.00	6821.89	7965.87	7008.82	8282.89	5245.41	5242.03	50758.87
Export of NR to WR (NR-WR)									
NR - WR 765 kV Gwalior - Agra 2xS/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 765 kV Gwalior - Phagi 2xS/C	0.00	5.41	0.00	0.00	0.00	0.00	0.00	0.00	5.41
NR - WR 765 kV Jabalpur-Orai D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 765 kV Satna-Orai	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 765 kV Gwalior-Orai	440.87	385.11	301.32	367.88	368.83	385.80	399.85	459.18	3108.84
NR - WR HVDC Champa - Kurukshetra	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR HVDC Vindhyachal	289.83	273.88	267.04	184.43	225.73	180.86	26.34	219.10	1667.21
NR - WR 400 kV Zerda Kankroli	185.75	94.43	108.34	24.78	31.17	41.36	96.48	75.31	657.62
NR - WR 400 kV Zerda Bhinmal	282.42	181.64	200.56	73.76	74.62	64.65	158.50	60.27	1096.42
NR - WR 400 kV RAPP C - Shujalpur D/C	85.74	45.64	45.64	0.00	3.36	0.00	46.93	23.25	250.56
NR - WR 220 kV Bhanpura - Ranpur	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 220 kV Bhanpura - Modak	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - WR 220 kV Auraiya - Malanpur/Mehgaon	56.56	28.58	50.82	47.16	97.78	68.03	111.08	74.65	534.66
NR-WR 400 kV Rihand-Vindhyachal D/C	644.30	672.49	642.32	664.33	653.25	658.85	669.33	655.41	5260.28
NR - WR HVDC Mundra - Mohindergarh	243.86	54.36	36.25	94.16	0.00	0.00	0.00	0.00	428.63
NR - WR 765 kV Banaskata-Chittorgarh D/C	976.83	428.19	421.52	94.53	51.63	180.84	597.81	524.56	3275.91
NR-WR 765 kV Varanasi-Vindhyachal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total NR - WR	3206.16	2169.73	2073.81	1551.03	1506.37	1580.39	2106.32	2091.73	16285.54

	Apr'23	May'23	June'23	July'23	Aug'23	Sep'23	Oct'23	Nov'23	Fin. Year 2023-24
Name of Line									
Import of NR from ER (ER-NR)									
ER - NR 765 kV Sasaram - Fatehpur	166.02	114.36	82.57	74.16	131.29	131.20	192.41	301.01	1193.02
ER - NR 765 kV Gaya - Varanasi 2*S/C	224.56	107.46	28.46	47.20	75.64	115.88	182.55	439.13	1220.88
ER - NR 765 kV Gaya - Ballia	246.38	335.46	321.28	300.74	224.31	332.43	250.12	325.32	2336.04
ER - NR HVDC Alipurduar - Agra	0.00	0.00	375.05	784.99	444.13	750.81	497.43	168.00	3020.41
ER-NR 400 kV Patna - Ballia D/C	219.16	246.32	242.35	242.23	300.66	262.01	203.70	186.43	1902.86
ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	127.03	223.63	277.35	376.44	415.21	330.20	255.04	312.10	2317.00
ER - NR 400 kV Biharshariff - Ballia D/C	22.11	72.32	102.97	92.77	101.30	90.47	56.79	138.97	677.70
ER - NR 400 kV Motihari - Gorakhpur D/C	156.88	188.70	180.09	228.59	251.16	190.46	142.10	216.07	1554.05
ER - NR 400 kV Biharshariff - Varanasi D/C	67.56	47.21	34.66	24.65	44.14	47.88	63.00	164.47	493.57
ER - NR 400 kV Sasaram -Varanasi	74.99	38.68	46.09	55.64	28.71	57.56	16.47	4.09	322.23
ER - NR 400 kV Sasaram -Allahabad	51.91	33.39	21.29	28.77	17.25	52.61	27.20	32.87	265.29
ER - NR 220 kV Sahupuri-Karamnasa	78.64	82.22	82.32	43.45	51.19	29.26	36.16	53.46	456.70
ER - NR 132 kV Sahupuri - Karamnasa	1.92	0.00	0.96	1.12	0.96	2.02	3.28	5.11	15.37
ER - NR 132 kV 132 kV Rihand - Nagar Untari	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ER - NR 132 kV Rihand - Garhwa	0.46	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.46
ER-NR 400 kV Naubatpur - Ballia D/C	211.13	243.59	239.56	236.53	300.12	264.57	198.04	328.08	2021.62
Total ER-NR	1648.75	1733.34	2035.00	2537.28	2386.07	2657.36	2124.29	2675.11	17797.20
Import of NR from NER (NER-NR)									
NER-NR HVDC Biswanath Chariali - Agra	0.00	0.00	153.62	339.28	465.76	447.38	470.05	190.40	2066.49
Total NER - NR	0.00	0.00	153.62	339.28	465.76	447.38	470.05	190.40	2066.49
Export of NR to ER (NR-ER)									
NR - ER 765 kV Sasaram - Fatehpur	0.00	2.65	7.74	3.45	0.00	4.62	0.00	0.00	18.46
NR - ER 765 kV Gaya - Varanasi 2*S/C	0.30	38.74	102.98	45.94	11.70	32.51	8.41	1.45	242.03
NR - ER 765 kV Gaya - Ballia	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER HVDC Alipurduar - Agra	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER 400 kV Patna - Ballia 2xD/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER 400 kV Muzaffarpur - Gorakhpur D/C	3.47	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3.47
NR - ER 400 kV Biharshariff - Ballia D/C	17.15	4.18	13.06	0.00	0.00	3.12	4.14	0.75	42.40
ER - NR 400 kV Motihari - Gorakhpur D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER 400 kV Biharshariff - Varanasi D/C	2.90	7.31	20.53	10.39	0.22	12.39	4.69	0.00	58.43
NR - ER 400 kV Sasaram -Varanasi	0.00	0.00	0.00	0.00	1.67	0.00	0.85	5.47	7.99
NR - ER 400 kV Sasaram -Allahabad	2.25	0.00	0.00	0.82	2.28	0.00	0.23	0.00	5.58
NR - ER 220 kV Sahupuri-Karamnasa	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NR - ER 132 kV Sahupuri - Karamnasa	0.01	0.96	0.48	0.96	0.00	0.58	0.48	0.00	3.47
NR - ER 132 kV 132 kV Rihand - Nagar Untari	1.74	2.20	1.08	3.37	2.39	3.46	1.80	2.75	18.79
ER - NR 132 kV Rihand - Garhwa	10.23	8.48	19.67	18.35	14.38	11.19	11.51	8.39	102.20
NR-ER 400 kV Ballia - Naubatpur D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total NR - ER	38.05	64.52	165.54	83.28	32.64	67.87	32.11	18.81	502.82
Export of NR to NER (NER-NR)									
NR-NER HVDC Biswanath Chariali - Agra	261.93	291.71	104.47	0.00	0.00	0.00	0.00	257.02	915.13
Total NR - NER	261.93	291.71	104.47	0.00	0.00	0.00	0.00	257.02	915.13

	Apr'23	May'23	June'23	July'23	Aug'23	Sep'23	Oct'23	Nov'23	Fin. Year 2023-24
Name of Line									
Export of WR to ER (WR-ER)									
WR - ER 765 kV Ranchi - Dharamjaygarh 2xS/C	139.81	504.96	673.15	734.00	331.39	537.29	228.34	42.77	3191.71
WR - ER 765 kV Dharamjaygarh - Jharsuguda 2xD/C	652.31	469.82	490.99	593.61	768.89	177.22	238.94	52.91	3444.69
WR - ER 765 kV Durg - Jharsuguda D/C	0.00	0.64	1.75	26.78	7.42	1.32	0.00	0.00	37.91
WR - ER 400 kV Sipat - Ranchi D/C	15.97	93.84	118.29	133.75	52.57	87.51	30.29	8.20	540.42
WR - ER 400 kV Raigarh - Jharsuguda 2xD/C	0.44	6.23	8.79	6.26	28.69	11.41	0.82	0.78	63.43
WR - ER 220 kV Korba - Budhipadar D/C	36.08	80.04	73.19	22.96	28.75	29.18	26.25	36.70	333.16
WR - ER 220 kV Raigarh - Budhipadar	0.15	0.57	1.17	0.00	0.35	0.34	0.00	0.08	2.65
Total WR - ER	844.75	1156.11	1367.33	1517.34	1218.07	844.29	524.64	141.44	7613.97
Import of WR from ER (ER - WR)									
ER - WR 765 kV Ranchi - Dharamjaygarh 2xS/C	144.10	11.95	15.57	10.81	83.69	14.84	62.72	294.10	637.78
ER - WR 765 kV Dharamjaygarh - Jharsuguda 2XD/C	25.46	58.29	33.08	15.65	2.33	214.83	131.20	338.83	819.68
ER - WR 765 kV Durg - Jharsuguda D/C	394.08	204.42	148.02	71.84	73.64	163.98	298.15	464.99	1819.11
ER - WR 400 kV Sipat - Ranchi D/C	55.56	5.71	9.07	7.22	29.65	8.91	40.59	105.31	262.01
ER - WR 400 kV Raigarh - Jharsuguda 2xD/C	248.78	148.94	125.69	125.00	127.87	110.94	267.35	221.93	1376.49
ER - WR 220 kV Korba - Budhipadar D/C	1.45	0.00	0.14	6.23	1.99	1.46	3.18	3.22	17.67
ER - WR 220 kV Raigarh - Budhipadar	55.98	38.46	41.41	66.29	74.76	49.37	81.54	41.32	449.12
Total ER - WR	925.41	467.76	372.98	303.04	393.92	564.32	884.73	1469.71	5381.87
Export of ER to NER (ER - NER)									
ER - NER 400 kV Binaguri - Bongaigaon D/C	6.99	16.15	105.81	142.19	71.19	215.10	134.98	60.68	753.09
ER - NER 400 kV Alipurduar - Bongaigaon D/C	4.62	34.31	135.94	83.03	84.82	161.59	95.15	85.48	684.94
ER - NER 220 kV Birpara - Salakati D/C	0.54	3.12	31.19	37.25	25.19	47.94	25.71	15.50	186.43
Total ER - NER	12.16	53.58	272.93	262.48	181.20	424.63	255.83	161.66	1624.46
Import of ER from NER (NER - ER)									
NER - ER 400 kV Binaguri - Bongaigaon D/C	52.83	40.19	37.75	2.10	10.62	0.07	5.53	48.07	197.16
NER - ER 400 kV Alipurduar - Bongaigaon D/C	182.37	135.22	72.81	20.59	29.76	3.08	23.52	172.08	639.43
NER - ER 220 kV Birpara - Salakati D/C	34.59	31.32	10.78	0.62	2.83	0.14	2.65	38.72	121.64
Total NER - ER	269.78	206.73	121.33	23.31	43.22	3.29	31.70	258.87	958.23
Export of ER to SR (ER - SR)									
ER - SR HVDC Gazuwaka	375.93	395.23	208.61	178.44	321.36	201.99	378.26	352.24	2412.06
ER - SR HVDC Talchar - Kolar	1233.20	1204.28	984.75	977.61	1284.58	1076.42	1294.48	1214.02	9269.33
ER - SR 765 kV Angul- Srikakulam D/C	1683.42	1561.33	1412.42	1184.37	1380.27	1160.32	1534.32	1466.97	11383.43
Total ER - SR	3292.55	3160.85	2605.77	2340.42	2986.21	2438.74	3207.06	3033.23	23064.82
Import of ER from SR (SR - ER)									
SR - ER HVDC Gazuwaka	0.00	0.00	55.34	114.11	32.48	0.00	0.00	6.03	207.97
SR - ER HVDC Talchar - Kolar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
SR - ER 765 kV Angul- Srikakulam D/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total SR - ER	0.00	0.00	55.34	114.11	32.48	0.00	0.00	6.03	207.97

	Apr'23	May'23	June'23	July'23	Aug'23	Sep'23	Oct'23	Nov'23	Fin. Year 2023-24
Name of Line									
Export of WR to SR (WR-SR)									
WR - SR HVDC Bhadrawati	544.41	411.52	171.68	139.64	464.81	222.80	593.89	435.49	2984.25
WR - SR 765 kV Sholapur - Raichur 2xS/C	417.69	222.76	146.07	83.93	451.70	125.10	226.89	116.70	1790.85
WR - SR 765 kV Wardha - Nizamabad D/C	1323.51	873.44	1018.25	852.05	1313.80	864.24	1099.04	930.66	8274.99
WR - SR 400KV Kolhapur-Kudgi D/C	0.00	0.00	0.00	0.00	0.06	0.00	0.00	0.00	0.06
WR-SR HVDC Raigarh-Pugalur	3004.35	1791.86	1182.91	885.62	2135.69	1450.16	2583.88	1858.37	14892.83
WR-SR 220kV Xeldem - Ambewadi S/C	77.16	70.18	68.17	61.74	68.61	67.55	69.49	67.29	550.18
WR-SR 220kV Ponda - Ambewadi S/C	0.29	1.54	0.28	0.16	5.23	0.04	0.08	0.97	8.58
Total WR - SR	5367.41	3371.30	2587.37	2023.14	4439.89	2729.89	4573.27	3409.48	28501.74
Import of WR from SR (SR - WR)									
SR - WR HVDC Bhadrawati	4.11	25.09	178.09	308.60	10.36	114.86	0.00	0.00	641.13
SR - WR 765 kV Raichur - Sholapur 2xS/C	29.13	205.27	344.82	478.51	54.56	420.76	215.80	334.15	2082.99
SR - WR 765 kV Wardha - Nizamabad D/C	0.00	3.90	0.32	18.33	0.00	19.87	0.03	0.89	43.34
SR - WR 400KV Kolhapur-Kudgi D/C	623.59	719.30	784.98	796.02	623.43	850.54	765.68	701.36	5864.91
SR-WR HVDC Pugalur-Raigarh	0.00	3.73	14.87	16.65	0.00	0.09	0.00	0.00	35.33
WR-SR 220kV Ambewadi - Xeldem S/C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
WR-SR 220kV Ambewadi-Ponda S/C	0.00	0.00	0.01	0.00	0.01	0.01	0.02	0.01	0.07
Total SR - WR	656.84	957.30	1323.09	1618.10	688.36	1406.13	981.53	1036.41	8667.76
TOTAL ALL INDIA	20628	19721	20060	20679	21383	21447	20437	19992	164347

*In case of mutple ckt / DC pole year corres. to the commissioning of final element

Date	15.1 Import-Export of NR with WR during November 2023																	
	Import of NR from WR (WR-NR)																	
	WR - NR 765 kV Gwalior - Agra 2x5/C	WR-NR 765 kV Gwalior-Phagi 2x5/C	WR-NR 765 kV Jabalpur- Orai D/C	WR-NR 765 kV Satna- Orai	WR-NR 765 kV Gwalior-Orai	WR - NR HVDC Champa- Kurukshetra	WR - NR HVDC Vindhyachal	WR - NR 400 kV Zerda- Kankroli	WR - NR 400 kV Zerda- Bhinmal	WR-NR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura - Ranpur	WR - NR 220 kV Bhanpura - Modak	WR - NR 220 kV Auraiya - Malanpur / Mehgaon	WR-NR 400 KV Rihand Vindhyachal D/C	WR - NR HVDC Mundra - M'garh	WR - NR 765 kV Banaskata - Chittorgarh D/C	WR-NR 765 kV Varanasi- Vindhyachal	Total WR - NR
1-Nov-23	15.65	33.60	18.90	19.94	0.00	15.34	5.98	0.00	0.00	0.00	1.60	2.80	0.00	0.00	22.12	0.00	33.46	169.39
2-Nov-23	9.84	29.30	13.80	17.13	0.00	18.38	6.00	0.00	0.00	0.00	1.72	2.90	0.00	0.00	21.18	0.00	30.20	150.45
3-Nov-23	11.40	31.64	18.20	19.10	0.00	19.58	5.98	0.00	0.00	0.00	1.73	2.24	0.00	0.00	19.40	0.00	29.53	158.80
4-Nov-23	11.28	30.89	9.54	19.65	0.00	18.83	6.02	0.00	0.00	0.00	1.66	2.83	0.00	0.00	21.63	0.00	28.40	150.73
5-Nov-23	16.89	31.18	19.00	19.49	0.00	18.84	4.35	0.00	0.00	0.00	1.49	2.86	0.00	0.00	23.64	0.00	31.91	169.65
6-Nov-23	23.50	34.37	22.03	20.32	0.00	19.27	0.00	0.00	0.00	0.74	1.08	2.78	0.00	0.00	12.16	0.00	38.11	174.36
7-Nov-23	22.82	35.35	25.40	21.50	0.00	18.83	0.00	0.00	0.00	0.75	1.78	2.89	0.00	0.00	24.12	0.00	44.24	197.68
8-Nov-23	22.29	36.05	25.60	21.40	0.00	18.91	0.00	0.00	0.00	0.98	1.75	2.83	0.00	0.00	12.21	0.00	46.04	188.06
9-Nov-23	23.47	36.87	27.55	22.33	0.00	19.65	0.00	0.00	0.00	1.14	1.22	2.72	0.00	0.00	11.65	0.00	42.21	188.81
10-Nov-23	18.62	30.20	22.05	19.98	0.00	29.58	0.00	0.00	0.00	0.00	1.22	2.53	0.00	0.00	11.94	0.00	45.35	181.47
11-Nov-23	22.20	27.09	25.06	20.88	0.00	21.46	0.00	0.00	0.00	0.00	1.84	2.27	0.00	0.00	12.18	0.00	49.15	182.13
12-Nov-23	16.30	19.31	16.26	16.77	0.00	18.67	0.00	0.00	0.00	0.00	1.22	1.83	0.00	0.00	12.16	0.00	49.87	152.39
13-Nov-23	13.70	20.67	15.15	16.10	0.00	18.90	0.00	0.00	0.00	0.00	1.64	1.75	0.00	0.00	12.16	0.00	40.08	140.15
14-Nov-23	19.03	24.78	18.12	18.20	0.00	18.80	0.00	0.00	0.00	0.00	1.59	1.79	0.00	0.00	12.16	0.00	40.34	154.81
15-Nov-23	24.69	29.65	20.10	20.80	0.00	18.80	0.00	0.00	0.00	0.00	1.40	1.84	0.00	0.00	12.16	0.00	30.30	159.74
16-Nov-23	19.40	26.82	17.28	19.63	0.00	18.39	0.00	0.00	0.00	0.00	1.68	2.16	0.00	0.00	19.10	0.00	23.93	148.39
17-Nov-23	23.44	32.41	20.10	22.06	0.00	19.43	0.00	0.00	0.00	0.68	1.65	2.35	0.00	0.00	24.21	0.00	29.47	175.80
18-Nov-23	24.60	35.89	21.92	22.80	0.00	18.63	0.00	0.00	0.00	1.55	1.70	2.52	0.00	0.00	21.44	0.00	28.75	179.80
19-Nov-23	25.80	33.26	24.58	21.72	0.00	7.46	0.00	0.00	0.00	0.67	0.69	2.43	0.00	0.00	18.88	0.00	29.40	164.89
20-Nov-23	29.04	30.62	26.37	22.40	0.00	2.44	0.00	0.00	0.00	0.28	1.62	2.53	0.00	0.00	15.57	0.00	30.05	160.92
21-Nov-23	22.42	29.46	19.15	21.43	0.00	19.30	0.00	0.00	0.00	0.00	1.56	2.57	0.00	0.00	24.87	0.00	23.32	164.08
22-Nov-23	16.50	30.80	15.65	21.15	0.00	19.83	0.00	0.00	0.00	0.00	1.51	2.62	0.00	0.00	24.60	0.00	17.61	150.27
23-Nov-23	11.59	29.42	14.64	18.21	0.00	19.84	0.00	0.00	0.00	0.00	1.46	2.53	0.00	0.00	27.62	0.00	16.72	142.03
24-Nov-23	20.33	30.81	20.56	20.28	0.00	19.98	0.00	0.00	0.00	0.00	1.44	2.61	0.00	0.00	29.03	0.00	20.15	165.19
25-Nov-23	28.30	29.35	23.92	21.79	0.00	20.10	0.00	0.00	0.00	0.00	1.33	2.27	0.00	0.00	27.72	0.00	30.95	185.73
26-Nov-23	36.24	33.24	30.08	22.34	0.00	28.98	0.00	1.26	0.00	2.96	1.38	2.38	0.00	0.00	17.22	1.96	41.26	219.30
27-Nov-23	37.20	29.96	29.15	27.20	0.00	42.62	0.00	0.00	0.00	4.93	1.03	1.84	0.00	0.00	5.28	0.00	41.92	221.13
28-Nov-23	36.30	28.13	15.20	23.40	0.00	48.40	0.00	0.00		3.46	1.04	2.26	0.00	0.00	4.28	1.78	43.50	207.75
29-Nov-23	36.58	30.31	19.70	23.70	0.00	47.98	0.00	0.04	1.90	3.97	0.98	2.27	0.00	0.00	6.84	0.80	46.60	221.67
30-Nov-23	36.45	25.51	27.38	21.54	0.00	46.82	0.00	0.00	1.38	3.71	1.00	2.36	0.00	0.00	4.00	0.00	46.31	216.46
Total	675.87	906.94	622.44	623.24	0.00	674.04	28.33	1.30	3.28	25.82	43.01	72.56	0.00	0.00	511.53	4.54	1049.13	5242.03

Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day

Date	15.1 Import-Export of NR with WR during November 2023																	
	Export of NR to WR (NR- WR)																	
	NR - WR 765 kV Gwalior - Agra 2xS/C	NR-WR 765 kV Gwalior-Phagi 2xS/C	NR-WR 765 kV Jabalpur- Orai D/C	NR-WR 765 kV Satna- Orai	NR-WR 765 kV Gwalior-Orai	NR - WR HVDC Champa- Kurukshetra	NR - WR HVDC Vindhyachal	NR - WR 400 kV Zerda- Kankrol	NR - WR 400 kV Zerda- Bhinmal	NR-WR 400 kV RAPP C - Shujalpur D/C	WR - NR 220 kV Bhanpura - Ranpur	WR - NR 220 kV Bhanpura - Modak	NR - WR 220 kV Auraiya - Malanpur / Mehgaon	NR-WR 400 kV Rihand- Vindhyachal D/C	NR - WR HVDC Mundra - M'garh	NR - WR 765 kV Banaskata - Chittorgarh D/C	NR-WR 765 kV Varanasi- Vindhyachal	Total NR - WR
1-Nov-23	0.00	0.00	0.00	0.00	17.77	0.00	0.00	3.94	3.87	0.09	0.00	0.00	3.27	22.23	0.00	27.41	0.00	78.58
2-Nov-23	0.00	0.00	0.00	0.00	9.00	0.00	0.00	4.06	4.54	1.01	0.00	0.00	3.48	22.00	0.00	28.64	0.00	72.73
3-Nov-23	0.00	0.00	0.00	0.00	10.30	0.00	0.00	3.78	2.92	1.11	0.00	0.00	3.44	22.48	0.00	25.73	0.00	69.76
4-Nov-23	0.00	0.00	0.00	0.00	4.10	0.00	0.00	3.55	2.73	1.20	0.00	0.00	3.35	22.29	0.00	27.22	0.00	64.44
5-Nov-23	0.00	0.00	0.00	0.00	18.57	0.00	0.00	2.45	2.89	0.65	0.00	0.00	3.08	22.10	0.00	25.72	0.00	75.46
6-Nov-23	0.00	0.00	0.00	0.00	19.77	0.00	3.50	2.50	1.85	0.00	0.00	0.00	2.91	22.10	0.00	22.53	0.00	75.16
7-Nov-23	0.00	0.00	0.00	0.00	18.45	0.00	6.85	2.77	1.85	0.00	0.00	0.00	2.36	22.33	0.00	22.20	0.00	76.81
8-Nov-23	0.00	0.00	0.00	0.00	18.40	0.00	10.30	2.10	0.06	0.00	0.00	0.00	2.76	22.27	0.00	19.01	0.00	74.90
9-Nov-23	0.00	0.00	0.00	0.00	19.18	0.00	10.21	3.19	1.23	0.00	0.00	0.00	2.94	22.24	0.00	26.19	0.00	85.18
10-Nov-23	0.00	0.00	0.00	0.00	17.50	0.00	7.30	3.92	4.30	1.90	0.00	0.00	2.92	22.05	0.00	29.50	0.00	89.39
11-Nov-23	0.00	0.00	0.00	0.00	15.19	0.00	10.20	2.42	2.55	1.68	0.00	0.00	2.89	22.16	0.00	14.35	0.00	71.44
12-Nov-23	0.00	0.00	0.00	0.00	14.50	0.00	12.00	2.90	2.45	3.51	0.00	0.00	3.00	20.22	0.00	18.38	0.00	76.96
13-Nov-23	0.00	0.00	0.00	0.00	14.25	0.00	12.06	3.70	3.70	3.15	0.00	0.00	3.19	22.10	0.00	20.78	0.00	82.93
14-Nov-23	0.00	0.00	0.00	0.00	16.56	0.00	12.04	2.47	2.75	1.49	0.00	0.00	2.98	22.10	0.00	14.03	0.00	74.42
15-Nov-23	0.00	0.00	0.00	0.00	17.95	0.00	10.27	2.32	2.50	0.19	0.00	0.00	2.85	21.98	0.00	14.28	0.00	72.34
16-Nov-23	0.00	0.00	0.00	0.00	16.71	0.00	10.25	2.62	2.62	1.21	0.00	0.00	2.94	22.01	0.00	17.29	0.00	75.65
17-Nov-23	0.00	0.00	0.00	0.00	17.70	0.00	12.15	2.10	1.76	0.00	0.00	0.00	2.47	22.20	0.00	14.28	0.00	72.66
18-Nov-23	0.00	0.00	0.00	0.00	19.62	0.00	10.30	3.10	1.38	0.00	0.00	0.00	2.55	22.42	0.00	23.20	0.00	82.57
19-Nov-23	0.00	0.00	0.00	0.00	18.76	0.00	10.40	2.46	0.46	0.00	0.00	0.00	2.46	22.15	0.00	18.22	0.00	74.91
20-Nov-23	0.00	0.00	0.00	0.00	17.42	0.00	8.90	1.60	0.37	0.00	0.00	0.00	2.24	16.29	0.00	10.77	0.00	57.59
21-Nov-23	0.00	0.00	0.00	0.00	17.37	0.00	10.41	2.70	1.80	1.15	0.00	0.00	2.82	22.10	0.00	18.07	0.00	76.42
22-Nov-23	0.00	0.00	0.00	0.00	8.21	0.00	11.90	3.56	2.38	1.59	0.00	0.00	3.03	22.10	0.00	26.55	0.00	79.32
23-Nov-23	0.00	0.00	0.00	0.00	5.93	0.00	7.80	3.75	3.03	1.62	0.00	0.00	2.85	21.91	0.00	26.12	0.00	73.01
24-Nov-23	0.00	0.00	0.00	0.00	16.91	0.00	2.69	2.58	2.30	0.79	0.00	0.00	2.59	21.98	0.00	15.32	0.00	65.16
25-Nov-23	0.00	0.00	0.00	0.00	16.07	0.00	10.26	3.02	2.98	0.91	0.00	0.00	1.82	22.04	0.00	13.77	0.00	70.87
26-Nov-23	0.00	0.00	0.00	0.00	17.40	0.00	12.36	0.00	0.64	0.00	0.00	0.00	1.13	21.96	0.00	0.00	0.00	53.49
27-Nov-23	0.00	0.00	0.00	0.00	15.80	0.00	6.32	1.52	0.36	0.00	0.00	0.00	0.44	22.46	0.00	3.72	0.00	50.62
28-Nov-23	0.00	0.00	0.00	0.00	12.60	0.00	3.64	0.07		0.00	0.00	0.00	0.63	20.94	0.00	0.00	0.00	37.88
29-Nov-23	0.00	0.00	0.00	0.00	12.28	0.00	3.36	0.00	0.00	0.00	0.00	0.00	0.64	22.10	0.00	0.00	0.00	38.38
30-Nov-23	0.00	0.00	0.00	0.00	14.91	0.00	3.63	0.16	0.00	0.00	0.00	0.00	0.62	22.10	0.00	1.28	0.00	42.70
Total	0.00	0.00	0.00	0.00	459.18	0.00	219.10	75.31	60.27	23.25	0.00	0.00	74.65	655.41	0.00	524.56	0.00	2091.73
Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day																		

Date	15.2 Import-Export of NR with ER & NER during November 2023																		
	Import of NR from ER (ER-NR)																	Import of NR from NER (NER-NR)	
	ER - NR 765 kV Sasaram - Fatehpur	ER - NR 765 kV Gaya - Varanasi 2*5/C	ER - NR 765 kV Gaya Balia	ER - NR HVDC Alipurduar - Agra	ER-NR 400 kV Patna - Balia D/C	ER - NR 400 kV Muzaffarpur - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Balia D/C	ER - NR 400 kV Motihari - Gorakhpur D/C	ER - NR 400 kV Biharshariff - Varanasi D/C	ER-NR 400 kV Sasaram - Varanasi	ER-NR 400 kV Sasaram - Allahabad	ER - NR 220 kV Sahupuri - Karamnasa	ER - NR 132 kV Sahupuri - Karamnasa	ER - NR 132 kV Rihand - Nagar Untari	NR - ER 132 kV Rihand - Garhwa	ER-NR 400 kV Naubatpur - Balia D/C	Total ER - NR	NER-NR HVDC Biswanath Charialli - Agra	Total NER-NR
1-Nov-23	9.15	12.39	10.10	14.42	6.87	5.03	1.52	3.87	3.31	0.35	0.82	1.06		0.00		6.95	75.84	13.93	13.93
2-Nov-23	9.38	12.24	9.62	11.65	6.98	4.93	1.63	3.75	3.55	0.30	0.88	1.10		0.00		7.07	73.08	12.04	12.04
3-Nov-23	9.98	14.70	10.20	12.65	8.80	7.49	2.40	7.20	4.50	0.00	1.20		1.49		0.00	8.98	89.59	12.06	12.06
4-Nov-23	10.02	14.31	9.92	12.14	9.83	5.78	1.86	5.64	2.87	0.16	0.94	1.30	1.30		0.00	10.30	86.37	11.44	11.44
5-Nov-23	8.68	13.01	10.02	12.50	9.82	7.40	2.50	4.80	3.50	0.38	0.80	1.30			0.00	8.05	82.76	11.46	11.46
6-Nov-23	9.42	14.00	11.60	12.56	8.70	8.57	3.75	5.27	5.13	0.00	1.20	1.90	0.48		0.00	8.42	91.00	11.95	11.95
7-Nov-23	7.89	13.23	11.90	11.30	7.20	9.60	3.34	5.46	4.60	0.00	0.93	1.42			0.00	7.32	84.19	11.85	11.85
8-Nov-23	7.24	10.65	11.62	12.66	6.05	8.08	3.88	4.47	4.48	0.00	0.72	1.27		0.00	0.00	6.16	77.28	17.04	17.04
9-Nov-23	7.61	14.84	12.68	11.76	6.69	10.31	4.08	6.10	5.25	0.00	1.34	2.59	0.00		0.00	8.24	91.49	16.68	16.68
10-Nov-23	5.12	12.33	11.98	12.67	7.80	8.58	2.72	5.93	4.98	0.08	0.90	2.52	0.00		0.00	8.15	83.76	17.27	17.27
11-Nov-23	5.83	5.60	11.01	12.00	5.92	6.70	1.29	4.26	2.44	0.46	0.78	1.65			0.00	5.56	63.50	16.91	16.91
12-Nov-23	2.64	0.00	9.06	11.98	3.25	3.24	0.00	1.97	0.10	0.77	0.33	1.02		0.00		3.29	37.65	16.91	16.91
13-Nov-23	3.15	1.60	7.87	12.00	4.10	2.80	0.00	2.27	0.57	0.72	0.36	0.87		0.00	0.00	3.90	40.21	16.17	16.17
14-Nov-23	7.05	8.84	9.77	7.71	5.62	7.27	2.47	4.08	2.14	0.52	0.56	1.30	0.00	0.00	0.00	5.79	63.12	4.69	4.69
15-Nov-23	11.05	13.62	10.48	0.00	9.17	13.13	5.40	7.38	5.61	0.00	1.20	1.78	1.36		0.00	9.62	89.80	0.00	0.00
16-Nov-23	10.63	17.21	9.98		9.92	13.41	5.80	7.59	6.14	0.35	0.74	1.91			0.00	10.64	94.32	0.00	0.00
17-Nov-23	12.60	15.19	10.96		12.30	14.73	6.27	9.25	6.80	0.00	1.26	2.20		0.00	0.00	12.25	103.81	0.00	0.00
18-Nov-23	13.98	20.36	9.82		13.82	16.60	7.82	11.16	8.52	0.00	1.32	2.58		0.00	0.00	14.62	120.60	0.00	0.00
19-Nov-23	14.86	18.10	12.78		14.40	17.10	7.88	11.40	9.88	0.00	1.76	2.94			0.00	15.15	126.25	0.00	0.00
20-Nov-23	13.27	21.33	11.23		12.20	15.38	6.70	10.44	8.10	0.00	1.51	2.42			0.00	13.00	115.58	0.00	0.00
21-Nov-23	13.40	17.20	10.86		11.19	13.70	4.83	9.62	8.10	0.00	1.03	2.24			0.00	11.95	104.12	0.00	0.00
22-Nov-23	14.67	24.41	10.61	0.00	5.80	14.97	6.65	10.91	9.52	0.00	1.13	2.71	0.00		0.00	15.40	116.78	0.00	0.00
23-Nov-23	14.95	25.31	11.03			14.13	6.95	11.25	9.81	0.00	1.24	2.72			0.00	17.82	115.21	0.00	0.00
24-Nov-23	15.27	25.40	11.10			14.87	7.00	11.02	9.45	0.00	1.29	2.31		0.00		16.86	114.57	0.00	0.00
25-Nov-23	12.73	19.55	10.99		0.00	13.42	6.93	9.19	7.40	0.00	1.59	2.09	0.00	0.00		15.33	99.22	0.00	0.00
26-Nov-23	11.24	18.00	11.46		0.00	12.92	8.21	9.22	6.62	0.00	1.52	1.99			0.00	15.50	96.68	0.00	0.00
27-Nov-23	10.56	15.79	12.62			12.88	8.36	9.35	5.92	0.00	1.42	1.92			0.00	17.05	95.87	0.00	0.00
28-Nov-23	9.79	14.60	8.90			10.34	6.40	8.50	4.35	0.00	1.47	1.74			0.00	15.90	81.99	0.00	0.00
29-Nov-23	10.10	13.06	12.62			10.32	6.79	7.44	5.80	0.00	1.50	1.30			0.00	14.39	83.32	0.00	0.00
30-Nov-23	8.75	12.26	12.53			8.42	5.54	7.28	5.03	0.00	1.13	1.31	0.48		0.00	14.42	77.15	0.00	0.00
Total	301.01	439.13	325.32	168.00	186.43	312.10	138.97	216.07	164.47	4.09	32.87	53.46	5.11	0.00	0.00	328.08	2675.11	190.40	190.40
Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day																			

Date	15.2 Import-Export of NR with ER & NER during November 2023																		
	Export of NR to ER (NR- ER)																	Export of NR to NER (NR-NER)	
	NR - ER 765 kV Sasaram - Fatehpur	NR - ER 765 kV Gaya - Varanasi 2*5/C	NR - ER 765 kV Gaya Ballia	NR - ER HVDC Alipurduar - Agra	NR-ER 400 kV Patna - Ballia D/C	NR - ER 400 kV Muzaffarpur - Gorakhpur D/C	NR - ER 400 kV Biharsharif - Ballia D/C	NR - ER 400 kV Motihari - Gorakhpur D/C	NR - ER 400 kV Biharsharif - Varanasi D/C	NR-ER 400 kV Sasaram - Varanasi	NR-ER 400 kV Sasaram - Allahabad	NR - ER 220 kV Sahupuri - Karamnasa	NR - ER 132 kV Sahupuri - Karamnasa	NR - ER 132 kV Rihand - Nagar Untari	NR - ER 132 kV Rihand - Garhwa	NR-ER 400 kV Ballia-Naubatpur D/C	Total NR - ER	NR-NER HVDC Biswanath Chariali - Agra	Total NR-NER
1-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.42		0.00	0.42	0.00	0.00
2-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.48		0.00	0.48	0.00	0.00
3-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.65	0.00		0.00		0.40	0.00	1.05	0.00	0.00
4-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.43	0.00	0.43	0.00	0.00
5-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.41	0.00	0.41	0.00	0.00
6-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.05	0.00	0.00	0.00		0.39	0.00	0.44	0.00	0.00
7-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.39	0.00	0.00			0.39	0.00	0.78	0.00	0.00
8-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.14	0.00	0.00		0.08	0.34	0.00	0.56	0.00	0.00
9-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.20	0.00	0.00	0.00		0.37	0.00	0.57	0.00	0.00
10-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00		0.38	0.00	0.38	0.00	0.00
11-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.38	0.00	0.38	0.00	0.00
12-Nov-23	0.00	1.45	0.00	0.00	0.00	0.00	0.38	0.00	0.00	0.00	0.00	0.00		0.50		0.00	2.33	0.00	0.00
13-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.37	0.00	0.00	0.00	0.00	0.00		0.45	0.39	0.00	1.21	0.00	0.00
14-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.49	0.49	0.00	0.98	0.00	0.00
15-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00		0.39	0.00	0.43	14.29	14.29
16-Nov-23	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00			0.38	0.00	0.38	16.99	16.99
17-Nov-23	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.13	0.00	0.00		0.08	0.29	0.00	0.50	17.06	17.06
18-Nov-23	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.28	0.00	0.00		0.00	0.33	0.00	0.61	15.96	15.96
19-Nov-23	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.65	0.00	0.00			0.41	0.00	1.06	16.79	16.79
20-Nov-23	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.35	0.00	0.00			0.38	0.00	0.73	15.39	15.39
21-Nov-23	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00			0.39	0.00	0.40	16.76	16.76
22-Nov-23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.00	0.00	0.00		0.38	0.00	0.41	15.73	15.73
23-Nov-23	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.22	0.00	0.00			0.18	0.00	0.40	15.44	15.44
24-Nov-23	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.27	0.00	0.00		0.07		0.00	0.34	16.33	16.33
25-Nov-23	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.29	0.00	0.00	0.00	0.18		0.00	0.47	16.74	16.74
26-Nov-23	0.00	0.00	0.00		0.00	0.00	0.00	0.00	0.00	0.48	0.00	0.00			0.08	0.00	0.56	16.31	16.31
27-Nov-23	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.40	0.00	0.00			0.09	0.00	0.49	15.64	15.64
28-Nov-23	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.24	0.00	0.00			0.10	0.00	0.34	13.82	13.82
29-Nov-23	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.43	0.00	0.00			0.13	0.00	0.56	16.89	16.89
30-Nov-23	0.00	0.00	0.00			0.00	0.00	0.00	0.00	0.22	0.00	0.00	0.00		0.49	0.00	0.71	16.88	16.88
Total	0.00	1.45	0.00	0.00	0.00	0.00	0.75	0.00	0.00	5.47	0.00	0.00	0.00	2.75	8.39	0.00	18.81	257.02	257.02
Disclaimer:- Blank entry if the line under outage/shutdown on corresponding day																			

Date	15.3 Import and Export of ER with WR & NER during November 2023																							
	Export of WR to ER (WR-ER)								Import of WR from ER (ER - WR)								Export of ER to NER (ER-NER)				Import of ER from NER (NER-ER)			
	WR - ER 765 kV Dharamjaygarh - Ranchi 2x5/C	WR - ER 765 kV Dharamjaygarh - Jharsuguda Q/C	WR - ER 765 kV Durg - Jharsuguda D/C	WR - ER 400 kV Sipat - Ranchi D/C	WR - ER 400 kV Raigarh - Jharsuguda-2x5/D/C	WR - ER 220 kV Korba - Budhipadar D/C	WR - ER 220 kV Raigarh - Budhipadar	Total WR - ER	ER - WR 765 kV Dharamjaygarh - Ranchi 2x5/C	ER-WR 765 kV Dharamjaygarh - Jharsuguda D/C	ER-WR 765 kV Durg - Jharsuguda D/C	ER - WR 400 kV Sipat - Ranchi D/C	ER - WR 400 kV Raigarh - Jharsuguda-2x5/D/C	ER - WR 220 kV Korba - Budhipadar D/C	ER - WR 220 kV Raigarh - Budhipadar	Total ER - WR	ER - NER 400 kV Binaguri - Bongaigaon D/C	ER - NER 400 kV Alipurduar - Bongaigaon D/C	ER - NER 220 kV Birpara - Salakati D/C	Total ER - NER	NER - ER 400 kV Binaguri - Bongaigaon D/C	NER - ER 400 kV Alipurduar - Bongaigaon 2x5/D/C	NER - ER 220 kV Birpara - Salakati D/C	Total NER - ER
1-Nov-23	0.1	0.0	0.0	0.0	0.0	0.6	0.0	0.74	7.9	16.6	13.9	3.7	11.7	0.2	2.4	56.39	3.76	5.60	1.09	10.45	0.00	0.14	0.01	0.15
2-Nov-23	0.0	0.0	0.0	0.0	0.0	0.6	0.0	0.57	10.2	21.6	15.6	4.8	12.7	0.2	2.5	67.48	3.02	4.12	0.80	7.93	0.00	0.27	0.03	0.30
3-Nov-23	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.17	11.0	17.2	16.2	5.2	14.0	0.4	2.6	66.57	3.58	3.90	0.86	8.35	0.00	0.29	0.03	0.32
4-Nov-23	0.0	0.0	0.0	0.0	0.0	0.7	0.0	0.69	13.5	16.0	15.6	5.3	10.7	0.2	1.9	63.13	4.20	5.47	0.85	10.53	0.00	0.21	0.01	0.22
5-Nov-23	1.2	3.7	0.0	0.0	0.0	1.2	0.0	6.21	8.0	2.7	13.9	3.7	8.6	0.1	1.5	38.42	3.34	3.36	0.45	7.16	0.00	0.57	0.08	0.65
6-Nov-23	1.5	1.6	0.0	0.1	0.0	1.4	0.0	4.55	5.4	5.5	14.3	3.1	9.2	0.0	1.5	38.97	3.21	3.21	0.68	7.10	0.00	0.58	0.06	0.64
7-Nov-23	1.5	0.2	0.0	0.1	0.0	1.5	0.0	3.23	4.3	11.8	14.1	2.4	8.5	0.0	1.7	42.86	3.53	3.92	0.85	8.31	0.00	0.43	0.02	0.45
8-Nov-23	2.3	0.3	0.0	0.3	0.0	1.1	0.0	4.02	2.3	8.2	12.7	1.4	5.3	0.0	2.1	32.09	4.82	7.10	1.45	13.37	0.00	0.03	0.00	0.03
9-Nov-23	1.1	0.6	0.0	0.0	0.0	0.3	0.0	1.99	3.1	13.4	14.6	2.3	8.1	0.1	2.2	43.76	4.65	6.47	1.22	12.35	0.00	0.19	0.02	0.21
10-Nov-23	1.8	0.2	0.0	0.0	0.0	0.7	0.0	2.80	2.1	11.8	14.0	1.8	8.3	0.1	1.8	39.92	5.51	8.83	1.60	15.94	0.00	0.18	0.02	0.20
11-Nov-23	7.4	1.9	0.0	1.0	0.0	1.6	0.0	11.95	0.4	9.6	11.7	0.7	7.4	0.1	1.0	30.91	6.32	10.49	1.79	18.60	0.00	0.06	0.01	0.07
12-Nov-23	9.6	1.7	0.0	1.9	0.0	2.2	0.0	15.37	0.2	4.4	10.3	0.3	4.1	0.0	0.7	19.91	5.92	9.39	1.59	16.89	0.00	0.00	0.00	0.00
13-Nov-23	6.8	0.8	0.0	1.7	0.0	2.5	0.0	11.82	0.8	7.1	11.2	0.6	4.0	0.0	0.7	24.44	5.48	8.27	1.39	15.14	0.00	0.09	0.01	0.10
14-Nov-23	0.9	0.4	0.0	0.3	0.0	2.0	0.0	3.59	3.6	13.0	13.7	2.0	5.4	0.0	0.9	38.58	3.28	5.32	0.89	9.49	0.94	3.56	0.74	5.24
15-Nov-23	0.3	1.1	0.0	0.0	0.0	1.7	0.0	3.14	11.3	13.3	15.4	3.7	5.0	0.0	1.1	49.89	0.01	0.00	0.00	0.01	2.24	8.20	1.75	12.19
16-Nov-23	0.0	0.3	0.0	0.0	0.1	1.6	0.0	1.94	16.0	16.0	15.5	4.6	5.6	0.0	1.4	59.16	0.00	0.00	0.00	0.00	2.73	9.40	2.02	14.15
17-Nov-23	0.0	0.3	0.0	0.0	0.0	2.1	0.0	2.35	18.8	16.9	14.8	5.6	6.3	0.0	0.7	63.16	0.00	0.00	0.00	0.00	4.03	13.22	2.84	20.08
18-Nov-23	0.0	3.8	0.0	0.0	0.0	1.0	0.0	4.80	16.9	15.7	15.5	6.5	9.8	0.1	0.8	65.27	0.00	0.00	0.00	0.00	4.86	15.48	3.20	23.54
19-Nov-23	0.0	3.2	0.0	0.0	0.0	1.1	0.0	4.28	14.7	13.1	15.6	6.1	11.0	0.1	0.7	61.40	0.00	0.00	0.00	0.00	3.54	11.80	2.52	17.86
20-Nov-23	0.7	0.2	0.0	0.1	0.0	1.2	0.0	2.25	14.3	18.0	15.3	5.0	8.6	0.1	1.2	62.40	0.01	0.00	0.00	0.01	3.01	10.34	2.20	15.55
21-Nov-23	0.0	1.8	0.0	0.0	0.0	1.8	0.0	3.61	21.7	16.2	14.1	6.1	7.1	0.0	1.1	66.13	0.00	0.00	0.00	0.00	3.15	10.90	2.38	16.43
22-Nov-23	0.0	7.2	0.0	0.0	0.0	1.7	0.0	8.95	25.5	8.8	17.8	7.5	6.3	0.0	1.0	66.96	0.02	0.01	0.00	0.03	2.97	10.34	2.21	15.51
23-Nov-23	0.0	6.9	0.0	0.0	0.0	1.0	0.0	7.95	23.6	7.4	22.4	7.1	7.2	0.1	1.9	69.67	0.00	0.00	0.00	0.00	2.73	9.72	2.62	15.07
24-Nov-23	0.0	3.1	0.0	0.0	0.0	0.9	0.0	4.06	19.2	10.0	23.4	5.8	7.9	0.2	1.7	68.18	0.00	0.00	0.00	0.00	2.72	9.93	2.31	14.96
25-Nov-23	0.0	1.7	0.0	0.0	0.0	1.3	0.0	2.95	11.1	8.6	21.9	3.2	5.9	0.1	1.3	52.08	0.00	0.00	0.00	0.00	3.10	10.93	2.51	16.53
26-Nov-23	0.6	5.1	0.0	0.2	0.0	0.9	0.0	6.84	6.9	9.9	19.2	2.2	5.8	0.3	1.6	45.90	0.00	0.00	0.00	0.00	2.84	9.98	2.42	15.24
27-Nov-23	0.5	2.8	0.0	0.1	0.2	0.8	0.0	4.37	9.2	12.3	17.8	2.0	4.6	0.1	1.4	47.40	0.00	0.00	0.00	0.00	2.82	9.87	2.32	15.01
28-Nov-23	0.8	0.7	0.0	0.6	0.3	1.3	0.0	3.68	5.5	5.7	16.8	1.1	3.2	0.0	1.0	33.29	0.01	0.00	0.00	0.01	1.60	6.90	1.79	10.30
29-Nov-23	2.2	1.7	0.0	0.6	0.1	1.0	0.0	5.65	4.0	5.1	15.5	0.8	3.3	0.1	0.6	29.49	0.00	0.00	0.00	0.00	2.70	9.92	2.40	15.03
30-Nov-23	3.4	1.7	0.0	1.1	0.1	0.7	0.0	6.90	2.6	3.0	12.0	0.9	6.5	0.4	0.4	25.92	0.00	0.00	0.00	0.00	2.09	8.54	2.20	12.83
Total	42.77	52.91	0.00	8.20	0.78	36.70	0.08	141.44	294.10	338.83	464.99	105.31	221.93	3.22	41.32	1469.71	60.68	85.48	15.50	161.66	48.07	172.08	38.72	258.87

Date	15.4 Import and Export of SR with ER & WR during November 2023																							
	Export of ER to SR (ER - SR)				Import of ER from SR (SR - ER)				Export of WR to SR (WR-SR)								Import of WR from SR (SR - WR)							
	ER - SR HVDC Gazuwaka	ER - SR HVDC Talchar -Kolar	ER-SR 765 kV Angul- Srikakulam D/C	Total ER - SR	SR - ER HVDC Gazuwaka	SR - ER HVDC Talchar -	SR-ER 765 kV Angul- Srikakulam D/C	Total SR - ER	WR - SR HVDC Bhadravati	WR - SR 765 kV Sholapur - Raichur 2x5/C	WR-SR 765 kV Wardha - Nizamabad D/C	WR-SR 400kV Kolhapur-Kudgi D/C	WR-SR HVDC Raigarh- Puglur	WR-SR 220kV Xeldem - Ambewadi S/C	WR-SR 220kV Ponda - Ambewadi S/C	Total WR - SR	SR - WR HVDC Bhadravati	SR - WR 765 kV Raichur - Sholapur 2x5/C	SR-WR 765 kV Wardha - Nizamabad D/C	SR-WR 400kV Kolhapur- Kudgi D/C	SR-WR HVDC Raigarh- Puglur	WR-SR 220kV Xeldem - Ambewadi S/C	WR-SR 220kV Ponda - Ambewadi S/C	Total SR - WR
1-Nov-23	12.23	24.21	46.84	83.28	0.00	0.00	0.00	0.00	20.78	0.00	28.69	0.00	84.76	2.08	0.00	136.31	0.00	21.26	0.00	29.68	0.00	0.00	0.00	50.95
2-Nov-23	11.93	24.22	45.61	81.76	0.00	0.00	0.00	0.00	24.07	0.00	26.42	0.00	81.03	2.17	0.00	133.70	0.00	17.43	0.00	30.19	0.00	0.00	0.00	47.62
3-Nov-23	12.46	24.24	50.21	86.92	0.00	0.00	0.00	0.00	24.01	0.10	33.23	0.00	68.44	2.28	0.00	128.07	0.00	12.85	0.00	27.57	0.00	0.00	0.00	40.43
4-Nov-23	12.22	30.26	47.86	90.35	0.00	0.00	0.00	0.00	22.58	0.99	30.76	0.00	92.92	2.39	0.00	149.64	0.00	12.92	0.00	25.59	0.00	0.00	0.00	38.52
5-Nov-23	11.85	41.46	48.64	101.95	0.00	0.00	0.00	0.00	12.75	2.81	36.34	0.00	41.06	2.22	0.00	95.19	0.00	8.30	0.00	21.19	0.00	0.00	0.00	29.50
6-Nov-23	11.86	44.07	47.50	103.43	0.00	0.00	0.00	0.00	20.36	1.37	33.40	0.00	38.35	2.24	0.00	95.72	0.00	11.85	0.00	19.30	0.00	0.00	0.00	31.16
7-Nov-23	11.86	42.32	40.95	95.13	0.00	0.00	0.00	0.00	24.10	0.05	25.41	0.00	55.65	2.31	0.00	107.51	0.00	19.91	0.00	23.15	0.00	0.00	0.00	43.06
8-Nov-23	11.87	47.89	39.88	99.64	0.00	0.00	0.00	0.00	24.11	0.21	23.36	0.00	58.19	2.16	0.00	108.04	0.00	18.50	0.00	24.57	0.00	0.00	0.00	43.06
9-Nov-23	11.87	44.62	47.10	103.59	0.00	0.00	0.00	0.00	15.27	1.25	27.54	0.00	43.80	2.22	0.00	90.07	0.00	17.82	0.00	27.57	0.00	0.00	0.00	45.39
10-Nov-23	12.22	39.76	46.34	98.32	0.00	0.00	0.00	0.00	14.48	1.77	26.02	0.00	44.75	2.29	0.00	89.30	0.00	18.87	0.00	28.32	0.00	0.00	0.00	47.19
11-Nov-23	12.56	39.77	47.39	99.72	0.00	0.00	0.00	0.00	10.16	4.66	25.98	0.00	32.24	2.29	0.00	75.33	0.00	18.21	0.00	25.43	0.00	0.00	0.00	43.64
12-Nov-23	12.57	39.76	44.54	96.86	0.00	0.00	0.00	0.00	10.16	4.48	27.69	0.00	17.17	2.05	0.00	61.55	0.00	13.25	0.87	20.91	0.00	0.00	0.00	35.03
13-Nov-23	12.57	39.76	46.02	98.35	0.00	0.00	0.00	0.00	19.32	4.19	28.34	0.00	26.83	2.10	0.00	80.77	0.00	9.69	0.00	20.17	0.00	0.00	0.00	29.87
14-Nov-23	12.57	37.67	49.01	99.24	0.00	0.00	0.00	0.00	19.31	6.18	32.66	0.00	51.04	2.22	0.00	111.42	0.00	3.42	0.00	17.92	0.00	0.00	0.00	21.34
15-Nov-23	5.78	40.63	56.05	102.46	0.00	0.00	0.00	0.00	7.84	11.73	37.91	0.00	59.88	2.16	0.00	119.53	0.00	4.80	0.00	16.14	0.00	0.00	0.00	20.95
16-Nov-23	5.71	38.19	54.55	98.46	0.00	0.00	0.00	0.00	10.45	8.58	37.09	0.00	75.96	2.16	0.00	134.24	0.00	6.10	0.00	20.15	0.00	0.00	0.00	26.25
17-Nov-23	12.61	45.77	49.30	107.69	0.00	0.00	0.00	0.00	19.95	4.89	32.43	0.00	94.25	2.17	0.00	153.70	0.00	9.94	0.01	25.06	0.00	0.00	0.00	35.01
18-Nov-23	12.61	48.12	54.77	115.50	0.00	0.00	0.00	0.00	13.73	6.60	38.26	0.00	58.40	2.26	0.00	119.25	0.00	6.09	0.00	22.79	0.00	0.00	0.00	28.89
19-Nov-23	12.61	43.40	55.10	111.12	0.00	0.00	0.00	0.00	7.28	4.97	32.99	0.00	49.16	2.21	0.00	96.61	0.00	7.36	0.00	24.00	0.00	0.00	0.00	31.36
20-Nov-23	12.61	42.86	49.38	104.86	0.00	0.00	0.00	0.00	11.40	1.96	26.66	0.00	91.47	2.22	0.00	133.71	0.00	13.12	0.00	27.20	0.00	0.00	0.00	40.32
21-Nov-23	12.62	48.13	48.24	108.99	0.00	0.00	0.00	0.00	11.56	1.70	27.92	0.00	107.23	2.22	0.00	150.63	0.00	11.76	0.01	27.04	0.00	0.00	0.00	38.81
22-Nov-23	12.33	43.98	51.87	108.18	0.00	0.00	0.00	0.00	11.55	3.54	32.06	0.00	86.59	2.29	0.00	136.04	0.00	7.21	0.00	26.00	0.00	0.00	0.00	33.21
23-Nov-23	12.47	39.09	51.19	102.75	0.00	0.00	0.00	0.00	11.55	2.90	32.76	0.00	81.46	2.34	0.93	131.95	0.00	6.36	0.00	24.85	0.00	0.00	0.00	31.21
24-Nov-23	12.49	37.88	50.43	100.80	0.00	0.00	0.00	0.00	11.36	1.48	30.76	0.00	76.42	2.37	0.00	122.38	0.00	9.17	0.00	24.47	0.00	0.00	0.00	33.65
25-Nov-23	12.49	44.92	46.68	104.10	0.00	0.00	0.00	0.00	11.49	0.98	26.22	0.00	75.91	2.58	0.00	117.18	0.00	15.50	0.00	25.78	0.00	0.00	0.00	41.27
26-Nov-23	5.65	35.25	50.59	91.49	3.85	0.00	0.00	3.85	9.11	6.68	31.15	0.00	37.30	2.21	0.00	86.46	0.00	15.61	0.00	23.38	0.00	0.00	0.00	38.99
27-Nov-23	8.47	41.49	51.21	101.17	2.19	0.00	0.00	2.19	7.12	6.41	35.92	0.00	74.31	2.25	0.00	126.01	0.00	5.30	0.00	19.50	0.00	0.00	0.00	24.79
28-Nov-23	12.44	48.08	51.77	112.29	0.00	0.00	0.00	0.00	7.12	8.85	37.75	0.00	60.24	2.34	0.00	116.30	0.00	2.77	0.00	17.74	0.00	0.00	0.00	20.51
29-Nov-23	14.48	48.11	49.82	112.41	0.00	0.00	0.00	0.00	6.98	9.10	35.26	0.00	51.79	2.29	0.00	105.42	0.00	4.55	0.00	18.16	0.00	0.00	0.00	22.71
30-Nov-23	16.20	48.10	48.12	112.41	0.00	0.00	0.00	0.00	15.55	8.23	29.68	0.00	41.75	2.22	0.00	97.43	0.00	4.20	0.00	17.52	0.00	0.00	0.00	21.73
TOTAL	352.24	1214.02	1466.97	3033.23	6.03	0.00	0.00	6.03	435.49	116.70	930.66	0.00	1858.37	67.29	0.97	3409.48	0.00	334.15	0.89	701.36	0.00	0.00	0.01	1036.41

16. भूटान , नेपाल, बांग्लादेश एवं म्यान्मार के साथ अंतरराष्ट्रीय विद्युत विनिमय

अप्रैल 2023 से मार्च 2024 April 2023 to March 2024

अंतरराष्ट्रीय विद्युत विनिमय [भारत 'से निर्यात'/'को आयात']

Transnational Exchange ('Export from'/'Import to' India)

माह MONTH	भूटान BHUTAN		नेपाल NEPAL		बांग्लादेश BANGLADESH		म्यान्मार MYANMAR	
	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)	Energy Exported (In MU)	Energy Imported (In MU)
अप्रैल 2023 APR'23	19.40	41.40	357.99	0.00	687.30	0.00	0.73	0.00
मई 2023 MAY'23	2.27	117.99	271.80	0.34	775.10	0.00	0.61	0.00
जून 2023 JUN'23	0.00	626.15	80.00	88.86	743.92	0.00	0.77	0.00
जुलाई 2023 JUL'23	0.00	1291.66	0.00	223.72	780.77	0.00	1.02	0.00
अगस्त 2023 AUG'23	0.00	1244.52	0.00	281.61	767.79	0.00	0.94	0.00
सितंबर 2023 SEP'23	0.00	924.25	0.00	323.59	754.88	0.00	0.77	0.00
अक्टूबर 2023 OCT'23	0.00	588.64	0.00	460.97	755.92	0.00	0.84	0.00
नवंबर 2023 NOV'23	51.41	36.77	0.00	271.72	628.24	0.00	0.72	0.00
कुल Total	73.08	4871.38	709.78	1650.81	5893.92	0.00	6.41	0.00

* Based on daily operational data

Date	16.1 Import from neighbouring countries during November 2023													
	(All figures in MU)													
	Import from Bhutan								Import from Bangladesh		Import from Nepal			Import from Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurwar D/C*	400 kV Jigmeling-Alipurwar D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-Nov-23	5.96	0.00	0.00	0.00	0.14	0.35	0.74	0.01	0.00	0.00	1.66	8.26	0.00	0.00
2-Nov-23	5.59	0.00	0.00	0.00	0.15	0.34	0.67	0.00	0.00	0.00	1.62	8.49	0.00	0.00
3-Nov-23	5.33	0.08	0.00	0.00	0.11	0.29	0.64	0.00	0.00	0.00	1.63	8.04	0.00	0.00
4-Nov-23	4.61	0.77	0.00	0.00	0.06	0.23	0.56	0.00	0.00	0.00	1.68	8.88	0.00	0.00
5-Nov-23	4.12	0.61	0.00	0.00	0.05	0.19	0.49	0.00	0.00	0.00	1.69	8.17	0.00	0.00
6-Nov-23	7.36	0.00	0.00	0.00	0.16	0.39	0.26	0.00	0.00	0.00	1.64	8.16	0.00	0.00
7-Nov-23	8.54	0.00	0.00	0.00	0.15	0.33	0.24	0.00	0.00	0.00	1.57	7.81	0.00	0.00
8-Nov-23	8.92	0.00	0.00	0.00	0.17	0.38	0.52	0.12	0.00	0.00	1.60	6.93	0.00	0.00
9-Nov-23	8.14	0.00	0.00	0.00	0.01	0.04	0.58	0.03	0.00	0.00	0.98	7.31	0.00	0.00
10-Nov-23	8.07	0.00	0.00	0.00	0.08	0.11	0.45	0.01	0.00	0.00	1.56	7.52	0.00	0.00
11-Nov-23	8.41	0.00	0.00	0.00	0.00	0.00	0.45	0.02	0.00	0.00	1.63	8.41	0.00	0.00
12-Nov-23	8.50	0.00	0.00	0.00	0.00	0.00	0.43	0.00	0.00	0.00	1.66	10.34	0.00	0.00
13-Nov-23	7.49	0.00	0.00	0.00	0.00	0.00	0.37	0.00	0.00	0.00	1.68	10.86	0.00	0.00
14-Nov-23	7.50	0.00	0.00	0.00	0.00	0.00	0.38	0.00	0.00	0.00	1.70	10.78	0.00	0.00
15-Nov-23	8.11	0.00	0.00	0.00	0.00	0.00	0.23	0.00	0.00	0.00	1.73	11.16	0.00	0.00
16-Nov-23	7.63	0.00	0.00	0.00	0.00	0.00	0.27	0.00	0.00	0.00	1.69	10.42	0.00	0.00
17-Nov-23	7.18	0.00	0.00	0.00	0.00	0.00	0.18	0.00	0.00	0.00	1.66	9.51	0.00	0.00
18-Nov-23	7.13	0.00	0.00	0.00	0.00	0.00	0.08	0.00	0.00	0.00	1.67	9.65	0.00	0.00
19-Nov-23	7.13	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00	1.66	10.64	0.00	0.00
20-Nov-23	7.14	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.00	0.00	1.69	8.72	0.00	0.00
21-Nov-23	7.12	0.00	0.00	0.00	0.00	0.00	0.24	0.00	0.00	0.00	1.73	7.91	0.00	0.00
22-Nov-23	6.80	0.00	0.00	0.00	0.00	0.00	0.23	0.00	0.00	0.00	1.71	6.63	0.00	0.00
23-Nov-23	6.70	0.00	0.00	0.00	0.00	0.00	0.17	0.00	0.00	0.00	1.71	5.81	0.00	0.00
24-Nov-23	6.87	0.00	0.00	0.00	0.00	0.00	0.08	0.00	0.00	0.00	1.68	5.73	0.00	0.00
25-Nov-23	6.64	0.00	0.00	0.00	0.00	0.00	0.16	0.00	0.00	0.00	1.62	5.34	0.00	0.00
26-Nov-23	6.69	0.00	0.00	0.00	0.00	0.00	0.18	0.00	0.00	0.00	1.27	5.05	0.00	0.00
27-Nov-23	6.69	0.00	0.00	0.00	0.00	0.00	0.18	0.00	0.00	0.00	1.06	3.48	0.00	0.00
28-Nov-23	6.45	0.00	0.00	0.00	0.00	0.00	0.26	0.00	0.00	0.00	0.99	3.10	0.00	0.00
29-Nov-23	6.91	0.00	0.00	0.00	0.00	0.00	0.06	0.00	0.00	0.00	0.95	2.44	0.00	0.00
30-Nov-23	5.89	0.00	0.00	0.00	0.00	0.00	0.06	0.00	0.00	0.00	0.55	1.92	0.00	0.00
Total	209.61	1.47	0.00	0.00	1.08	2.65	9.24	0.19	0.00	0.00	45.66	227.48	0.00	0.00

Based on SEM/Energy meter data for links where available (*Mangdechu generation receipt at APD through a bypassed arrangement at 400kV Punatsanchu station)

Date	16.2 Export to neighbouring countries during November 2023													
	(All figures in MU)													
	Export to Bhutan								Export to Bangladesh		Export to Nepal			Export to Myanmar
	400 kV Tala-Binaguri I,II & IV	400 kV Binaguri-Malbase	220 kV Birpara-Chuka D/C	220 kV Birpara-Malbase	400 kV Punatsanchu-Alipurwar D/C*	400 kV Jigmeling-Alipurwar D/C	132 kV Rangia-Motanga	132 kV Salakati-Gelephu	400 kV Behrampur-Bheramara 1,2,3&4	132 kV Surjyamaninagar-Comilla D/C	132 kV Tanakpur-Mahendranagar	400 kV Muzaffarpur-Dhalkebar	From BIHAR Source	11 kV Moreh-Tamu
1-Nov-23	0.00	0.00	1.56	1.61	0.00	0.00	0.00	0.00	21.68	3.12	0.00	0.00	0.00	0.03
2-Nov-23	0.00	0.00	1.59	1.62	0.00	0.00	0.00	0.03	21.72	2.90	0.00	0.00	0.00	0.03
3-Nov-23	0.00	0.00	1.71	1.64	0.00	0.00	0.00	0.04	21.71	2.79	0.00	0.00	0.00	0.03
4-Nov-23	0.00	0.00	1.76	1.63	0.00	0.00	0.00	0.05	19.09	3.12	0.00	0.00	0.00	0.03
5-Nov-23	0.00	0.00	1.86	1.72	0.00	0.00	0.00	0.06	18.57	3.34	0.00	0.00	0.00	0.03
6-Nov-23	0.00	1.24	1.81	1.63	0.00	0.00	0.00	0.04	18.63	3.15	0.00	0.00	0.00	0.03
7-Nov-23	0.00	2.79	1.59	1.09	0.00	0.00	0.00	0.05	18.49	2.91	0.00	0.00	0.00	0.03
8-Nov-23	0.00	2.75	1.32	1.51	0.00	0.00	0.00	0.00	18.79	2.77	0.00	0.00	0.00	0.03
9-Nov-23	0.00	2.95	1.89	1.66	0.00	0.00	0.00	0.00	18.72	2.87	0.00	0.00	0.00	0.03
10-Nov-23	0.00	3.09	1.96	1.62	0.00	0.00	0.00	0.00	18.84	2.37	0.00	0.00	0.00	0.02
11-Nov-23	0.00	3.46	2.25	2.07	0.24	0.37	0.00	0.00	20.23	2.78	0.00	0.00	0.00	0.02
12-Nov-23	0.00	3.40	2.34	2.24	0.27	0.40	0.00	0.00	22.41	2.87	0.00	0.00	0.00	0.02
13-Nov-23	0.00	3.41	2.57	2.35	0.30	0.43	0.00	0.06	21.04	2.86	0.00	0.00	0.00	0.02
14-Nov-23	0.00	3.13	2.63	2.55	0.24	0.35	0.00	0.20	21.03	2.94	0.00	0.00	0.00	0.02
15-Nov-23	0.00	2.75	2.81	2.85	0.20	0.31	0.00	0.43	20.95	2.99	0.00	0.00	0.00	0.03
16-Nov-23	0.00	2.74	2.28	3.04	0.08	0.12	0.00	0.43	21.06	2.87	0.00	0.00	0.00	0.02
17-Nov-23	0.00	2.65	2.94	2.82	0.27	0.39	0.00	0.60	11.21	1.50	0.00	0.00	0.00	0.02
18-Nov-23	0.00	2.60	2.94	2.85	0.23	0.35	0.00	0.65	15.10	1.87	0.00	0.00	0.00	0.03
19-Nov-23	0.00	2.64	2.95	2.73	0.18	0.31	0.00	0.63	19.48	2.66	0.00	0.00	0.00	0.02
20-Nov-23	0.00	2.65	3.29	2.55	0.42	0.09	0.00	0.68	20.10	2.78	0.00	0.00	0.00	0.03
21-Nov-23	0.00	2.90	2.79	3.05	0.51	0.13	0.00	0.60	20.22	2.61	0.00	0.00	0.00	0.02
22-Nov-23	0.00	3.23	2.86	2.89	0.24	0.82	0.00	0.56	20.12	2.71	0.00	0.00	0.00	0.02
23-Nov-23	0.00	3.18	2.90	3.09	0.33	0.48	0.00	0.48	20.44	2.58	0.00	0.00	0.00	0.02
24-Nov-23	0.00	3.05	2.98	3.14	0.46	0.64	0.00	0.56	11.77	2.24	0.00	0.00	0.00	0.02
25-Nov-23	0.00	3.01	3.27	3.03	0.36	0.48	0.00	0.53	10.18	2.38	0.00	0.00	0.00	0.02
26-Nov-23	0.00	2.99	3.34	3.05	0.22	0.29	0.00	0.44	10.77	2.51	0.00	0.00	0.00	0.02
27-Nov-23	0.00	2.91	3.22	2.94	0.20	0.31	0.00	0.43	10.25	2.64	0.00	0.00	0.00	0.02
28-Nov-23	0.00	3.11	3.29	2.94	0.14	1.09	0.00	0.44	13.73	2.54	0.00	0.00	0.00	0.02
29-Nov-23	0.00	3.01	3.25	2.96	0.00	1.18	0.00	0.52	20.49	2.48	0.00	0.00	0.00	0.02
30-Nov-23	0.00	1.85	3.48	3.34	0.18	1.46	0.00	0.51	18.41	2.55	0.00	0.00	0.00	0.03
Total	0.00	71.49	75.42	72.18	5.08	9.98	0.00	9.02	545.21	80.70	0.00	0.00	0.00	0.72

Based on SEM/Energy meter data for links where available (*Mangdechu generation receipt at APD through a bypassed arrangement at 400kV Punatsanchu station)

17. एसटीओए (द्विपक्षीय एवं सामूहिक) एवं डी एस एम बिलिंग का ब्योरा – नवंबर 2023

STOA (BILATERAL AND COLLECTIVE) & DSM BILLING DETAILS - NOVEMBER 2023

द्विपक्षीय एसटीओए BILATERAL SHORT TERM OPEN ACCESS

नवंबर 2023	NOVEMBER 2023		Apr'23- Nov'23	
नोडल क्षे.भा.प्रे.के. का नाम Name of Nodal RLDC	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Energy Approved(MU)	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Energy Approved (MU)
ऊ. क्षे. NR	166	1012	7971	31570
प.क्षे. WR	313	1038	7994	16050
द.क्षे. SR	140	649	7236	18740
पू.क्षे. ER	99	188	5123	6165
पूर्वोत्तर क्षे. NER	55	4	1017	468
कुल TOTAL	773	2892	29341	72992

एसटीओए SHORT TERM OPEN ACCESS

	सामूहिक एसटीओए Collective STOA		द्विपक्षीय एसटीओए Bilateral STOA	
माह MONTH	क्रेताओं / विक्रेताओं की सं. NO. of Buyers/Sellers	अनुमोदित ऊर्जा (मि.यु.) Approved Energy (MU)	अनुमोदित लेन – देनों की संख्या No. of Approved Transactions	अनुमोदित ऊर्जा (मि.यु.) Approved Energy (MU)
अप्रैल 2023 Apr'23	5580	6941	3322	9714
मई 2023 May'23	5798	6626	3469	8418
जून 2023 Jun'23	5783	6966	3925	10583
जुलाई 2023 Jul'23	6428	6719	3872	10780
अगस्त 2023 Aug'23	5954	6805	4841	12338
सितंबर 2023 Sep'23	6440	6506	4481	12402
अक्टूबर 2023 Oct'23	6571	7388	4658	5866
नवंबर 2023 Nov'23	7848	7741	773	2892
कुल TOTAL	50402	55692	29341	72992

मासिक डी एस एम बिलिंग का ब्योरा* 2023-24
MONTHLY DSM BILLING DETAILS* 2023-24

अनंतिम आँकड़े
Provisional data
subject to change
करोड़ रु. में (RS. IN CRORES)

क्षेत्र REGION →	उत्तरी क्षेत्र NORTH	पश्चिमी क्षेत्र WEST	दक्षिणी क्षेत्र SOUTH	पूर्वी क्षेत्र EAST	पूर्वोत्तर क्षेत्र NORTH EAST
सप्ताह WEEK ↓					
30.10.23 to 05.11.23	148.57	179.71	187.89	242.05	53.12
06.11.23 to 12.11.23	150.36	248.65	286.33	276.65	48.19
13.11.23 to 19.11.23	166.67	248.99	257.82	293.14	34.10
20.11.23 to 26.11.23	195.22	260.05	224.68	259.44	53.43

* Amount shown is Payable to DSM pool ^Provisional Data

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एस. एच.पी.एस.आई.एच.) POWER MARKET INFORMATION (Source IEX & PXL) पावर एक्स्चेंज के माध्यम से विनिमय - माह:- नवंबर 2023 EXCHANGES THROUGH POWER EXCHANGES - NOVEMBER 2023				
क्र. सं. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
1	Chandigarh (UT)	उत्तरी क्षेत्र NR	89.68	0.88
2	Delhi		312.05	412.88
3	Haryana		355.36	173.98
4	Himachal Pradesh		471.02	10.99
5	Jammu Kashmir		63.40	2183.49
6	North Central Railway Prayagraj		0.00	137.50
7	Punjab		184.41	651.77
8	Rajasthan		1592.57	348.34
9	SHREE CEMENT LIMITED TPS		84.80	0.00
10	Uttar Pradesh		2721.92	1851.46
11	Uttarakhand		60.05	1047.97
12	NEPAL ELECTRICITY AUTHORITY TANAKPUR		0.00	0.00
13	AD HYDRO POWER LIMITED		69.36	0.00
14	ADANI SOLAR ENERGY JAISALMER TWO PRIVATE LIMITED(Project-2)		56.35	0.00
15	ADANI SOLAR ENERGY JODHPUR TWO LIMITED		33.93	0.00
16	Budhil HEP (Greenko Budhil Hydro Power Pvt. Ltd.)		3.42	0.00
17	IGSTPS JHABAR		27.11	0.00
18	KARCHAM WANGTOO HYDRO ELECTRIC PLANT.		103.86	0.00
19	NEA Nepal Upper Chameliya Hydropower Project		85.07	0.00
20	NEA Nepal Upper Kalanggad Hydropower Project		92.17	0.00
21	NTPC Dadri Stage I		14.61	0.00
22	NTPC Dadri Stage II		6.71	0.00
23	NTPC Rihand stage I		65.27	0.00
24	NTPC Rihand stage II		61.39	0.00
25	NTPC Rihand stage III		58.98	0.00
26	NTPC Singrauli		121.24	0.00
27	NTPC Tanda Stage II		55.45	0.00
28	NTPC Unchahar Stage I		4.06	0.00
29	NTPC Unchahar Stage II		8.92	0.00
30	NTPC Unchahar Stage III		4.00	0.00
31	NTPC Unchahar Stage IV		22.23	0.00
32	PARBATI II HE PROJECT		14.00	0.00
33	ReNew Surya Ravi Private Limited		146.10	0.00
34	Sainj HEP		34.60	0.00
35	Singoli Bhatwari HEP		61.64	0.00
36	Sorang HEP (Himachal Sorang Power Pvt. Ltd.)		44.10	0.00

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एस., एच.पी.एस.,आई.एच.) POWER MARKET INFORMATION (Source IEX & PXL) पावर एक्स्चेंज के माध्यम से विनिमय - माह:- नवंबर 2023 EXCHANGES THROUGH POWER EXCHANGES - NOVEMBER 2023			
क्र. सं. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU
			(DAM+HP DAM+RTM)
37	ACB (INDIA) LIMITED	पश्चिमी क्षेत्र WR	276.44
38	Adani Power Limited-Raipur TPP		1243.42
39	ArcelorMittal Nippon Steel India Limited		0.00
40	BHARAT ALUMINIUM COMPANY LTD (Bulk Consumer)		0.00
41	Chhattisgarh		2578.80
42	Daman and Diu - Dadra and Nagar Haveli		2.44
43	Goa WR		11.55
44	Gujarat		392.40
45	Jhabsua Power Limited		58.04
46	Jindal Power Limited, Stage-1		914.26
47	Jindal Power Limited, Stage-2		1016.68
48	Jindal Steel & Power Ltd , DCPP		304.21
49	Madhya Pradesh		1038.13
50	MAHAN ENERGEN LIMITED.		765.05
51	Maharashtra		383.54
52	Maruti Clean Coal and Power Limited		59.45
53	MB POWER (MADHYA PRADESH) LIMITED		115.03
54	NTPC Jhanor Gandhar GPS		0.00
55	NTPC Kawas GPS		0.00
56	Ratnagiri Gas & Power Private Limited		3.34
57	SJS Power Generation Chhattisgarh Limited		506.10
58	Adani Power Limited - Raigarh TPP		573.28
59	ADANI WIND ENERGY KUTCHH FOUR Ltd. Naahatrana		143.83
60	BHARAT ALUMINIUM COMPANY LTD		8.32
61	CONTINUUM POWER TRADING (TN) PRIVATE LIMITED		36.58
62	D B Power Limited		1170.42
63	Dharawal ISTS		5.96
64	DHARIWAL STU OTH		6.94
65	GMR Warora Energy Limited		16.60
66	INOX GREEN ENERGY SERVICES LIMITED		3.68
67	Jaypee Nigrie Super Thermal Power Plant		435.13
68	NREL-Dayapar Wind		18.77
69	NTPC Gadawara		11.30
70	NTPC khargone		16.53
71	NTPC Korba Stage I & II		4.51
72	NTPC Korba Stage III		1.73
73	NTPC Lara Stage I		5.45
74	NTPC Mouda Stage I		15.69
75	NTPC Mouda Stage II		9.52
76	NTPC SAIL POWER COMPANY LIMITED		0.57
77	NTPC Sipat Stage I		24.21
78	NTPC Sipat Stage II		15.78
79	NTPC Solapur		12.69
80	NTPC Vindhyachal Stage I		24.35
81	NTPC Vindhyachal Stage II		19.87
82	NTPC Vindhyachal Stage III		21.24
83	NTPC Vindhyachal Stage IV		27.73
84	NTPC Vindhyachal Stage V		17.52
85	R.K.M POWERGEN PRIVATE LIMITED		38.77
86	Sasan Power Limited		45.47
87	The Tata Power Co Ltd (MTPS)		6.65
88	TRN ENERGY PRIVATE LIMITED		38.98
89	WIND FIVE RENERGY LIMITED		20.65

18. पावर मार्केट की सूचना (स्रोत : आई ई एक्स. एन पी एक्स.आई एन.) POWER MARKET INFORMATION (Source IEX & PEX) पावर एक्स्चेंज के माध्यम से विनिमय - माह:- नवंबर 2023 EXCHANGES THROUGH POWER EXCHANGES - NOVEMBER 2023			
क्र. सं. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU
			(DAM+HP DAM+RTM)
90	Andhra Pradesh	दक्षिणी क्षेत्र SR	106.75
91	Goa SR		0.43
92	Karnataka		87.39
93	Kerala		29.90
94	NLC INDIA LIMITED NEYVELI NEW THERMAL POWER STATION		24.12
95	SIMHAPURI ENERGY LIMITED		19.34
96	Tamil Nadu		283.38
97	Telangana		106.82
98	COASTAL ENERGEN PRIVATE LIMITED		114.80
99	IL&FS TAMIL NADU POWER COMPANY LIMITED		21.19
100	NLC INDIA LIMITED THERMAL POWER STATION I EXPANSION		18.46
101	NLC INDIA LIMITED THERMAL POWER STATION II EXPANSION		17.83
102	NLC INDIA LIMITED THERMAL POWER STATION II STAGE I		22.24
103	NLC INDIA LIMITED THERMAL POWER STATION II STAGE II		17.19
104	NLC Tamilnadu Power Limited		88.53
105	NTECL VALLUR		30.08
106	NTPC KUDGI		6.47
107	NTPC Ramagundam Stage I & II		34.75
108	NTPC Ramagundam Stage III		6.27
109	NTPC Simhadri Stage I		0.21
110	NTPC Simhadri Stage II		11.49
111	NTPC Talcher Super Thermal Power Station Stage II		14.69
112	OSTRO KANNADA POWER PRIVATE LIMITED		69.32
113	Pondicherry UT		214.02
114	Ramagundam Floating solar		28.14
115	RENEW SURYA ROSHNI PRIVATE LIMITED Koppal PS		29.51
116	SEIL ENERGY INDIA LIMITED		124.95
117	SEIL Energy India Limited Project II		335.31
118	Simhadri Floating solar (10 MW)		5.80
119	Simhadri FSP 15 MW		8.60
120	Bihar	पूर्वी क्षेत्र ER	2549.53
121	Damodar Valley Corporation		89.34
122	Jharkhand		323.68
123	Odisha		1408.44
124	Sikkim		118.65
125	West Bengal		1397.17
126	ADHUNIK POWER & NATURAL RESOURCES LIMITED		170.66
127	Basochhu Hydropower Plant Bhutan		57.69
128	Chilime Hydro Power Project NEPAL ELECTRICITY AUTHORITY		57.02
129	Chusachen HEP		0.41
130	DEVIGHAT NEPAL ELECTRICITY AUTHORITY		35.08
131	GMR KAMALANGA ENERGY LTD-CTU		1.77
132	Jindal India Thermal Power Limited		660.16
133	JORETHANG LOOP HEP, DANS ENERGY PRIVATE LIMITED		4.10
134	Druk Green Power Corporation Limited (DGPC-L Bhutan)		0.00
135	Kabeli B-1 Hydro Power Project NEPAL ELECTRICITY AUTHORITY		17.43
136	KALI GANDAKI NEPAL ELECTRICITY AUTHORITY		223.86
137	KANTI BULEE UTPADAN NIGAM LIMITED		12.25
138	LIKHU-IV NEA		121.83
139	Lower Modi Hydro Power Project NEPAL ELECTRICITY AUTHORITY		17.08
140	MARSYANGDI NEPAL ELECTRICITY AUTHORITY		161.49
141	NABINAGAR POWER GENERATING COMPANY LIMITED		98.52
142	NEA Nepal Upper Dardl A Hydropower Project		8.08
143	NEPAL ELECTRICITY AUTHORITY-MIDDLE MARSYANGDI		164.62
144	NEPAL ELECTRICITY AUTHORITY MUZAFFARPUR		0.00
145	NTPC BARH Stage I		46.39
146	NTPC BARH Stage II		119.69
147	NTPC Dardipati		22.51
148	NTPC Farakka stage I		180.48
149	NTPC Farakka stage III		31.45
150	NTPC Kahalgaoon stage I		443.22
151	NTPC Kahalgaoon stage II		80.70
152	NTPC North Karanpura STPS		2.60
153	NTPC Talcher Stage I		9.93
154	Solu Hydropower Project NEPAL ELECTRICITY AUTHORITY		53.96
155	Tashiding HEP, Shiga Energy Private Limited		2.50
156	TRISHULI NEPAL ELECTRICITY AUTHORITY		55.12

18. पावर मार्केट की सूचना (स्रोत : आई.ई.एस., एच.पी.एस.,आई.एल.) POWER MARKET INFORMATION (Source IEX & PXL) पावर एक्स्चेंज के माध्यम से विनिमय – माह:- नवंबर 2023 EXCHANGES THROUGH POWER EXCHANGES - NOVEMBER 2023				
क्र. सं. S. No.	क्षेत्रीय इकाई Regional Entity	क्षेत्र Region	पावर एक्स्चेंज के माध्यम से (मि.यु. मे) Through Power Exchange in MU	
			(DAM+HP DAM+RTM)	
157	AGARTALA GAS BASED POWER STATION	पूर्वोत्तर क्षेत्र NER	0.18	0.00
158	Arunachal Pradesh		9.41	1.71
159	Assam		322.91	31.30
160	KAMENG HYDRO POWER STATION		53.56	0.00
161	Manipur		12.88	10.83
162	Meghalaya		33.50	111.39
163	Tripura		126.44	2.37
164	ASSAM GAS BASED POWER STATION		1.39	0.00
165	Bongaigaon Thermal Power Station NTPC		29.79	0.00
166	Doyang HPS, NEEPCO		0.89	0.00
167	Kopili 1 IN		13.66	0.00
168	Kopili Power Station NEEPCO		2.76	0.00
169	Mizoram		18.50	0.00
170	Nagaland		26.85	0.00
171	Palatana Plant		86.95	0.00
172	Pare Hydro Power Station	0.34	0.00	
173	Ranganadi Hydro Power Station	0.06	0.00	
			30963	30963

19. INFORMATION ABOUT RENEWABLE ENERGY
CERTIFICATE MECHANISM
MONTH : NOVEMBER 2023

RE Source & Unit wise break up (01.11.2023-30.11.2023)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	18	7	11	5
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	150	1	2	1
5	Small Hydro	59	1	0	0
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	0	0	0	0
9	Bio-fuel cogeneration	0	0	0	0
	Total	227	9	13	6

RECs Issued (01.11.2023-30.11.2023)

Sr.No.	Non Solar	Solar	Total
1	3065585	73001	3138586

Redemption of REC (01.11.2023-30.11.2023)

Sr.No.	Non Solar	Solar	Total
1	1331192	33173	1364365

19. INFORMATION ABOUT RENEWABLE ENERGY
CERTIFICATE MECHANISM
MONTH : NOVEMBER 2023

RE Source & Unit wise break up (Apr'23-Nov'23)

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	366	21	449	19
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	316	51	68	5
5	Small Hydro	101	8	42	7
6	Others	0	0	0	0
7	Geothermal	0	0	0	0
8	Biomass	5	1	5	1
9	Bio-fuel cogeneration	8	2	23	5
	Total	796	83	587	37

RECs Issued (Apr'23-Nov'23)

Sr.No.	Non Solar	Solar	Total
1	5710845	785902	6496747

Redemption of REC (Apr'23-Nov'23)

Sr.No.	Non Solar	Solar	Total
1	4540979	615372	5156351

19. INFORMATION ABOUT RENEWABLE ENERGY
CERTIFICATE MECHANISM

MONTH : NOVEMBER 2023

RE Source & Unit wise break up Since Inception to Nov'23

Sr.No	Source Wise	Accreditation		Registration	
		Capacity (MW)	Unit	Capacity (MW)	Unit
1	Wind	3079	526	2903	521
2	Urban or Municipal Waste	0	0	0	0
3	Solar Thermal	0	0	0	0
4	Solar PV	1585	543	1253	470
5	Small Hydro	745	41	698	42
6	Others	4	2	3	1
7	Geothermal	0	0	0	0
8	Biomass	402	37	378	35
9	Bio-fuel cogeneration	817	90	383	55
	Total	6632	1239	5618	1124

RECs Issued since Inception to Nov'23

Sr.No.	Non Solar	Solar	Total
1	93590985	13196640	106787625

Redemption of REC since Inception to Nov'23

Sr.No.	Non Solar	Solar	Total
1	72069164	12243760	84312924

REC Closing balance as on 30.11.2023

Sr.No.	Non Solar	Solar	Total
1	21521821	952880	22474701

20.Details of Grid Events during the Month of November 2023 in Northern Region

Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI for GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GI-2	Rajasthan	03-Nov-2023 09:16	03-Nov-2023 09:59	00:43	0	335	0.000	0.623	46716	53744	i) During antecedent condition, 220kV Hindaun220-Sikrai(Dausa)[Raj] ckt was not in service and MVA loading of 400/220 KV 315 MVA ICT-1 & 2 at Hindaun(Raj) was 269 and 259 MVA respectively. ii) As reported, at 09:16 hrs, 400/220KV 315MVA ICT-1 & 2 both tripped due to overloading. iii) As per DR, current in three phases in LV side of 400/220 KV 315 MVA ICT-1 at Hindaun(Raj) are I _R =936A, I _Y =966A and I _B =938A. iv) As per PMU at Bassi(PG), no fault is observed in the system. v) As per SCADA, load loss of approx. 355MW is observed in Rajasthan control area.	1) 400/220 KV 315 MVA ICT-1 at Hindaun(Raj) 2) 400/220 KV 315 MVA ICT-2 at Hindaun(Raj)
2	GI-2	Uttar Pradesh	04-Nov-2023 04:03	04-Nov-2023 05:48	01:45	455	0	1.492	0.000	30487	39273	i) During antecedent condition, interconnectors were in opened condition. 490MW Unit-5 was not running and 490MW Unit-6 was generating approx. 455MW. HVDC Rihand-Dadri Bipole was carrying total ~600MW. ii) As reported, at 04:03:05:240 hrs, B-N phase to earth fault occurred on 400KV Dadri-Mandola ckt-1. Fault distance was approx. 100meter from Dadri TPS end. This fault was sensed by both the ends in 2-1. After ~160msec (08 cycles) of fault, B-ph pole of CB at both then ends opened and A/R started. Further after ~1sec (dead time), line successfully autoreclosed due to transient nature of fault. Delayed tripping initiation in 2-1 was due to 2-1 time delay setting which was kept as 100msec instead of instantaneous. As informed by NTPC Dadri, 2-1 time delay has been set as 0 sec (instantaneous). iii) As per PMU & DR of 400KV Dadri-Mandola ckt-1, B-N phase to earth fault with successful A/R operation is observed. Steady state fault current was approx. 35kA, during transient fault current magnitude was ~52kA. iv) On this fault, commutation failure at HVDC Rihand-Dadri occurred and power order dropped to zero [0]. v) Distance protection relay at Harshwar end of 400KV Dadri-Harshwar ckt-2 sensed the fault on 400KV Dadri-Mandola ckt in 2-1 and successful autoreclosed from Harshwar end. Dadri end relay sensed fault in 2-4 as fault was in reverse direction however as informed by Dadri, instant three phase tripping occurred on DT received from Harshwar end. Reason of DT received at Dadri end is yet to be identified. vi) During fault time, over voltage of the magnitude of approx. 723kV in 400KV Dadri-Mandola ckt-2 and Dadri end and approx. 560kV in 400KV Bus-2 at Dadri TPS is observed (as per PMU at Dadri TPS). Over voltage sustained for approx. 100msec. On this over voltage, 400KV Dadri-Mandola ckt-2 tripped on over voltage stage-2 protection operation at Dadri end. vii) At the same time, all three filter banks connected at Dadri HVDC tripped on over voltage protection operation. As reported by POWERGRID, over voltage protection of filter banks is 480-80kV with 20msec pickup time delay. Due to tripping of filter banks, HVDC Rihand-Dadri Bipole got blocked. viii) From DR & PMU voltage plots, over voltage didn't occur in other 400KV elements at Dadri TPS. On overvoltage in Mandola ckt-2, Dadri TPS informed that neutral of CVT at Main 1 relay found opened at Dadri end which led to rise in voltage at secondary side. Reason of over voltage in 400KV Bus and Dadri HVDC bus is yet to be identified.	1) 400 KV Dadri(NT)-Mandola(PG) (PG) Ckt-1 2) 400 KV Dadri(NT)-Mandola(PG) (PG) Ckt-2 3) 400 KV Dadri(NT)-Loni Harsh Vihar(DV) (NT) Ckt-2 4) 490 MW Dadri-II TPS- UNIT 2 5) 500 KV HVDC Rihand-Dadri (PG) Ckt-1 6) 500 KV HVDC Rihand-Dadri (PG) Ckt-2
3	GI-1	Himachal Pradesh	10-Nov-2023 13:54	10-Nov-2023 14:30	00:36	0	90	0.000	0.181	45307	49690	i) During antecedent condition, 400/220KV 315MVA ICT at Dehar(BB) was carrying 90MW among which 220KV Dehar(BB)-Kangoo(HP) (HPPTCL) Ckt was carrying 48MW and 220/132KV 40MVA ICT-1 & 2 were carrying 22MW and 20MW respectively and the total of 42MW was evacuating through 132 KV Dehar(BB)-Kangoo(HP) (HPPTCL) Ckt. ii) As reported, at 13:54 hrs, 220KV Dehar(BB)-Kangoo(HP) (HPPTCL) Ckt tripped on R-N phase to earth fault (Exact reason and location of fault yet to be shared). iii) During the same time, 132 KV Dehar(BB)-Kangoo(HP) (HPPTCL) Ckt also tripped due to over-current (as reported by SLDC-HP). iv) As per PMU at Panchkula(PG), B-N phase to earth fault is observed with fault clearing time of 80ms. v) As per SCADA, change in demand of approx. 100MW is observed in HP control area. vi) As reported by SLDC-HP, load loss of approx. 90MW is observed in HP control area.	1) 220 KV Dehar(BB)-Kangoo(HP) (HPPTCL) Ckt 2) 132 KV Dehar(BB)-Kangoo(HP) (HPPTCL) Ckt
4	GI-2	Himachal Pradesh	16-Nov-2023 02:01	16-Nov-2023 08:14	06:13	0	0	0.000	0.000	27333	33193	i) 400/220KV Gumma(HP) has one and half breaker scheme at 400KV level. ii) As reported, at 02:01hrs, bus bar protection operated at both 400KV Bus 1 & 2 at Gumma(HP) due to malfunction of relay P746 (Y-phase relay of Bus-1). Due to this, both the 400KV buses at Gumma(HP) became dead and 400/220 KV 315 MVA ICT-1 & 2 at Gumma(HP) also tripped. iii) Though the main CBs at Gumma(HP) of 400KV Gumma(HP)-Nahpa Jhark Ckt-18 & 2 and 400KV Gumma(HP)-Panchkula(PG) Ckt-1 & 2 tripped due to bus bar protection operation, but the CBs of the said four circuits remained closed. iv) As per PMU at Panchkula(PG), no fault is observed in the system. v) As per SCADA, no change in demand in observed in HP control area.	1) 400 KV Gumma (HP) - Bus 1 2) 400 KV Gumma (HP) - Bus 2 3) 400/220 KV 315 MVA ICT 1 at Gumma (HP) 4) 400/220 KV 315 MVA ICT 2 at Gumma (HP)
5	GI-2	Rajasthan	17-Nov-2023 20:36	17-Nov-2023 23:11	02:35	0	80	0.000	0.179	34411	44672	i) 400KV Suratgarh(RS) has one and half breaker bus scheme. 400KV Suratgarh SCTPS-Suratgarh ckt-18.2 acts as interconnector between Suratgarh SCTPS and Suratgarh S/L, having three CBs Unit-3, 4, 5 & 6 are connected at 400KV Suratgarh(RS) and Unit-7 & 8 of SCTPS are further connected via two interconnectors. ii) During antecedent condition, Unit-3, 4, 5 & 6 were already boxed up and Unit-7 & 8 of SCTPS were running at 488MW and 247MW load respectively. Power imported by SCTPS to Suratgarh(RS) via interconnectors were 245MW and 241MW respectively. 400 KV Suratgarh(RVUN)-Bikaner(RS) (RS) Ckt and 400 KV Suratgarh(RVUN)-Ratangarh(RS) (RS) Ckt-1 were carrying approx. 113MW and 144MW respectively. 400 KV Suratgarh(RVUN)-Ratangarh(RS) (RS) Ckt-2 was already tripped manually at 19:42 hrs. iii) As reported, at 20:36hrs, 400 KV Suratgarh(RVUN)-Bikaner(RS) (RS) Ckt tripped on B-N phase to earth fault with fault distance of 28km from Suratgarh(RS) end. As per DR at Suratgarh(RS) end, B-N phase to earth fault occurred and zone-1 distance protection operated with fault current of ~9.05kA from Suratgarh(RS) end and fault clearing time of ~68ms (DR time sync issue observed at Suratgarh(RS) end). As per DR at Bikaner(RS) end, R-N phase to earth fault occurred and zone-1 distance protection operated with fault current of ~1.69kA from Bikaner(RS) end and fault clearing time of ~48ms (Phase sequence issue observed at Suratgarh(RS) and Bikaner(RS) end). iv) During the same time, over-current protection of B-phase operated at 400KV Suratgarh SCTPS-Suratgarh ckt-18.2 (as reported). v) 400 KV Suratgarh(RVUN)-Ratangarh(RS) (RS) Ckt-1 tripped at the same time due to over-voltage stage-1 protection operation. As per DR at Suratgarh(RS) end, phase voltages of R, Y and B phase were respectively 236.44kV, 244.83kV and 254.83kV depicting B phase voltage ~110.34% (DR time sync issue observed at Suratgarh(RS) end). vi) As per PMU at Bikaner(RS) (RS), Y-N phase to earth fault with fault-clearing time of 80ms is observed. vii) As per SCADA, change in demand of approx. 80MW is observed Rajasthan control area. No change in generation is observed at Suratgarh(RS).	1) 400 KV Suratgarh(RVUN)-Ratangarh(RS) (RS) Ckt-1 2) 400 KV Suratgarh SCTPS(RVUN)-Suratgarh(RS) (RS) Ckt-1 3) 400 KV Suratgarh SCTPS(RVUN)-Suratgarh(RS) (RS) Ckt-2 4) 400 KV Suratgarh(RVUN)-Bikaner(RS) (RS) Ckt
6	GI-1	Punjab	30-Nov-2023 06:51	30-Nov-2023 08:14	01:23	463	60	1.331	0.140	34790	43000	i) 220/132KV Ropar GGSTP(RS) has main and transfer bus scheme at 220KV level. ii) During antecedent condition, 210 MW Guru Gobind Singh TPS (Ropar) - UNIT 4 (carrying ~164MW), UNIT 5 (carrying ~148MW) & UNIT 6 (carrying ~151MW) and 220KV feeders to Kharar, Mohali & Gobindgarh-2 were connected to 220KV main bus section-III. Rest of the elements were connected to main bus section-I & II. iii) As reported, at 06:51 hrs, 220KV GGSTP-Kharar Ckt tripped on R-N phase to earth fault [zone-1 distance protection operated] with fault current of 4.071kA and fault distance of 33.91km from GGSTP end. Fault occurred due to heavy lightning. iv) On this fault, all other elements connected to 220KV main bus section-III tripped. (Exact reason yet to be shared) v) As reported by GGSTP Ropar, 220KV GGSTP-Bassi Pathana Ckt (connected to 220KV main bus section-I) and 220KV GGSTP-Gobindgarh Ckt-1 (connected to 220KV main bus section-II) also tripped during the same time. (Exact reason yet to be shared) vi) As per SCADA SOE, 66kV Morinda-Kharar(PG) Ckt also tripped at the same time. (Exact reason yet to be shared) vii) As per PMU at Jalandhar(PG), R-N phase to earth fault is observed with delayed fault clearance time of 440ms. viii) As per SCADA, generation loss of approx. 463MW occurred at Ropar GGSTP. ix) As per SCADA load loss of approx. 60MW is observed in Punjab control area.	1) 210 MW Guru Gobind Singh TPS (Ropar) - UNIT 4 2) 210 MW Guru Gobind Singh TPS (Ropar) - UNIT 5 3) 210 MW Guru Gobind Singh TPS (Ropar) - UNIT 6 4) 220KV GGSTP-Kharar Ckt 5) 220KV GGSTP-Mohali Ckt 6) 220KV GGSTP-Bassi Pathana Ckt 7) 220KV GGSTP-Gobindgarh Ckt-1 8) 220KV GGSTP-Gobindgarh Ckt-2
7	GD-1	Uttarakhand	30-Nov-2023 17:31	30-Nov-2023 19:17	01:46	500	0	1.371	0.000	36478	47744	i) 400KV Tehri(THDC) has double main bus scheme. ii) During antecedent condition, 250 MW TEHRI HPS - UNIT 2 was running at approx. 250MW and 250 MW TEHRI HPS - UNIT 3 was synchronized to grid at 17:24 hrs and was increasing the total generation upto approx. 500MW. iii) As reported, at 17:31 hrs, 400 KV Tehri(THDC)-Koteshwar(PG) (PG) Ckt-1 tripped from both the ends and 400 KV Tehri(THDC)-Koteshwar(PG) (PG) Ckt-2 tripped from Tehri end only on line bus duct differential protection operation. iv) On tripping of both 400 KV Tehri(THDC)-Koteshwar(PG) (PG) Ckt-1 & 2, 250 MW TEHRI HPS - UNIT 1, 2 & 3 also tripped due to loss of evacuation path. v) 400 KV Tehri(THDC)-Koteshwar(PG) (PG) Ckt-2, was synchronized at 18:01hrs. Subsequently, 250 MW TEHRI HPS - UNIT 1, 2 & 3 were also synchronized to the grid and load was gradually increased. vi) At 18:15 hrs, 400 KV Tehri(THDC)-Koteshwar(PG) (PG) Ckt-2 tripped again from Tehri end on line bus duct differential protection operation. On this, 250 MW TEHRI HPS - UNIT 1, 2 & 3 also again tripped due to loss of evacuation path. vii) As reported by TehriHEP, on analysis it was found that the differential current is being measured as CT of line bus duct differential relay was erroneously shorted instead of CT of bus bar protection relay due to mismatch of drawing and actual field connections during Main-II bus bar protection related works at Tehri HEP. However, the issue is addressed and resolved. viii) As per PMU at Koteshwar(PG), Y-B phase to phase fault is observed with fault clearance time of 80ms at 17:31 hrs and no fault is observed in the system at 18:15 hrs. ix) As per SCADA, generation loss at Tehri HEP of approx. 500MW and 205MW are observed at 17:31hrs and 18:15hrs respectively.	1) 400 KV Tehri(THDC)-Koteshwar(PG) (PG) Ckt-1 2) 400 KV Tehri(THDC)-Koteshwar(PG) (PG) Ckt-2 3) 250 MW TEHRI HPS - UNIT 1 4) 250 MW TEHRI HPS - UNIT 2 5) 250 MW TEHRI HPS - UNIT 3

20.Details of Grid Events during the Month of November 2023 in Western Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI for GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	WR	03-Nov-23 17:52	03-Nov-23 18:14	00:22	-	200	-	0.30%	72953	66421	At 17:52 hrs/03-11-2023, 6 phase lightning arrestor of 220 kV Kaws-ichhapore-1 at ichhapore station failed and resulted Bus Bar protection operation and 220 kV ichhapore-Bus-1&2 and all connected elements tripped. Load loss of 200 MW occurred at ichhapore due to the event.	Tripping of following elements- 1. 220 kV ichhapore-Bus-1&2 2. 220 kV Kaws-ichhapore-1&2 3. 220 kV Sachin-ichhapore-1 4. 220 kV RIL-ichhapore-1&2 5. 220/66 kV ichhapore-ICT-1,2,3&4
2	GI-1	WR	11-Nov-23 06:54	11-Nov-23 12:00	05:06	152	-	0.23%	-	67080	60700	At 06:54 Hrs/11-11-2023, 220 kV Bus coupler of Dedhiya substation of Nakhatrana Wind PSS tripped on overcurrent protection resulting in tripping of 220 kV Dedhiya Bus-2 and 220/33 kV Dedhiya-ICT-1&2 (connected to Bus-2) became dead. 220/33 kV Dedhiya-ICT-1&2 were hand tripped at 08:24 hrs to prevent high inrush current during charging of bus coupler. Generation loss of 152 MW occurred at Dedhiya Wind power plant due to the event.	Tripping of following elements- 1. 220 kV Dedhiya-Bus coupler 2. 220/33 kV Dedhiya-ICT-1&2 3. 220 kV Dedhiya Bus-2
3	GD-1	WR	07-Nov-23 23:12	08-Nov-23 00:38	01:26	19.13	-	0.03%	-	67521	61977	At 23:12 Hrs/07-11-2023, 220 kV Indore-Pritamnagar tripped on R-E fault at Pritamnagar end only and auto recloser successful at Indore end, Pritamnagar Wind Power Plant became dead. Generation loss of 19.13 MW occurred Pritamnagar Wind Power Plant due to loss of evacuation path.	Tripping of following elements- 1. 220 kV Indore-Pritamnagar-1 2. 220/33 kV Pritamnagar-ICT-1,2&3 3. 220 kV Pritamnagar Bus-1&2
4	GD-1	WR	13-Nov-23 11:11	13-Nov-23 12:18	01:07	-	167	-	0.28%	57136	58781	At 11:10 Hrs /13.11.2023, opening of 400 kV Jetpur-Bus Coupler Bay-B phase Bus-2 side Dropper Jumper from Fly-Over Bus resulted in tripping of 400kV Jetpur Bus-1 and all connected elements on Bus Bar protection operation. 400/220 kV Jetpur ICT-2 tripped on over current protection operation and 400 kV TPCL Mundra-Jetpur-2 tripped on overvoltage stage-1&2 operation. 400 kV Jetpur-Anneil-2 and 400/220 kV Jetpur ICT-3 (500 MVA) were under outage due to high voltage. Load loss of 167 MW occurred due to SP5 operation on overloading of 400/220 kV Jetpur ICT-2.	Tripping of following elements- 1. 400 kV TPCL Mundra-Jetpur-1&2 2. 400 kV Jetpur-Anneil-1 3. 400/220 kV Jetpur ICT-1,2 (315 MVA) 4. 400kV Jetpur Bus-1 5. 400/220 kV Jetpur ICT-4 (500 MVA)
5	GI-2	WR	14-Nov-23 05:00	14-Nov-23 09:45	04:45	-	-	-	-	53156	48265	At 05:00 Hrs /14-11-2023, fault in R-phase Disconnector switch (DS1) of 400 kV Vav-Kosamba resulted in tripping of 400 kV Vav Bus-1 and all the connected elements on Bus bar protection operation. 400 kV Vav-Kosamba was under outage due to high voltage but its Disconnector switch (DS1) was closed. 400 kV Ukai-Vav connected to 400 kV Vav Bus-2 also tripped during the event on Zone-4 protection operation and Carrier received. No load loss occurred due to the event.	Tripping of following elements- 1. 400/220 kV Vav-ICT-3 (500 MVA) 2. 400 kV Vav Bus Reactor 3. 400 kV Vav Bus Coupler 4. 400 kV Ukai-Vav (Connected to 400 kV Vav Bus- 2) 5. 400 kV Vav-Kosamba-1
6	GD-1	WR	25-Nov-23 18:20	25-Nov-23 19:37	01:17	35	-	0.05%	-	70810	65221	At 18:20 Hrs/25-11-2023, 220kV Kotda Madh-Bhuj-1 tripped on R-E fault from Kotda Madh end and Auto Recloser successfully operated at Bhuj end. Kotda madh end AR blocked due to unhealthy PLC. Generation loss of 40 MW reported at Kotda Madh wind power plant due to loss of evacuation path.	Tripping of following elements- 1. 220 kV Bhuj-Kotda Madh-1
7	GD-1	WR	27-Nov-23 02:20	27-Nov-23 18:52	16:32	115	-	0.24%	-	48200	42027	At 02:20 Hrs / 27-11-2023, 220 kV Indore(PG)-Pritamnagar-1 tripped on Y-E fault due to Y phase jumper open at fault location. Generation loss of 115 MW occurred at Pritamnagar (AWEMP1PL) wind power plant due to loss of evacuation path.	Tripping of following elements- 1. 220 kV Indore(PG)-Pritamnagar-1
8	GD-1	WR	27-Nov-23 13:55	27-Nov-23 16:03	02:08	17	-	0.03%	-	61545	52483	At 13:55 Hrs / 27-11-2023, 220 kV Bhuj-Kotda Madh-1 tripped due to false DT signal received from Bhuj end. Generation loss of 17 MW occurred at Kotda Madh (Alfanar) wind power plant due to loss of evacuation path.	Tripping of following elements- 1. 220 kV Bhuj-Kotda Madh-1
9	GI-1	WR	28-Nov-23 16:27	28-Nov-23 18:15	01:48	135	-	0.002	-	63658	53346	At 16:27 Hrs / 29-11-2023, 220 kV Bhuj-Nanavalka-1 auto reclosed successfully at both end on R-E fault. At same time 220/33 kV Nanavalka-ICT-1&2 tripped on Over Current Earth Fault protection operation. No fault found during inspection. Generation loss of 135 MW occurred at Nanavalka (Alfanar) wind power plant due to the event.	Tripping of following elements- 1. 220/33 kV Nanavalka-ICT-1&2
10	GD-1	WR	29-Nov-23 16:06	29-Nov-23 18:09	02:03	-	487	-	0.009	62831	52678	At 16:04 Hrs / 29-11-2023, R phase CT of Bus coupler burst at 220 kV Kharadpada. 220 kV Kharadpada-New Kharadpada-1&2 tripped on Over Current Earth Fault, 200 kV Vapi-Bhilosa-1 and 220 kV Vapi-Kharadpada-1 tripped from Vapi end only on zone-2 R-E fault. Due to these trippings powerflow to 220 kV Kharadpada-Bus-1&2 became zero (Bus became Dead). Load loss of 487 MW occurred at Kharadpada and Bhilosa due to the event.	Tripping of following elements- 1. 220 kV Kharadpada-New Kharadpada-1&2 2. 200 kV Vapi-Bhilosa-1 3. 220 kV Vapi-Kharadpada-1

20. Details of Grid Events during the Month of November 2023 in Southern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI for GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-1	Tamil Nadu	03-Nov-23 03:19	03-Nov-23 07:12	3hrs 53mins	0	200	0.00%	0.57%	30368	34874	Complete Outage of 230kV/110kV Koyambedu SS, 230kV Guindy SS, 230kV/33kV Mambalam SS and 230kV/33kV RA Puram SS of TANTRANSCO: During antecedent conditions, were being radially fed from 230kV/110kV Koyambedu SS. As per the reports submitted the triggering incident was B-N fault in 230kV Koyambedu Guindy line.Koyambedu end failed to clear the fault and the fault was cleared but remote ends of other connected lines in zone-3. This resulted in complete outage of 230kV/110kV Koyambedu SS, 230kV Guindy SS, 230kV/33kV Mambalam SS and 230kV/33kV RA Puram SS	1. 230kV Koyambedu Guindy 2. 230kV Koyambedu Alamathy 3. 230kV Koyambedu Siperumbudur 4. 230kV Koyambedu Ambathur 5. 230kV Koyambedu Thiruvarkadu
2	GD-1	Kerala	08-Nov-23 13:35	08-Nov-23 14:12	37mins	25	420	0.06%	0.91%	43336	45916	Complete Outage of 220kV/110kV/11kV Mylatty SS, 220kV/110kV Ambalathara SS, 220kV/110kV/11kV Kanhirode SS, 220kV/110kV/33kV Talparamba SS and 220kV/110kV Talassery SS of KSEB: As per the reports submitted, 220kV Orkattery Areakode was under outage. The triggering incident was fault in 220kV Areakode Kanhirode Line-1 and the line tripped. Since 220kV/110kV/11kV Mylatty SS, 220kV/110kV Ambalathara SS, 220kV/110kV/11kV Kanhirode SS, 220kV/110kV/33kV Talparamba SS and 220kV/110kV Talassery SS are radially connected through 220kV Areakode Kanhirode Line-1, tripping of the 220kV Line resulted in complete outage of the stations.	220kV Areakode Kanhirode Line-1
3	GD-1	Karnataka	08-Nov-23 22:14	09-Nov-23 01:20	3hrs 06mins	1089	260	3.15%	0.65%	34624	40171	Complete Outage of 400kV/220kV UPCL, 220kV MSEZ SS, 220kV Kavor and Tripping of 220kV Bus-1 of 220kV Kemar SS: During antecedent conditions, 220kV Kemar SS was operating with a bus split condition with 220kV Puttur, 220kV Varahi, and 220kV Heggur on 220kV Bus-2 and 220kV UPCL, 220kV MSEZ and 220kV Kavor on 220kV Bus-1. 400kV UPCL was generating approximately 1000MW. The triggering incident was a B-N fault on 400kV UPCL Hebbanahalli Line-1&2 during heavy rain. The fault was sensed in zone-2 at both ends and the lines tripped with a delay of approximately 400ms. Immediately, UPCL Unit-2 tripped on SPS operation. Tripping of both these lines resulted in an unstable island with 400kV/220kV UPCL, 220kV MSEZ SS, 220kV Kavor SS, 220kV Kemar Bus-1 where UPCL Unit-1 generation was more than load. Immediately, under frequency protection operated and UPCL Unit-1 tripped. This resulted in a complete outage of 400kV/220kV UPCL, 220kV MSEZ SS, 220kV Kavor, and 220kV Kemar Bus-1.	1. 400kV UPCL Hebbanahalli Line-1&2
4	GD-1	Tamil Nadu	11-Nov-23 03:21	11-Nov-23 04:05	44mins	100	0	0.33%	0.00%	30478	34994	Complete Outage of 230kV JSW_Vlathikulam_Wind Station: As per the reports submitted, the triggering incident was Y-N fault in 230kV TTGS JSW_Vlathikulam_Wind line and the line tripped. Tripping of the only connected line resulted in complete outage of 230kV JSW_Vlathikulam_Wind Station.	230kV TTGS JSW_Vlathikulam_Wind
5	GD-1	Andhra Pradesh	13-Nov-23 07:59	13-Nov-23 10:26	2hr 27mins	0	0	0.00%	0.00%	34550	39401	Complete Outage of 220kV Talapally SWS of APTRANSCO: As per the reports submitted, the triggering incident is the R-phase CVT failure for 220kV Talapally-Srisailem RB-2 at Talapally end. Since VT fuse fail was observed before the eventual failure of the CVT in the line, the zone protection was blocked in the line. The fault in R-phase was observed to have a fault current of 320A and developed into a R-Y fault after around 250ms. Since the zone protection was blocked in the line of 220kV Talapally-Srisailem RB-2, the line tripped at remote end on Z2 or in Z4 at Talapally end. The 400/220kV N'Sagar ICTs were feeding the fault while sensing Z2 (Back up impedance) and have sensed after around 600ms, the ICTs tripped on Z2. With the tripping of all the elements connected to 220kV Talapally SS led to complete outage of the station	1. 220kV Talapally Nsagar-1,2&3 2. 220kV Talapally Srisailem-1&2 3. 220kV Talapally Rentachintala 4. 220kV Talapally Piduguralla-1&2 5. 220kV Talapally VTPS 6. 400kV/220kV Nsagar ICT-1,2&3
6	GD-1	Karnataka	20-Nov-23 10:19	20-Nov-23 11:05	46mins	14	363	0.03%	0.70%	50677	51988	Tripping of 220kV Bus-1 of 400kV/220kV Guttur SS, 220kV Bus-2 of 220kV/66kV Honnali SS and complete outage of 220kV/110kV Ranebenur SS, 220kV/66kV Davanagere SS, 220kV/66kV Hosadurga SS, 220kV/66kV Benikere SS & 220kV/66kV Guttur SAS SS of KPTCL. During antecedent conditions, 220kV Guttur Haveri Line-2 was under outage. 220kV Bus-2 of 220kV/66kV Honnali SS, 220kV/110kV Ranebenur SS, 220kV/66kV Davanagere SS, 220kV/66kV Hosadurga SS, 220kV/66kV Benikere SS, and 220kV/66kV Guttur SAS SS were being radially fed through 220kV Bus-1 of 400kV/220kV Guttur SS. As per the reports submitted, triggering incident was an R-phase jumper failure in 220kV Guttur Ranebenur Line bay causing an RY fault in 220kV Bus-1 of 400kV/220kV Guttur SS. Immediately BBP operated and all the elements connected to the bus tripped including LV side of 400kV/220kV Guttur ICT-1. This resulted in a complete outage of these stations and 220kV Bus-2 of 220kV/66kV Honnali SS.	1. 400/220 kv ICT-1 AT GUTTUR
7	GD-1	Tamil Nadu	22-Nov-23 18:22	23-Nov-23 20:17	1hr 55mins	0	0	0.00%	0.00%	39153	47552	Complete Outage of 230kV SPRNG_Pugalur: As per the reports submitted, the triggering incident was R-N fault in 230kV Pugalur Sprng_Pugalur line and the line tripped. Tripping of the only connected line resulted in complete outage of 230kV SPRNG_Pugalur.	1. 230kV Pugalur Sprng_Pugalur line
8	GD-1	Andhra Pradesh	22-Nov-23 00:39	24-Nov-23 09:39	9hrs	0	0	0.00%	0.00%	32560	32548	Complete outage of 400kV RYTPP Generating station of APGENCO: As per the reports submitted, the triggering incident was Over voltage at RYTPP end. Immediately Over voltage protection of 400kV RYTPP Kalkiri Line-2&2 operated at RYTPP end and DT was sent to Kalkiri. Tripping of both lines resulted in complete outagae of 400kV RYTPP Generating station .	1. 400kV RYTPP Kalkiri Line-1&2
9	GD-1	Karnataka	24-Nov-23 08:05	24-Nov-23 09:17	1hr 12 mins	0	159	0.00%	0.38%	41846	41991	Complete Outage of 220kV/66kV Ancheplaya SS, 220kV/66kV Magadi SS and 220kV/66kV Naga Mangala SS of KPTCL: As per the reports submitted, the triggering incident was tripping of 400/220kV Nelamangala 500MVA ICT 2 on operation of WTL. Immediately, SPS operated tripping the 220kV Nelamangala-Ancheplaya and 220kV Nelamangala-Magadi lines. Tripping of these lines resulted in a complete outage of 220kV/66kV Ancheplaya SS, 220kV/66kV Magadi SS and 220kV/66kV Naga Mangala SS. At the same time, 400kV Nelamangala-Hassan line, whose main breaker was under outage and was charged through Tie of 400kV/220kV Nelamangala ICT-2, also tripped along with the ICT-2	1. 400kV Nelamangala-Hassan 2. 220kV Nelamangala - Magadi 3. 220kV Nelamangala - Ancheplaya 4. 220kV Ancheplaya - Nagamangala 1&2 5. 400/220kV Nelamangala 500MVA ICT 2
10	GD-1	Karnataka	29-Nov-23 12:15	29-Nov-23 12:25	10mins	50	270	0.12%	0.63%	40890	42824	Complete Outage of 220kV/66kV Begur SS, 220kV/66kV TK Halli SS, 220kV/66kV Chamrajnagar SS, 220kV/66kV Madhuvanahalli SS and 220kV/66kV Hebbani SS and 220kV Bus-2 of 220kV/66kV Vajamangala SS of KPTCL. During antecedent conditions, 220kV TK Halli Kankapura line was under outage. As per the reports submitted, the triggering incident was a phase jumper failure in 220kV Mysore Kadakola line. At the same time, 220kV Kadakola Adakamahally Line-1&2 tripped on OEF protection at Kadakola end. Subsequently, Vajamangala bus coupler tripped on over current protection. This resulted in complete outage of 220kV/66kV Begur SS, 220kV/66kV TK Halli SS, 220kV/66kV Chamrajnagar SS, 220kV/66kV Madhuvanahalli SS and 220kV/66kV Hebbani SS and 220kV Bus-2 of 220kV/66kV Vajamangala SS.	1. 220kV Kadakola Adakamahally Line-1&2 2. 220kV Kadakola Mysore line
11	GI-1	Kerala	07-Nov-23 13:02	07-Nov-23 14:33	1hr 31mins	0	0	0.00%	0.00%	42815	47556	Tripping of 220kV/110kV Pothencode SS of KSEB: As per the reports submitted, the triggering incident was tension insulator failure in 220kV Bus-1 causing an R-N fault in the bus. Immediately BBP operated and all the elements connected to the bus tripped.	1. 220kV Edamon Pothencode Line-1 2. 220kV Pothencode Kattakada Line-1 3. 220kV Pothencode Trivandrum Line-1 4. 220kV/110kV Pothencode Transformer Line-1&3
12	GI-2	Tamil Nadu	12-Nov-23 02:23	12-Nov-23 08:19	5hr 56mins	267	0	0.95%	0.00%	28206	33659	Tripping of 400kV Bus-1 of 400kV NTECL Vallur Generating station: As per the reports submitted, while withdrawing NTECL Vallur Unit-2, LBB operated and all the elements connected to the 400kV NTECL Bus-1 tripped.	1. 400kV Vallur Kalvendapattu Line-1 2. 400kV Vallur Alamathi Line-1 3. 400kV Vallur NCTPS line-1 4. 400kV Vallur Unit-3

20. Details of Grid Events during the Month of November 2023 in Eastern Region													
Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
	NIL												

20.Details of Grid Events during the Month of November 2023 in North Eastern Region



Sl No.	Category of Grid Event	Affected Area	Time and Date of occurrence of Grid Event	Time and Date of Restoration	Duration (HH:MM)	Loss of generation / loss of load during the Grid Event		% Loss of generation / loss of load w.r.t Antecedent Generation/Load in the Regional Grid during the Grid Event		Antecedent Generation/Load in the Regional Grid*		Brief details of the event (pre fault and post fault system conditions)	Elements Tripped
	(GI 1or GI 2/ GD-1 to GD-5)					Generation Loss(MW)	Load Loss (MW)	% Generation Loss (MW)	% Load Loss (MW)	Antecedent Generation (MW)	Antecedent Load (MW)		
1	GD-I	Tinsukia & Margherita area of Assam Power system	07-Nov-23 12:48	07-Nov-23 13:38	00:50	98	53	4.29%	2.78%	2287	1905	Tinsukia along with Margherita areas of Assam Power System were connected with the rest of the grid by 220 kV Kathalguri-Tinsukia 2 line, 220 kV NTPS-Tinsukia line, 220 kV NRPP-Tinsukia line, 132 kV Dibrugarh-Tinsukia line, 132 kV Bordubi-Tinsukia line, 132 kV Rupai-Tinsukia line & 132 kV Rupai-Margherita lines. 220 kV Kathalguri-Tinsukia 1 line was under planned shutdown prior to the event. At 12:48 Hrs of 07-11-2023, 220 kV Kathalguri-Tinsukia 2 line, 220 kV NTPS-Tinsukia line, 220 kV NRPP-Tinsukia line, 132 kV Dibrugarh-Tinsukia line, 132 kV Bordubi-Tinsukia line & 132 kV Rupai-Tinsukia line & 132 kV Rupai-Margherita lines tripped. Due to tripping of this elements, Tinsukia along with Margherita area of Assam Power System got separated from rest of NER grid and collapsed due to no source available in these areas. Power was extended to Tinsukia and Margherita areas of Assam Power system by charging 220 kV Tinsukia-NTPS line & 132 kV Rupai-Margherita line at 13:38 Hrs and 14:02 Hrs of 07-11-2023 respectively.	220 kV Kathalguri-Tinsukia 2 line, 220 kV NTPS-Tinsukia line, 220 kV NRPP-Tinsukia line, 132 kV Dibrugarh-Tinsukia line, 132 kV Bordubi-Tinsukia line, 132 kV Rupai-Tinsukia line & 132 kV Rupai-Margherita lines
2	GD-I	Rokhia area of Tripura Power system	17-Nov-23 14:12	17-Nov-23 15:41	01:29	20	6	1.18%	0.37%	1698	1642	Rokhia area of Tripura Power System were connected with the rest of the grid by 132 kV Rokhia-Agartala I & II lines and 132 kV Rokhia-Monarchak lines. At 14:12 Hrs of 17-11-2023, 132 kV Rokhia-Agartala D/C & 132 kV Rokhia-Monarchak lines tripped. Due to tripping of this elements, Rokhia area of Tripura Power System got separated from rest of NER grid and collapsed due to load-generation mismatch in this area. Power was extended to Rokhia area of Tripura Power system by charging 132 kV Rokhia-Agartala I line at 15:41 Hrs of 17-11-2023.	132 kV Agartala-Rokhia I & II and 132 kV Rokhia-Monarchak
3	GD-I	Kohima area of Nagaland Power System	25-Nov-23 11:51	25-Nov-23 12:21	00:30	17	0	0.92%	0.00%	1853	1874	Kohima area of Tripura Power System were connected with the rest of the grid by 132 kV Dimapur- Kohima, 132 kV Karong- Kohima and 132 kV Chiempobozou- Kohima lines. At 11:51 Hrs of 25-11-2023, 132 kV Dimapur- Kohima, 132 kV Karong- Kohima and 132 kV Chiempobozou- Kohima lines tripped. Due to tripping of this elements, Kohima area of Nagaland Power System got separated from rest of NER grid and collapsed due to load-generation mismatch in this area. Power was extended to Kohima area of Tripura Power system by charging 132 kV Karong- Kohima line at 12:21 Hrs of 25-11-2023.	132 kV Dimapur- Kohima, 132 kV Karong- Kohima and 132 kV Chiempobozou- Kohima lines.

21. NEW ELEMENTS COMMISSIONED* IN NOVEMBER 2023

COMMISSIONING OF NEW GENERATING UNITS							
REGION	S.NO.	ENTITY	UNIT NAME / LOCATION	ADD. CAPACITY (MW)	UNIT NO.	INS. CAPACITY (MW)	DATE®
NR	1	NUPPL	NUPPL_Stage-I,Ghatampur(Thermal)/Uttar Pradesh	660	I	1980	04-11-2023
WR	2	NTPC Renewable Energy Limited	NTPC_REL_DYPR_Bhuj(Wind)/Gujarat	50	I	50	04-11-2023
SR	3	RENEW SURYA ROSHINI	Renew Surya Roshini(Solar)/Karnataka	72.77	I	72.77	09-11-2023
	4			62.37	II	135.14	10-11-2023
	5			69.3	III	204.44	10-11-2023
	6	AYANA RENEWABLE SIX	Ayana Renewable Six(Wind)/Karnataka	50.4	I	50.4	15-11-2023
	7			50.4	II	100.8	30-11-2023
ER	8	NTPC	North Karanpura(Thermal)/Jharkhand	660	II	1320	09-11-2023
NER	9	NEEPCO	Kopili(Hydro)/Assam	50	II	200	03-11-2023

®RE (Solar & Wind generation added capacity FTC is in phases. Date mentioned is the date by which complete (Added) capacity

COMMISSIONING OF NEW ICTs							
REGION	S.NO.	ENTITY	SUB-STATION	VOLTAGE (kV)	ICT NO.	CAPACITY (MVA)	DATE
NR	1	PRTL	Fatehgarh	400/220	IV	500	17-11-2023
	2	PGCIL	Kurukshetra	400/220	III	500	20-11-2023
SR	3	KARUR	Karur	400/230	II	500	29-11-2023
	4	KARUR	Karur	400/230	I	500	29-11-2023
NER	5	TPTL	Gournagar	132/33	I	50	06-11-2023
	6	ASSAM	Tezpur	132/33	II	50	09-11-2023
	7	ASSAM	Narengi	132/33	I	50	17-11-2023
	8	ASSAM	Ghoramari	132/33	I	31.5	17-11-2023

COMMISSIONING OF NEW TRANSMISSION LINES						
REGION	S.NO.	ENTITY	LINE NAME	CKT NO.	VOLTAGE (kV)	DATE
NR	1	GTL,NUPPL	765kV Agra Fatehbad-Ghatampur_TPS	I	765	01-11-2023
	2	UPPTCL	400kV Simbholi-Muradnagar_2	I	400	03-11-2023
	3	UPPTCL	400kV Simbholi-Muradnagar_2	II	400	03-11-2023
	4	NRSS36 TRANSMISSION LTD	400kV Babai-Bhiwani	II	400	08-11-2023
	5	NRSS36 TRANSMISSION LTD	400kV Babai-Bhiwani	I	400	08-11-2023

LILO OF TRANSMISSION LINES						
REGION	S.NO.	ENTITY	LINE NAME (LILOed LINE)	LILO AT	VOLTAGE (kV)	DATE

COMMISSIONING OF BUS/LINE REACTORS						
REGION	S.NO.	ENTITY	SUB-STATION / LINE (@SUB-STATION)	VOLTAGE (kV)	CAPACITY (MVAR)	DATE
NR	1	GTL,NUPPL	Agra(Fatehabad)- Ghatampur TPS@Ghatampur TPS	765	189	01-11-2023
ER	2	WBSETCL	Bus Reactor-I@Chanditala	400	125	09-11-2023
NER	3	PGCIL	Bus Reactor-I@Balipara	400	63	06-11-2023
	4	PGCIL	Lower Subansiri IV-Biswanath Chariali@Biswanath Chariali	400	63	01-11-2023

COMMISSIONING OF HVDC/FACTS DEVICE/SYSTEM						
REGION	S.NO.	NAME	SUB-STATION / CONSTITUENT	RATINGS	VOLTAGE (kV)	DATE

*Includes first time charging (FTC) as well

22. Voltage Profile

विद्युत विभव रूपरेखा माह - नवम्बर 2023

VOLTAGE PROFILE - NOVEMBER 2023

Date of reporting: 05-Dec-23

क्र.स. No.	क्षेत्र REGION	उपकेंद्र SUBSTATION	VOLTAGE < V(lower)* (V=380,728 kV)	V(lower) < VOLTAGE < V(upper)*	VOLTAGE > V(upper)* (V=420,800 kV)	Voltage Deviation Index (%age of time voltage is outside range)	Number of hours voltage was outside IEGC band during month	उच्चतम (कि.वो.) Maximum(kV)	निम्नतम(कि.वो.) Minimum(kV)	औसत (कि.वो.) Average(kV)
1	पूर्वी क्षेत्र ER	अंगुल ANGUL	0%	100%	0%	0%	0	789	752	773
2		दालिपली DARPILALI	0%	100%	0%	0%	0	790	764	777
3		गया GAYA	0%	100%	0%	0%	0	785	746	771
4		जीरत JEERAT	0%	100%	0%	0%	0	781	736	758
5		झारसूड़ा JHARSUGUDA	0%	100%	0%	0%	1	801	771	787
6		मिदनापुर MEDINIPUR	0%	100%	0%	0%	0	787	749	768
7		रांची RANCHI	0%	100%	0%	0%	0	796	764	781
8		सासाराम SASARAM	0%	99%	1%	1%	9	802	741	764
1	उत्तरी क्षेत्र NR	आगरा AGRA	0%	100%	0%	0%	0	799	741	777
2		आगरा (फतेहाबाद) AGRA (FATEHABAD)	0%	100%	0%	0%	1	802	738	778
3		अजमेर AJMER	0%	100%	0%	0%	1	801	749	778
4		अलीगढ़ ALIGARH	0%	100%	0%	0%	0	794	736	773
5		अनपरा सी ANPARA-C	0%	100%	0%	0%	0	781	757	770
6		अनपरा डी ANPARA-D	0%	100%	0%	0%	0	779	756	768
7		अंता ANTA	0%	100%	0%	0%	0	796	756	778
8		बालिया BALLIA	0%	100%	0%	0%	0	767	755	762
9		बारा BARA	0%	100%	0%	0%	0	791	756	773
10		भाड़ला BHADLA	0%	100%	0%	0%	0	791	751	773
11		भाड़ला-2 BHADLA-2	0%	100%	0%	0%	0	796	754	775
12		भिवानी BHIWANI	0%	100%	0%	0%	2	803	756	782
13		बिकानेर BIKANER	0%	100%	0%	0%	0	794	751	774
14		बरेली BAREILLY	0%	100%	0%	0%	0	801	735	777
15		चित्तौड़गढ़ CHITTORGARH	0%	100%	0%	0%	0	801	744	779
16		फतेहगढ़-2 FATEHGARH-2	0%	100%	0%	0%	0	790	752	774
17		फतेहपुर FATEHPUR	0%	100%	0%	0%	0	796	738	767
18		घाटमपुर GHATAMPUR	0%	99%	1%	1%	6	806	747	783
19		ग्रेटर नोएडा GREATER NOIDA	0%	100%	0%	0%	0	799	746	780
20		हापड़ HAPUR	0%	100%	0%	0%	0	802	744	778
21		जवाहरपुर JAWAHARPUR	0%	100%	0%	0%	0	794	739	774
22		झाटिकरा JHATIKARA	0%	100%	0%	0%	0	796	750	777
23		कानपुर जीआईएस KANPUR GIS	0%	100%	0%	0%	0	793	738	772
24		खैती KHETRI	0%	100%	0%	0%	0	801	754	779
25		कोटेश्वर KOTESHWAR	0%	100%	0%	0%	0	799	746	775
26		ललितपुर LALITPUR	0%	100%	0%	0%	0	778	734	761
27		लखनऊ LUCKNOW	0%	100%	0%	0%	0	796	742	772
28		मैनपुरी MAINPURI	0%	100%	0%	0%	0	795	737	765
29		मेरठ MEERUT	0%	100%	0%	0%	2	804	750	780
30		मेरठ(UP) MEERUT (UP)	0%	100%	0%	0%	0	801	743	777
31		मोहा MOHA	0%	100%	0%	0%	0	803	733	770
32		ओबरा सी OBRA C	0%	100%	0%	0%	0	782	752	766
33		उरई ORAI	0%	100%	0%	0%	1	803	749	784
34		फागी PHAGI	0%	100%	0%	0%	0	798	750	778
35		रामपुर RAMPUR	0%	99%	1%	1%	5	807	722	764
36		उन्नाव UNNAO	0%	100%	0%	0%	0	795	735	764
37		वाराणसी VARANASI	0%	100%	0%	0%	0	792	751	775
1	पश्चिमी क्षेत्र WR	अकोला AKOLA	0%	100%	0%	0%	0	794	752	774
2		औरंगाबाद AURANGABAD	0%	100%	0%	0%	0	797	746	777
3		भोपाल (बीडीटीसीएल) BHOPAL (BDTCL)	0%	100%	0%	0%	0	792	742	770
4		भुज BHUJ	0%	100%	0%	0%	0	792	752	775
5		भुज-2 BHUJ-II	0%	100%	0%	0%	0	786	748	771
6		बिलासपुर BILASPUR	0%	100%	0%	0%	0	788	757	772
7		बेना BENA	0%	100%	0%	0%	0	801	750	779
8		बनारस BANASKANTHA	0%	100%	0%	0%	1	803	760	783
9		चापा CHAMPA	0%	93%	7%	7%	48	808	773	790
10		धुले (बीडीटीसीएल) DHULE (BDTCL)	0%	100%	0%	0%	0	797	745	775
11		धर्मजगढ़ DHARAMJAGARH	0%	100%	0%	0%	0	792	756	776
12		रायपुर पूलिंग RAIPUR POOLING	0%	100%	0%	0%	0	794	765	782
13		एकतुनी EKTUNI	0%	100%	0%	0%	0	795	744	770
14		गदरवाड़ा GADARWARA	0%	99%	1%	1%	4	802	765	784
15		ग्वालियर GWALIOR	0%	100%	0%	0%	0	801	747	780
16		इंदौर INDORE	0%	100%	0%	0%	0	797	743	773
17		जबलपुर JABALPUR	0%	99%	1%	1%	10	805	760	785
18		खडवा KHANDWA	0%	100%	0%	0%	0	798	747	775
19		कोरडी KORADI	0%	100%	0%	0%	0	784	752	769
20		लकाडिया LAKADIYA	0%	100%	0%	0%	1	802	760	782
21		रायगढ़ पूलिंग RAIGARH POOLING	0%	100%	0%	0%	0	800	772	786
22		पड़घे PADGHE	0%	99%	1%	1%	4	806	743	779
23		परली PARLI	0%	98%	2%	2%	14	805	750	784
24		पुणे PUNE	0%	99%	1%	1%	5	805	740	778
25		राजनन्दागांव RAJNANDGAON	0%	96%	4%	4%	29	810	765	788
26		सासन SASAN	0%	100%	0%	0%	0	789	760	777
27		सतना SATNA	0%	100%	0%	0%	0	800	758	782
28		सिवनी SEONI	0%	100%	0%	0%	0	797	751	775
29		सोपट SIPAT	0%	100%	0%	0%	0	788	758	773
30		सोलापुर SOLAPUR	0%	98%	2%	2%	14	806	752	785
31		तिरोड़ा TIRODA	0%	100%	0%	0%	0	793	750	766
32		ताम्रार TAMARAR	0%	100%	0%	0%	0	798	768	784
33		वडोदरा VADODARA	0%	100%	0%	0%	3	803	740	779
34		विन्ध्यवत पूलिंग VINDHYACHAL PS	0%	100%	0%	0%	0	793	767	780
35		वर्धा WARDHA	0%	100%	0%	0%	0	796	750	777
36		वरौर WARORA	0%	99%	1%	1%	4	803	755	784
1	दक्षिणी क्षेत्र SR	कडपा CUDDAH	0%	93%	7%	7%	54	812	756	785
2		चिल्कलुरिपेट CHILAKALURIPETA	0%	93%	7%	7%	53	811	756	786
3		कुरुनैल KURNOOL	0%	100%	0%	0%	0	799	747	776
4		महेश्वर MAHESHWARAM	0%	81%	19%	19%	134	812	757	790
5		निजामाबाद NIZAMABAD	0%	85%	15%	15%	106	813	758	790
6		नेल्लूर पूलिंग NELLORE PS	0%	100%	0%	0%	0	795	750	774
7		रायचूर RAICHUR	0%	100%	0%	0%	0	801	752	780
8		श्रीकाकुलम SRIKAKULAM	0%	97%	3%	3%	24	806	757	785
9		तिरुवलम THIRUVALEM	0%	100%	0%	0%	0	796	738	770
10		वेमगिरी VEMAGIRI	0%	86%	14%	14%	103	810	760	788
11		वारंगल WARANGAL	0%	85%	15%	15%	109	810	758	789
1	पूर्वोत्तर क्षेत्र NER	बिस्वनाथ चरिआलिस BISWANATH CHARIALI (400 kV)	0%	100%	0%	0%	0	419	393	408
2		बोंगाईगांव BONGAIGAON (400 kV)	0%	100%	0%	0%	0	417	400	409
3		बोंगाईगांव टीपीएस BONGAIGAON TPS (400 kV)	0%	100%	0%	0%	0	420	401	411
4		इम्फाल IMPHAL (400 kV)	0%	100%	0%	0%	0	416	396	406
5		बर्निहाट BYRNHAT (400 kV)	0%	99%	1%	1%	5	421	402	412
6		कामेंग KAMENG (400 kV)	0%	100%	0%	0%	0	414	392	404
7		अज़ारा AZARA (400 kV)	0%	100%	0%	0%	0	402	391	397
8		मिसा MISA (400 kV)	0%	100%	0%	0%	0	413	393	404
9		न्यू मारियानी NEW MARIANI (400 kV)	0%	100%	0%	0%	0	416	396	406
10		न्यू कोहिमा NEW KOHIMA (400 kV)	0%	100%	0%	0%	0	419	395	407
11	पूर्वोत्तर क्षेत्र NER	पालाताना PALATANA (400 kV)	0%	100%	0%	0%	0	413	402	409
12		पूर्वकचनबारी PK BARI (400 kV)	0%	100%	0%	0%	0	412	393	405
13		रंगानादी RANGANADI (400 kV)	0%	99%	1%	1%	7	423	396	411
14		सिलचर SILCHAR (400 kV)	0%	100%	0%	0%	0	418	399	409
15		सूर्यमणिनगर SURJYAMANINAGAR (400 kV)	0%	100%	0%	0%	0	410	394	404
16		थोबाल THOUBAL (400 kV)	0%	100%	0%	0%	0	413	380	398

All listed stations are 765 kV stations unless otherwise mentioned
 *Percentage is calculated w.r.t. Time of one month.

24. ALL TIME HIGHEST

30-11-2023

	Maximum Demand Met during the day (MW)	Demand Met during Evening Peak hrs(MW)	Energy Met (MU)	Hydro Gen. (MU)	Wind Gen. (MU)	Solar Gen. (MU)
NR	81220 04-09-2023	76274 04-09-2023	1796 04-09-2023	443 01-08-2023	86 07-08-2023	147 10-05-2023
WR	74035 01-09-2023	65518 13-10-2023	1618 02-09-2023	167 18-12-2014	273 07-08-2023	70.69 12-05-2023
SR	64314 15-03-2023	52366 18-04-2023	1329 19-04-2023	208 31-08-2018	298 21-07-2023	142.7 21-03-2023
ER	29641 28-07-2023	28579 18-04-2023	639 08-06-2023	157 14-09-2022	-	7 23-03-2023
NER	3712 04-09-2023	3510 12-08-2022	72 15-09-2023	41 29-08-2023	-	2 22-06-2022
All India	239978 01-09-2023	215592 28-08-2023	5224 02-09-2023	877 30-08-2022	611 02-08-2023	351 10-05-2023
Regions	States	Max. Demand Met during the day (MW)		Energy Consumption (MU)		
			As on date		As on date	
NR	Punjab	15237	23-06-2023	343.5	24-06-2023	
	Haryana	12692	18-08-2023	273.1	18-08-2023	
	Rajasthan	17772	02-09-2023	371.6	04-09-2023	
	Delhi	7770	29-06-2022	153.5	28-06-2022	
	UP	28553	25-07-2023	580.0	03-09-2023	
	Uttarakhand	2594	14-06-2022	55.8	17-06-2023	
	HP	2071	06-01-2023	40.7	22-08-2023	
	J&K(UT) and Ladakh(UT)	3636	29-06-2023	65.4	20-01-2023	
	Chandigarh	426	08-07-2021	8.4	08-07-2021	
Railways NR ISTS		-	-	-	-	
WR	Chhattisgarh	5832	02-09-2023	133.5	28-08-2023	
	Gujarat	24544	01-09-2023	511.6	02-09-2023	
	MP	17238	13-01-2023	335.9	10-11-2023	
	Maharashtra	28846	14-04-2022	636.1	21-06-2023	
	Goa	776	07-06-2023	17.1	08-06-2023	
	DD & DNH	1327	17-08-2023	30.9	10-08-2023	
	AMNSIL	977	18-08-2021	21.0	31-05-2022	
Balco		-	-	-	-	
SR	Andhra Pradesh*	13039	19-10-2023	263.8	16-06-2023	
	Telangana*	15497	30-03-2023	303.4	03-03-2023	
	Karnataka	16950	25-08-2023	309.5	20-04-2023	
	Kerala	5024	18-04-2023	103.3	19-04-2023	
	Tamil Nadu	19387	20-04-2023	415.4	20-04-2023	
	Pondy	512	17-05-2023	11.6	18-05-2023	
ER	Bihar	7578	27-07-2023	161.3	28-07-2023	
	DVC	3751	18-04-2023	81.2	22-04-2022	
	Jharkhand	1923	16-06-2023	43.3	17-06-2023	
	Odisha	7192	19-04-2023	145.5	25-09-2022	
	West Bengal	11868	18-04-2023	245.7	18-04-2023	
	Sikkim	137	03-02-2022	2.5	28-01-2020	
	Railways ER ISTS		-	-	-	-
NER	Arunachal Pradesh	179	16-09-2023	3.2	04-09-2023	
	Assam	2498	16-09-2023	50.0	15-09-2023	
	Manipur	258	31-01-2022	3.7	10-02-2022	
	Meghalaya	415	09-02-2022	7.8	31-01-2022	
	Mizoram	157	16-01-2023	2.3	18-01-2023	
	Nagaland	179	28-09-2023	3.29	05-09-2023	
	Tripura	370	07-06-2023	7.0	06-06-2023	
*SR	Andhra Pradesh (Undivided)	13162	23-03-2014	284.8	22-03-2014	

25. System Reliability Indices Report for the month of November 2023

Percentage (%) of times ATC was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	0	0.00	0.00
2	ER-NR	1	0.25	0.03
3	Import of NR	0	0.00	0.00
4	NEW-SR	0	0.00	0.00
5	NER Import	0	0.00	0.00

Percentage(%) of times (N-1) Criteria was violated

S.No.	Corridor	Number of Blocks Violated	Number of Hours Violated	%Violation
1	WR-NR	0	0.00	0.00
2	ER-NR	0	0.00	0.00
3	Import of NR	0	0.00	0.00
4	NEW-SR	0	0.00	0.00
4	NER Import	0	0.00	0.00

Remarks: Flows crossing Total Transfer Capability (TTC) on interregional corridors has been worked out as a proxy for (N-1) violation.



POWER SYSTEM OPERATION CORPORATION LIMITED

पावर सिस्टम ऑपरेशन कारपोरेशन लिमिटेड

NATIONAL LOAD DESPATCH CENTRE – राष्ट्रीय भार प्रेषण केन्द्र

First floor, B-9, Qutab Institutional Area,
Katwaria Sarai, New Delhi-110016.